

UNDERGRADUATE FINAL YEAR PROJECT REPORT

Department of Mathematics

NED University of Engineering and Technology



Assessing Dynamic Hedge Effectiveness and Statistical Arbitrage with Conventional and Deep Learning Models in Equity Futures

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Statement of Contributions

Throughout this project, our team embraced both unity and healthy specialization to achieve a robust and insightful analysis of minimum variance hedge ratios for the KSE-30 Index and its futures contracts. From the outset, every member—Ms. Hanadi Sabir, Ms. Zaara Asim Mirza, Mr. Hussain Aftab, and Mr. Ammar Sheikh—came together to collect and adjust futures data, diligently filling missing values and ensuring data continuity. This foundational work was carried out as a cohesive unit, reflecting our shared commitment and collective effort without which none of our subsequent achievements would have been possible.

As we progressed to the core modeling phase, we honored individual strengths and interests by dividing tasks between mathematical/statistical and deep learning approaches. Ms. Hanadi Sabir and Ms. Zaara Asim Mirza applied their exceptional analytical skills to develop and refine the conventional DCC-GARCH and Copula-GARCH models, demonstrating both precision and creativity in their statistical formulations. Meanwhile, Mr. Hussain Aftab and Mr. Ammar Sheikh harnessed their deep learning expertise to build and optimize our LSTM-CNN and FT-Net Hybrid architectures, driving innovation in self-supervised learning for dynamic hedge ratio estimation.

When we reached the stage of evaluating Fuzzy TOPSIS and Statistical Arbitrage, we deliberately structured our work to maximize shared learning. First, Ms. Zaara Asim Mirza and Mr. Hussain Aftab conducted a thorough study of the Fuzzy TOPSIS method and presented their findings to the group, while Mr. Ammar Sheikh and Ms. Hanadi Sabir simultaneously investigated statistical arbitrage opportunities and explained their conclusions. After these initial presentations, we held a group discussion to ensure everyone fully understood both techniques. To reinforce this collective knowledge and test our grasp of each other's insights, we then swapped implementation responsibilities: Ms. Zaara Asim Mirza and Mr. Hussain Aftab took on the statistical arbitrage coding based on the group's shared understanding, and Mr. Ammar Sheikh together with Ms. Hanadi Sabir implemented the Fuzzy TOPSIS ranking procedure using the insights provided by their teammates. This deliberate role reversal guaranteed that each team member not only taught but also applied both methodologies, fostering deeper comprehension and stronger collaboration across the entire project.



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Finally, inspired by our model comparisons, we collectively decided to extend the FT-Net Hybrid model by incorporating macroeconomic variables, with Mr. Ammar Sheikh and Mr. Hussain Aftab leading its advanced development and Ms. Hanadi Sabir and Ms. Zaara Asim Mirza meticulously managing the economic data collection and transformation. Throughout the report writing, all four of us collaborated seamlessly, merging our analyses, interpretations, and reflections into a coherent and polished document. It has been an honor to work alongside such dedicated and talented colleagues, whose enthusiasm, expertise, and mutual respect have driven this project to its successful conclusion.

Executive Summary

This project addresses the pressing challenge of managing risk and uncovering arbitrage opportunities in the Pakistani equity futures market, particularly for the KSE-30 index. In emerging economies like Pakistan, financial markets are often marked by heightened volatility, limited liquidity, and frequent inefficiencies that undermine the effectiveness of conventional static hedging strategies. Against this backdrop, our study aims to devise and assess dynamic hedging methodologies that adapt in real time to market conditions, thereby delivering superior risk mitigation and the potential for statistical arbitrage.

To achieve these objectives, we utilized daily data for the KSE-30 index and its corresponding futures, supplemented by weekly macroeconomic variables including GDP growth, SPI-based inflation, and SBP policy rates. This comprehensive dataset, sourced from the Pakistan Stock Exchange and the Pakistan Bureau of Statistics, enabled the development and rigorous testing of both econometric and deep learning models for dynamic hedge ratio estimation.

The methodology centers on four advanced frameworks: the Dynamic Conditional Correlation GARCH model, an enhanced Copula-DCC–GARCH model incorporating Student-t copula for improved tail dependence, a hybrid LSTM–CNN deep learning model adept at capturing both short-term and sequential dependencies in returns, and the novel FT-Net Hybrid, which uniquely combines Fourier-based spectral analysis with temporal convolution to recognize complex, non-stationary market patterns. Each model's hedging effectiveness was assessed using variance reduction, RMSE, MAD, directional accuracy, Sharpe ratio, average hedged return, VaR and CVaR reductions via Extreme Value Theory, and time complexity. These measures were synthesized through the Fuzzy TOPSIS multi-criteria decision-making process to provide an integrated model ranking.

Empirical findings consistently highlight the FT-Net Hybrid as the most effective model for risk minimization, excelling in variance reduction, tail-risk control, and predictive accuracy. Moreover, the model exhibited mean-reversion and statistical arbitrage properties, outperforming traditional methods in exploiting market inefficiencies. When further extended to incorporate macroeconomic variables, the FT-Net Hybrid demonstrated enhanced hedge stability and adaptability to macro shocks, underscoring its practical value.



Acknowledgments

We would like to express our most sincere gratitude to our Project Advisor, Miss Komal Batool, for her invaluable guidance, insightful suggestions, and constant encouragement throughout the course of our project, “Assessing Dynamic Hedge Effectiveness and Statistical Arbitrage with Conventional and Deep Learning Models in Equity Futures.” Her unwavering support was instrumental in shaping the direction and successful completion of our research. We are deeply thankful to our Co-Supervisor, Miss Ubaidah Fatima, for her constructive feedback and continuous motivation, which played a significant role in refining our work and deepening our understanding of the subject. We also extend our heartfelt thanks to our External Supervisor, Mr. Asad Khan, whose industry expertise and practical insights provided us with a much broader perspective and helped us connect our academic work to real-world challenges.

We would also like to thank the FYDP Committee for giving us the opportunity to complete this project and for their guidance at every stage. Their valuable advice has been a tremendous help throughout our research journey.



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List of Abbreviations

Abbreviation	Full Form	Abbreviation	Full Form
ACF	Autocorrelation Function	HE	Hedge Effectiveness
ADCC	Asymmetric Dynamic Conditional Correlation	HFT	High Frequency Trading
ADF	Augmented Dickey-Fuller Test	IGC	Integrated GARCH
AHP	Analytic Hierarchy Process	IUP	Index Underlying Portfolio
API	Application Programming Interface	KIBOR	Karachi Interbank Offered Rate
ARCH	Autoregressive Conditional Heteroskedasticity	KPSS	Kwiatkowski-Phillips-Schmidt-Shin Test
ARCHT	Autoregressive Conditional Heteroskedasticity with t-distribution	KSE	Karachi Stock Exchange
ARMA	Autoregressive Moving Average	LM	Lagrange Multiplier
BEKK	Baba-Engle-Kraft-Kroner Model	LSTM	Long Short-Term Memory
CAPM	Capital Asset Pricing Model	M2	Money Supply (Broad)
CCC	Constant Conditional Correlation	MAD	Mean Absolute Deviation
CDF	Cumulative Distribution Function	MAE	Mean Absolute Error
CMES	Chicago Mercantile Exchange Swap	MAPE	Mean Absolute Percentage Error
CNN	Convolutional Neural Network	MCDM	Multi-Criteria Decision Making
CNX	CNX Nifty Index	MLE	Maximum Likelihood Estimation
CNY	Chinese Yuan	MSE	Mean Squared Error
CSI	China Securities Index	MVHR	Minimum Variance Hedge Ratio



Abbreviation	Full Form	Abbreviation	Full Form
CV	Coefficient of Variation	NCCPL	National Clearing Company of Pakistan Limited
CVAR	Conditional Value at Risk	NIS	Net Investment Strategy
DCC	Dynamic Conditional Correlation	OLS	Ordinary Least Squares
DFT	Discrete Fourier Transform	PACF	Partial Autocorrelation Function
DL	Deep Learning	PI	Price Index
ECM	Error Correction Model	PIS	Positive Ideal Solution
EVT	Extreme Value Theory	PSX	Pakistan Stock Exchange
EWMA	Exponentially Weighted Moving Average	RMSE	Root Mean Squared Error
FFT	Fast Fourier Transform	RNN	Recurrent Neural Network
FTNET	Fourier Transform-based Neural Network	SBP	State Bank of Pakistan
FTS	Financial Time Series	SIF	Statistical Information Function
GARCH	Generalized Autoregressive Conditional Heteroskedasticity	SPI	Sensitive Price Indicator
GBM	Geometric Brownian Motion	TFN	Triangular Fuzzy Number
GDP	Gross Domestic Product	TFT	Temporal Fusion Transformer
GRU	Gated Recurrent Unit	TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
GUI	Graphical User Interface	VAR	Vector Autoregression
XGB	Extreme Gradient Boosting		



United Nations Sustainable Development Goals

The Sustainable Development Goals (SDGs) are the blueprint to achieve a better and more sustainable future for all. They address the global challenges we face, including poverty, inequality, climate change, environmental degradation, peace and justice. There is a total of 17 SDGs as mentioned below. Check the appropriate SDGs related to the project.

- ☐ No Poverty
- ☐ Zero Hunger
- ☐ Good Health and Wellbeing
- ☐ Quality Education
- ☐ Gender Equality
- ☐ Clean Water and Sanitation
- ☐ Affordable and Clean Energy
- ☒ Decent Work and Economic Growth
- ☐ Industry, Innovation and Infrastructure
- ☐ Reduced Inequalities
- ☐ Sustainable Cities and Communities
- ☐ Responsible Consumption and Production
- ☐ Climate Action
- ☐ Life Below Water
- ☐ Life on Land
- ☐ Peace and Justice and Strong Institutions
- ☐ Partnerships to Achieve the Goals



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1 Introduction

1.1 Background Information

The management of portfolio risk is a cardinal concern of investors and financial institutions operating within the stock and derivative markets. Especially in the context of emerging economies such as Pakistan, the equity markets pose unique challenges and opportunities driven by market volatility, liquidity, and the evolution of regulatory landscapes. As these challenges pertain, the literature present on the Pakistani Markets also focuses on the impact of derivatives, especially the future contracts, on market volatility, efficiency, systematic risk of underlying stocks, and spot price discovery ((Khan , 2006); (Malik & Shah, 2016); (Khan & Abbas, 2013)) with scarce literature on effective hedging strategies using future contracts.

From a global perspective, a lot of research has focused on minimum variance hedging strategies derived from foundational models like the Ordinary Least Squares (OLS) technique (Ederington, 1979) and the Error Correction Models (ECM) (Ghosh, 1993) to estimate optimal static hedge ratios to offset the risk exposure in underlying stock portfolios. However, advanced research has identified that these methodologies fail to hedge risk effectively, due to their inability to capture autocorrelation, heteroskedasticity, and the time-varying relationship of variances and covariances of futures and spot return series. Therefore, academic research shifted from traditional econometric models to more sophisticated models such as the Autoregressive Conditional Heteroskedasticity (ARCH) (Engle R. F., Autoregressive Conditional Heteroscedasticity with Estimates of the Variance of United Kingdom Inflation, 1982) and Generalized Autoregressive Conditional Heteroskedasticity (GARCH) (Bollerslev T. , 1986) frameworks and their multiple variants. These models, although proven beneficial in capturing time-series data characteristics such as heteroskedasticity and autocorrelation, may lead to misspecification errors in the computation of conditional volatilities due to the underlying model assumptions such as normality, linearity, and symmetry of the time-series data, resulting in less efficient hedge ratios.

With recent advancements in computational finance, the risk management landscape has entered a novel era of deep learning and machine learning techniques to develop optimal hedging strategies. Early studies on machine learning models explored simple neural networks to

approximate option sensitivities, followed by advancements in deep learning led to the adoption of LSTM, hybrid LSTM–CNN models, and Fourier Transform-based feature modules with temporal convolutional blocks – the FT-Net Hybrid model capable of capturing complex, nonlinear patterns and explicit spectral analysis components in options. (Chen, Hussain, Cauteruccio, & Zhang, 2023) Although the majority of research lays emphasis on option hedging strategies, the implications of these techniques are immense in hedging through future contracts as well.

While there is a considerable volume of international research dedicated to the comparative analysis of various econometric models to ascertain which best estimates minimum-variance hedge ratios (Floros & Vougas, 2007), few studies emphasize the comparability between mathematical and machine learning models. The study by (François, Gauthier, Godin, & Mendoza, 2025) combines the mathematical delta hedging technique with deep hedging. Rather than evaluating the models through various performance metrics, it examines a distinctive proposition suggested by (Horikawa & Nakagawa, 2024). This proposition states that under a complete market admitting statistical arbitrage, the discrepancy between the hedging position derived by a deep hedging model and that of a traditional replicating portfolio, may provide a statistical arbitrage opportunity. These studies explore a unique perspective on the application of statistical arbitrage beyond asset mispricing. Conventionally, statistical arbitrage is primarily defined as the temporary pricing mismatches of an asset or related asset with strategies requiring buying undervalued assets and selling overvalued ones to profit from the anticipated price convergence. On the contrary, the concept of “statistical arbitrage between models” focuses on identifying and exploiting the consistent, predictable profit arising from how one model performs differently from the other based on underlying assumptions and mechanisms, potentially when turned into a zero-cost strategy. This evolving landscape accentuates the ongoing pursuit of more refined and adaptive hedging methodologies.

Therefore, in this project, we develop and evaluate dynamic hedging strategies using the KSE 30 Index future contracts by employing both econometric and cutting-edge machine learning frameworks. Specifically, we have used the DCC GARCH framework and extended it by layering it with a copula function using the Copula GARCH framework to model dynamic correlations and tail dependencies. Furthermore, we have leveraged the LSTM-CNN and FT-

Net Hybrid models to capture complex, non-linear, and cyclical patterns in Pakistani equity markets.

1.2 Significance and Motivation

The motivation for this research stems from critical gaps in the existing literature and the dynamic evolution of financial risk management practices. Firstly, despite the significant role derivatives play in managing risks in the equity markets, especially in the case of developing markets like Pakistan, specific research in this context is notably limited. Therefore, we focus our methodology on the KSE 30 index and its future contracts to fill this crucial regional void and offer practical insights to local investors and institutions.

Secondly, while there is significant research on developing sophisticated econometric models, their underlying assumptions often inhibit their effectiveness, especially in capturing non-linear and complex relationships in real-world data. The shift to machine learning models is a promising alternative as it models intricate relationships in data without fitting the data to a lot of the model's underlying assumptions. However, there is a need to study these methodologies together to gauge the relative strengths and weaknesses of the models. There is a distinct literature gap in this aspect, and our research focuses on this analytical gap by providing empirical evidence on the relative strengths and weaknesses of each framework.

Thirdly, this study is inspired by the pioneering work of (François, Gauthier, Godin, & Mendoza, 2025) and (Horikawa & Nakagawa, 2024), which introduces the concept of "statistical arbitrage between models." They test the proposition that the difference in the hedging performance between models can lead to exploitable arbitrage opportunities. This is a unique perspective that extends the traditional concept of statistical arbitrage and presents a novel research area. By investigating this phenomenon, we layer our objective with an added perspective on our models' efficiency and market equilibrium.

1.3 Aims and Objectives

In this research, we aim to:

- (i) formulate hedging strategies for a KSE 30 index portfolio using the KSE 30 futures contract through both sophisticated econometric models and advanced deep learning

- models to dynamically minimize risk in the equity futures market,
- (ii) utilize key comparison metrics to evaluate and contrast the effectiveness of hedging strategies derived from both econometric and machine learning models in minimizing risk, and
 - (iii) conduct an in-depth investigation into potential statistical arbitrage trading opportunities by all the models and analyze which model presents the best opportunities.

1.4 Methodology

To incorporate the time-varying nature of optimal hedge ratios, we have focused on models that dynamically compute the hedge ratios. In this research, we first use the (Engle R. , 2002) DCC-GARCH framework. This framework models each return series through an ARMA–eGARCH margin and updates the conditional correlation matrix, effectively capturing the time-varying dynamics of covariance. Acknowledging that linear correlation might underestimate joint extreme movements, we further enhance the DCC-GARCH model with a student-t copula, tailored to the probability integral transformations of the standardized residuals. This adjustment allows for symmetric tail dependence and improves hedging effectiveness during periods of market stress (Lai Y.-S., 2018). Concurrently, we also evaluate machine learning models that can capture intricate nonlinear patterns and long-term dependencies. The LSTM–CNN hybrid model integrates convolutional layers to extract local temporal features from returns and futures spreads, followed by Long Short-Term Memory (LSTM) layers that model the dynamics of sequential hedge ratios. Variants of this hybrid model have demonstrated ability to learn optimal hedging strategies directly from market data, efficiently adapting to transaction costs and market frictions in contexts involving illiquid assets (Wang et al., 2024) (Zhang & Huang, 2021). The innovative FT-Net Hybrid model combines Fourier Transform-based feature modules with temporal convolutional blocks. The spectral modules are designed to capture cyclical market patterns and non-stationarities that may be overlooked by networks operating solely in the time domain. Meanwhile, the convolutional blocks, along with a final recurrent or transformer layer, focus on learning dynamic relationships within the transformed data space. This hybrid approach enhances recent deep-hedging frameworks by incorporating explicit spectral analysis components.

Furthermore, we evaluate the hedging effectiveness of our econometric and deep learning

models through a total of nine performance metrics which include variance reduction percentages, Root Mean Square Errors (RMSE), average hedged returns, Sharpe ratios, directional accuracy, Value at Risk (VaR) reduction, Conditional Value at Risk (CvaR) reduction, mean absolute deviation, and time complexity as used in prior research. To assess which model performs the best given all these criteria, we use the Fuzzy TOPSIS multicriteria decision-making technique. Furthermore, to gauge how well a model performs, we analyze the hedged portfolio's Profit and Loss series for statistical arbitrage opportunities.

1.5 Contributions of the Study

This study makes several substantial and novel contributions to the fields of financial engineering and quantitative risk management, particularly within the context of emerging markets like Pakistan. The most significant contribution lies in the pioneering application of advanced machine learning models, along with systematically designed statistical arbitrage trading strategies, directly to equity futures contracts. While the existing global literature on dynamic hedging and statistical arbitrage has predominantly centered on options markets, this research extends these innovative methodologies to the relatively underexplored domain of index futures. Hence, it addresses a critical gap in both practical implementation and academic coverage.

Moreover, this work stands out for its specific focus on the KSE 30 Index, and its futures traded on the Pakistan Stock Exchange. There is a notable scarcity of quantitative research and deep learning applications targeting the Pakistani capital markets within existing academic literature. Through empirical evaluation, contrasting sophisticated econometric models (such as DCC-GARCH with Student-t copula) and cutting-edge deep learning models (notably the LSTM-CNN hybrid and FT-Net Hybrid), this study not only introduces methodological innovation but also serves as a foundational reference for subsequent research in the region. The research demonstrates how deep learning models can efficiently extract complex, nonlinear, and time-varying patterns in hedge ratios, which are often missed by traditional techniques.

A further methodological contribution is the integration of Fourier Transform-based spectral analysis into the deep learning pipeline through the FT-Net Hybrid model. This approach enhances the ability to capture cyclical and non-stationary characteristics in financial time series data features particularly relevant to the volatility and market structure of emerging economies.

Additionally, this research incorporates a multi-criteria decision-making framework based on fuzzy TOPSIS, allowing for a transparent comparison across multiple dimensions of hedge effectiveness, predictive accuracy, profitability, and computational efficiency.

The beneficiaries of this project include investors such as institutional portfolio managers and professional traders operating in the Pakistani or similar emerging markets to benefit from enhanced risk management tools and empirically validated trading strategies that adapt to market dynamics and exploit inefficiencies. Second, academic researchers gain a region-specific empirical foundation and a set of advanced modeling frameworks that can be extended to further research on financial innovation in underexplored markets.

1.6 Report Outline

This thesis is structured into seven chapters, each addressing a critical component of the research. Chapter 1 introduces the study, outlining the background, significance, research objectives, and the methodology employed. Chapter 2 offers a comprehensive review of relevant literature and is divided into two sections: the first presents a bibliometric analysis of existing research, while the second provides a descriptive assessment of the academic landscape. Chapter 3 details the processes of data collection and preprocessing, including the sources, selection criteria, and preparation techniques applied to the dataset. Chapter 4 elaborates on the research methodology, describing the theoretical framework, model architecture, and analytical techniques utilized throughout the study. Chapter 5 presents and discusses the results of model evaluation, including performance metrics and comparative analyses. Chapter 6 explores the statistical arbitrage trading opportunities uncovered by each model and discusses their practical implications. Furthermore, Chapter 7 explores the integration of macroeconomic variables into our highest-ranked model, as determined by the fuzzy TOPSIS decision-making criteria. Finally, Chapter 8 concludes the thesis by summarizing the key findings, highlighting the contributions of the research, and suggesting potential directions for future work.

2 Literature Review

2.1 Introduction

Hedging risk exposure in equity markets using derivatives, particularly stock index futures, is a fundamental concept in financial risk management. The development of an effective and robust hedging strategy is predicated on a comprehensive understanding of the theoretical grounds and practical advancements regarding the hedging framework. In consideration of the extreme importance of a thorough understanding, we contextualize our research with existing literature. This chapter presents a systematic literature review of the existing literature on the development and assessment of hedging strategies in equity futures markets.

To effectively traverse the diverse and rapidly evolving research landscape, we have divided our literature review into two mutually supportive sections. The first section is devoted to a bibliometric analysis of the literature mapping the major research streams, influential authors, publication trends, and thematic concentrations within the broader domain of equity market hedging and statistical arbitrage. This approach leads to an in-depth view of scholarly activity and key research streams and contributions. Expanding on these insights further, the second section delivers a systematic qualitative review, delving deeper into methodologies, findings, and limitations of relevant studies and research streams.

The primary objective of structuring the literature review in two sections is to achieve a refined understanding of the research landscape of this project. This chapter outlines the development of research on hedging strategies, especially those using advanced econometric and machine learning models, clarifying the bases of our study. Additionally, it aims to pinpoint essential research gaps and methodological flaws that our thesis intends to tackle, highlighting the importance and uniqueness of our research contributions.

2.2 Bibliometric Analysis

The bibliometric analysis is conducted on a sample of 588 articles imported from the Web of Science (WoS) database. The Clarivate's Web of Science database is one of the oldest and most reputable databases of research papers and citations. An article search technique was formulated based on author keywords to gather the most relevant sample of articles. To refine the search, a

keyword search criterion was defined which included the following keywords: “futures contracts”, “hedging effectiveness”, “minimum variance hedging”, “optimal hedge ratio”, “stock index futures”. The publication years were limited to 2014 to 2024. Furthermore, the most relevant WoS core collection subject categories were chosen, which were “Business Finance”, “Mathematics Applied”, “Statistics Probability”, “Mathematics”, and “Economics”. The meta literature review was conducted using VoS Viewer, Biblioshiny R Package, and Microsoft Power BI.

2.2.1 Keyword Analysis

The result of the keyword analysis performed using VoS Viewer is presented in Figure 1. The figure shows a cartography map that highlights clusters of different research streams within the broader literature of optimal dynamic hedging using stock index futures. These clusters are identified by VoS Viewer based on the close association of keywords depicting thematic groupings within the literature and are represented by different colors. The lines interconnecting the clusters amongst themselves depict the strength and frequency of the co-occurrence of various keywords within the same publications to capture the interdisciplinary and interconnected nature of research. The VoS Viewer software has a scale of co-occurrence that sets a lower limit to the number of co-occurrences of keywords. For this analysis, it was configured to a minimum of five co-occurrences of a keyword. This results in five main clusters in the cartography map, which represent five major research streams in the literature.

The red cluster is the most prominent and densely populated, with keywords “volatility”, “model”, “risk management”, and “hedging effectiveness”. This cluster represents the literature focusing on volatility modelling, assessment of hedge effectiveness, and development of frameworks for risk mitigation in financial markets. The second most prominent cluster is the blue one, which majorly shows keywords with the theme of “hedge effectiveness”; however, it includes other related topics such as “volatility spillovers”, “returns”, and “uncertainty”. The third cluster in green is centered on the research stream “stock index futures”. It represents the part of literature that focuses on price discovery, information transformation, lead-lag relationships, and the econometric modelling of future contracts.

Furthermore, we have two other clusters in purple and yellow. The purple cluster is characterized by keywords such as “volatility spillover”, “conditional heteroskedasticity”, and “structural breaks”, emphasizing the body of literature on advanced time series modelling and

structural changes in financial markets. Meanwhile, the yellow cluster is dominated by keywords such as “gold”, “bitcoin”, “diversification”, and “COVID-19”. This cluster encapsulates the body of literature that emphasizes alternative investments and their impact on periods of heightened volatility.

The cartography results on an overall level depicted the major research streams in our literature sample and highlighted the interconnected nature of the themes.

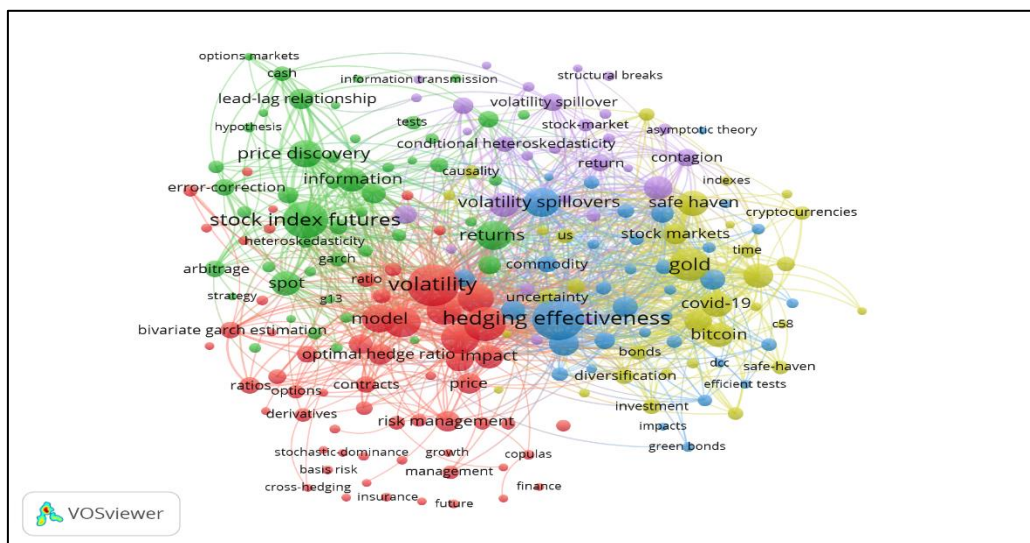


Figure 1: Cartography Results obtained from VoS Viewer

2.2.2 Yearly Research Trends and Publication Volume

Figure 2 below shows the yearly analysis of publication volume and citation trends of the literature from 2014 to 2024. The graph presents two key metrics: the number of papers published each year, represented by blue bars, and the cumulative number of times these works have been cited across all databases, depicted by the solid line.

The overall trend in the publication volume and the times cited score shows an increasing trend throughout the years. However, a notable increase in the volume of literature is evident from the graph after 2018, which reflects a positive development in the academic interest and engagement in research relevant to our project’s themes. The growth in research is particularly high after 2020, corresponding with increasing market uncertainty and the rapid development of new computational methodologies in finance.

The trend in citation counts simultaneously exhibits a positive growth pattern, although it is

somewhat more variable. A sharp increase in citations from 2018 onwards suggests that recent research has garnered significant attention and has played an instrumental role in influencing the current academic and research landscape. The steady upward movement in both the volume of published papers and citation scores in recent years is not only indicative of the vitality of this research field but also of its growing relevance to financial practitioners and policymakers.

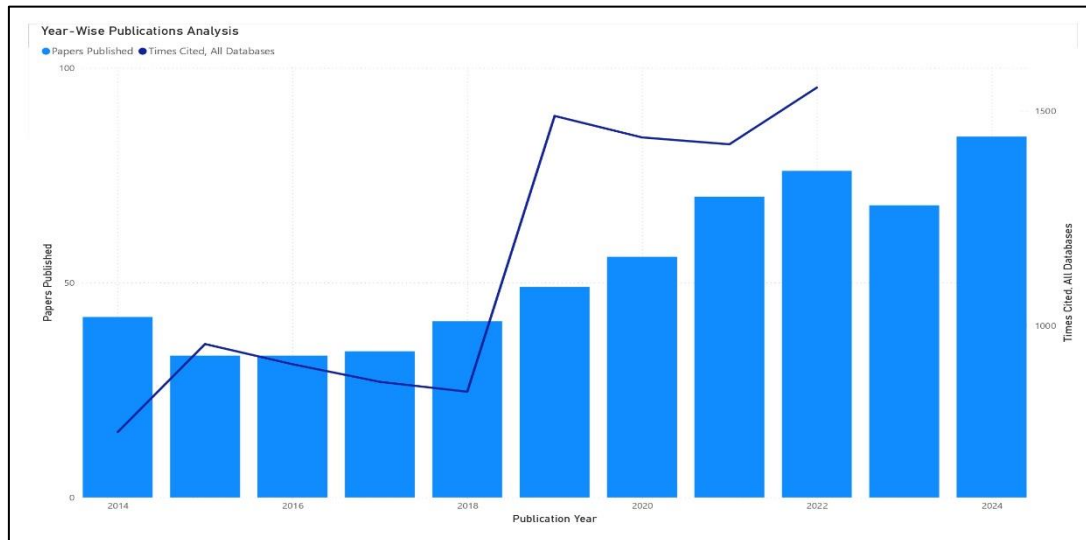


Figure 2: Yearly Research Trends

2.2.3 Most Influential Countries

Figure 3 presents the country-wise publication analysis. The map shows every country in a distinctive color and its corresponding publication volume through the size of the marker. This figure visualizes the regional distribution of literature and helps identify the leading countries in terms of publication in the context of the literature.

The map reveals a greater concentration of literature in developed markets such as China and the United States of America (USA). Moreover, other significant contributors to the risk management through stock index futures literature include the United Kingdom (UK), South Korea, Australia, and Germany.

In the South and Southeastern regions, India, Taiwan, and Vietnam dominate most of the publication volume. Moreover, there is some prominent research activity in France, Canada, Turkey, and Saudi Arabia which emphasizes the widespread growth of the literature in various parts of the world. There are smaller volumes of research in countries such as

Pakistan, Greece, Oman, Tunisia, and New Zealand; however, despite the lower volumes, they contribute important regional perspectives in the literature.

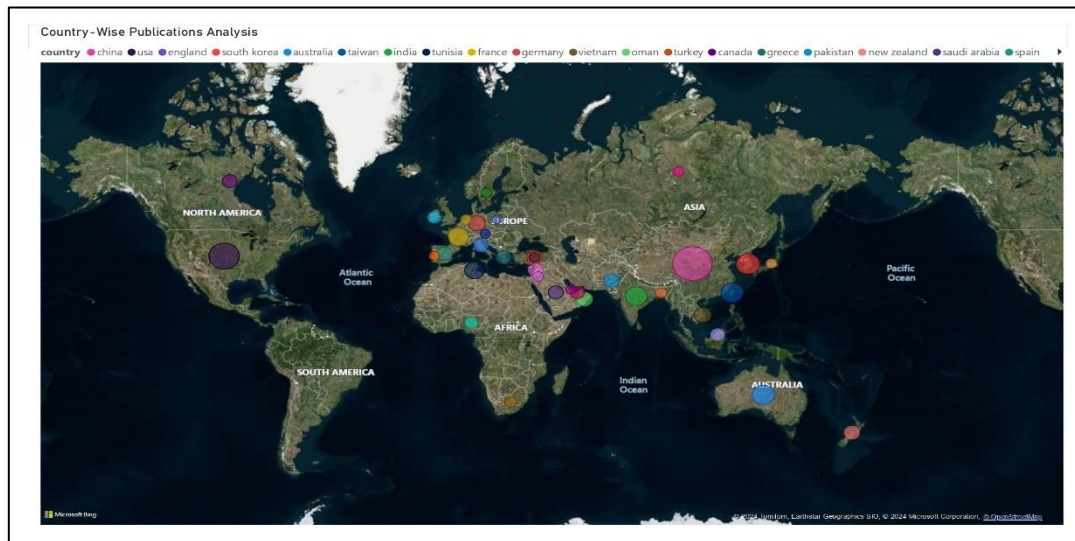


Figure 3: Number of publications in every country

2.2.4 Three Factor Analysis

Figure 4 shows the three-factor analysis diagram. The diagram combines three factors crucial in understanding the dynamics of the literature: the principal research keywords, most frequently occurring terms extracted from articles, abstracts, and titles, and the top 10 countries in terms of volume of research.

Previously, we viewed these factors separately; now we put them together to gauge how different countries focus on particular aspects of the broader research landscape. For instance, the linkage between volatility-focused research and the Chinese, UK, and US researchers shows that these markets are central to research in quantitative risk modelling. Moreover, countries like Tunisia and Vietnam contribute more research on themes such as market efficiency or commodity risk.

In the overall aspect, we can observe from the three-factor analysis the dynamic and diverse interplay between research themes and their global distribution.

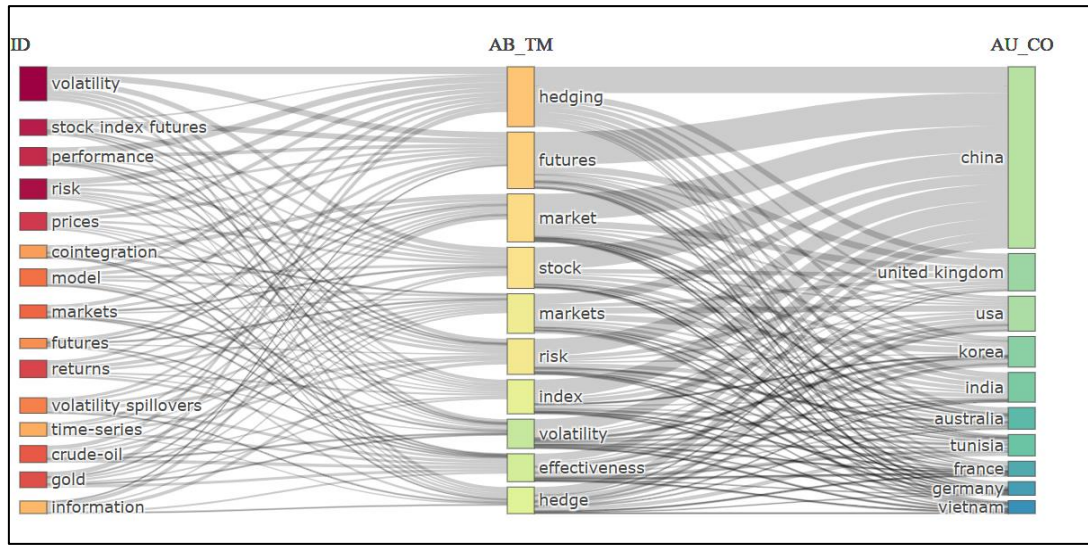


Figure 4: Three Factor Analysis

2.3 Systematic Review of Key Research Streams

The conceptual framework of this research is inspired from the work authored by (François, Gauthier, Godin, & Mendoza, 2025) which is based on option contracts that test the proposition made by (Horikawa & Nakagawa, 2024). This proposition states that under a complete market admitting statistical arbitrage, the difference between the hedging position provided by deep hedging and that of the replicating portfolio, is a statistical arbitrage. (François, Gauthier, Godin, & Mendoza, 2025) tests this proposition using GARCH based models for delta hedging and deep hedging. Therefore, in this paper, we test the same proposition in the context of hedging with future contracts, where, instead of delta hedging, we use minimum variance hedging, which is a popular choice in research. To estimate minimum variance hedge ratios (MVHRs), research has used multiple econometric models such as traditional Ordinary Least Squares (OLS) and Error Correction Models (ECM), only as in (Lin, 2002) or along with sophisticated models like Autoregressive Conditional Heteroscedasticity (ARCH) and Generalized Conditional Autoregressive Heteroscedasticity (GARCH) models and their variants (eGARCH, mGARCH, DCC, ADCC, GO GARCH, GARCH COPULA, BEKK GARCH, etc.) (Alvarez-Diez & Gonzalez, 2006) (Bodla & Jindal, 2006) (Floros & Vougas, 2007) (Wang & Zhang, 2019).

The conventional strategies for estimating MVHRs, such as the OLS method, estimate constant MVHRs and are criticized for their inability to account for cointegration, which

results in under-hedging. To incorporate cointegration, ECM methods were introduced; they capture short-term dynamics and the long-term equilibrium between the spot and futures returns, resulting in better performance. However, like the OLS technique, they also assume that the relationship between spot and futures is timeless. It is evident from various studies (Koutmos & Tucker, 1996) that there is a strong causal relationship between the spot and futures return that is time varying; therefore, dynamic hedge ratios are more effective for risk management. (Baillie & Myers, 1991) To compute dynamic hedge ratios, various extensions of the GARCH method (Bollerslev T. , 1986) are used in prior research. Some prominent variants used in research were BEKK GARCH (Baba, Engle , Kraft, & Kroner, 1990) which is computationally intensive as it allows for complex interaction between asset volatilities, CCC GARCH (Bollerslev T. , 1990) is simpler than BEKK GARCH. It assumes that although the variances are time varying, the conditional correlation between assets remains constant. DCC GARCH is an extension of this model by (Engle R. , 2002) which allows conditional correlations to vary with time. Furthermore, Copula GARCH models as in (Patton A. J., 2006), (Lai Y. S., 2018), (Lai, Chen, & Gerlach, 2009) estimate the marginal distribution using univariate GARCH models and then link them to a copula function to capture asymmetric and non-linear dependencies. These GARCH-type models, although outperform static models, might overestimate the persistence in volatilities since regime switches and unexpected changes in variance are often ignored. (Qu, Wang, Zhang, & Sun, 2019)

The recent developments in deep learning have led the estimation of minimum variance hedge ratios (MVHRs) in stock index futures markets to a new paradigm which overcomes the model assumption and specification challenges in traditional and sophisticated models by capturing complex relationships between time series through the power of neural networks. Many researchers have concluded that deep learning-based hedging strategies can understand complex, nonlinear features in high frequency data significantly better than conventional approaches while yielding higher economic benefits (Hu & Ni, 2024). The literature is especially dominated with Long Short-Term Memory (LSTM) architectures that have demonstrated superior performance in capturing time-varying dynamics in financial markets, outperforming static OLS and parametric GARCH models in both simulated and real-time data (Adams, Asemota, & Ibrahim, 2024). Hybrid models that integrate

Convolutional Neural Networks (CNNs) with LSTMs or more novel Frequency-Temporal (FT-Net) Hybrid architectures have greater potential to enhance hedge effectiveness by combining local feature extraction with long-term dependency modeling. It is evident from research that these deep and hybrid models produce lower hedged return variance and higher risk adjusted performance metrics across diverse asset classes in contrast with the hedge produced through dynamic mathematical models.

The theoretical basis of “deep hedging” was initialized in (Beuhler, Gonon, & Wood, 2019). In this research, it was highlighted that the deep learning framework does not require conventional finance basis such as the Greeks in finance or underlying assumptions regarding price behavior, however, the neural networks can be trained to directly minimize risk. Later studies including (François, Gauthier, Godin, & Mendoza, 2025) and (Hu & Ni, 2024) build upon this methodology and formulated a Deep Learning based Financial Hedging (DL-HE) strategy. This strategy was first implemented on commodity futures to learn complex patterns through nonlinear feature extraction for inventory and market risk management. The annualized gains reported from the use of this strategy were greater than traditional hedges by 1.2 million Chinese Yen (CNY). Subsequently, other researches explored implications of these similar deep learning models in illiquid markets. However, in illiquid markets LSTM based models outperform significantly than prior buy and hold strategies judged on the basis of improved Sharpe ratios and draw down control when forecasting futures prices for hedging purposes. Moreover, comprehensive comparisons of traditional versus deep MVHRs exhibited that deep models not only reduce hedged-position variance but also adapt more swiftly to regime shifts and volatility spikes than static OLS or ECM methodologies.

(Knoll, 2023) illustrate a “Deep Mean-Variance Hedging” approach wherein a quadratic risk objective is used to train an LSTM hedging agent, which then exhibits superior hedging performance across both exponential Lévy and stochastic volatility models in simulation experiments. This work is further extended by (Agram, Oksendal, & Rems, 2024), who propose a self-supervised learning approach for quadratic hedging in incomplete jump markets. The self-supervised learning network learns and adjusts for discontinuities and jumps in data, thereby performing better than GARCH-based hedges in synthetic and empirical tests.



Furthermore, hybrid deep learning models that layer two types of networks, CNN and LSTM, are being used in research due to their ability to capture spatial and temporal patterns in futures price series. Their application on EU Emissions Trading Scheme (EU FTS) futures realized up to a 43% reduction in mean absolute percentage error (MAPE) compared to pure LSTMS or Vector Autoregressive (VAR) models, leading to more efficient intraday trading and hedging strategies. Hybrid models such as CNN-LSTM and CNN-LSTM-AM proved to outperform single model baselines as they are able to focus on short term, sudden price changes, long term trends and patterns along with attention mechanisms to help the model concentrate on the most important data. This proven success of hybrid models led to the emergence of the FT-NET Hybrid architectures in 2024. These models consist of Fourier transformers that help detect cyclical patterns in the market and respond to sudden shifts in market behavior. Although FT-Net is a novel approach that has yet to be used widely in research some preliminary works have highlighted its potential to reduce hedged return variance by an additional 5-8% comparative to CNN-LSTM hybrids in the context of stock index futures.

3 Data Collection and Processing

This study hedges the risk of a spot equity portfolio of the KSE 30 index by conventional and deep learning hedging techniques using the KSE 30 stock index futures contract. It uses historical data of stock index and stock index futures contracts from January 2019 to June 2024. The frequency of the data recorded is daily to ensure accuracy. All data was collected from the PSX Data Portal (Daily Downloads page).

The data obtained includes the following fields:

- Index cash price: The current or spot price of the underlying index on the date of report.
- Interest/ Financing Rate: This is the interest rate used for calculating the fair value of the futures contract. It reflects the cost of carrying or financing the position.
- Index dividend yield: The expected annual dividend yield. of the index. This is used to adjust the futures price for the expected dividends that would be paid out over the life of the contract.
- Days to expire: The number of days remaining until the futures contract expires
- Indicative fair value: The Exchange calculates a reference price as fair value of the SIF Contract on which all outstanding SIF Contracts are settled. The fair value is a function of cash or underlying index value plus financing charges
- (r, determined as a function of KIBOR rates) less any dividends (d) that would accrue with the purchase and carry of all Index constituent until the final settlement date.

Equation 1

$$\text{Indicative fair value} = \text{Underlying Index} * \left(1 + r \left(\frac{x}{365} \right) \right) - d$$

The Pakistan Stock Exchange offers three stock index futures contracts from which we selected the KSE 30 Index Future contracts. The futures contracts have a maturity of 3 months. We have selected the contracts with the closest days to expire which will require rolling the contract every month for continuity.

3.1 Data Processing

After the data collection, the next step was to process the data, which included interpolating missing data. Our data had missing Indicative fair value of contracts for multiple days. To estimate the indicative fair value of those contracts, we used the equation provided in the PSX rule book (equation 1). However, a major problem was the absence of interest/financing rate data for multiple days. Therefore, the random forest model was used to interpolate missing values of the interest/financing rate. Furthermore, to ensure continuity and integrity of the futures price data as the nearest to expiry contracts were rolled over, we performed mean back adjustments.

3.1.1 Filling the Missing Values

One of the critical challenges encountered was the presence of missing data in our data set. Missing values are a pervasive issue in the financial time series, often arising due to data recording errors, market holidays, or irregular reporting. In our case, the most significant gaps were observed in the daily futures data. This posed a substantial barrier because, as stipulated by the Pakistan Stock Exchange (PSX) rulebook, the indicative fair value of the KSE 30 futures contract is derived from a precise mathematical formula (refer to formula 1).

As mentioned earlier too, the Underlying Index is the spot value of the KSE-30 index on a particular date, r represents the annualized interest or financing rate, x denotes the number of days remaining until the futures contract expires, and d is the value of the expected dividends over the contract period. The historical data of price of underlying asset and dividend rate were readily available on PSX website, and the number of days to expiry was directly computable from the date fields. The only true ambiguity was the missing financing or interest rate, which is pivotal for pricing futures accurately and, by extension, for all subsequent econometric and machine learning analysis that depend on continuous, high-quality price data.

To address this, we implemented a robust data imputation approach. Our first step was to gather the required auxiliary data from authoritative sources, particularly the 1-month KIBOR rates from the State Bank of Pakistan (SBP), since the financing rate, as defined in the NCCPL rulebook, is fundamentally a function of this short-term interbank offered rate. This step ensured that we had a reliable explanatory variable for predicting the missing

interest rates.

Given the non-linear and potentially complex relationship between the 1-month KIBOR and the futures contract's financing rate, we opted for a machine learning approach rather than a simple linear interpolation. We selected the Random Forest Regressor, an ensemble method capable of capturing intricate, non-linear dependencies without requiring explicit model specification or parametric assumptions. The Random Forest Regressor builds a number of decision trees during training. The prediction of each tree is averaged to get the final prediction output, which provides strong predictive accuracy and robustness against overfitting. Figure 5 below shows the architecture and pipeline used.

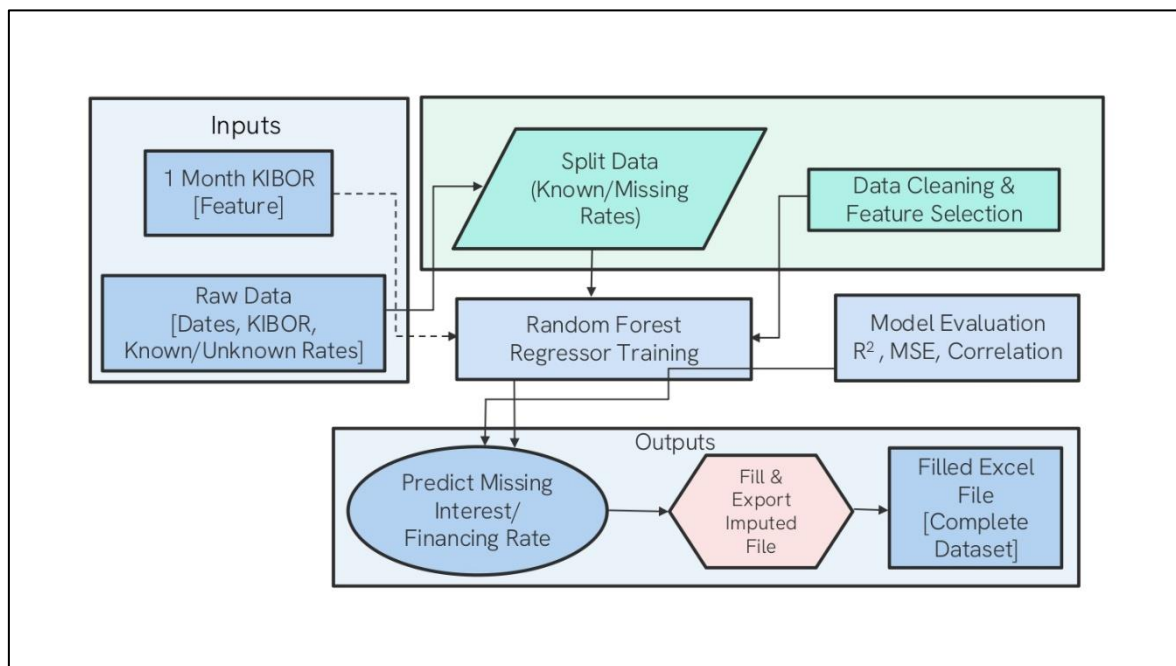


Figure 5: Architecture and Pipeline of the Random Forest Regressor

The implementation involved splitting the available data into two subsets: one containing rows where the interest rate was known (used for training), and one where it was missing (for prediction). Using the 1-month KIBOR as the sole predictor, the Random Forest model was repeatedly trained (across 1000 different random seeds) to maximize out-of-sample performance, assessed by metrics such as R-Squared (R^2), Mean Squared Error (MSE), and Pearson correlation coefficient between actual and predicted interest rates. This iterative process ensured that we identified the optimal model configuration, providing the highest explanatory power and the lowest prediction error. Ultimately, the best-performing model

achieved an exceptionally high R^2 value of 0.9092 and a remarkably low MSE of 0.0003, as highlighted in the output. This indicates that the model explained more than 90% of the variance in the observed interest rate values, instilling high confidence in the fidelity of the imputed values.

Once the missing interest rates were filled using the trained model, we were able to compute the indicative fair values for all missing entries using the aforementioned PSX formula. This seamless integration of machine learning-driven imputation with domain-specific financial formulas ensured the consistency, accuracy, and completeness of our futures dataset. Consequently, the resultant data was rendered suitable for rigorous econometric modeling and deep learning experiments, preserving the integrity of our research findings and allowing us to confidently pursue the subsequent stages of dynamic hedging and arbitrage analysis.

3.1.2 Adjustments of the Future Contracts

In the context of our research, we encountered a significant issue relating to the continuity and integrity of futures contract price data. The fundamental problem arose from the nature of futures contracts themselves: each contract has a defined expiration date, typically at the end of the month, after which trading transitions or "rolls over" to a new contract with a future expiration. Theoretically, as a futures contract approaches its expiry, its price is expected to converge with the spot price of the underlying asset, a principle rooted in the cost-of-carry model. However, in practice, and particularly in the dataset we obtained, the way continuous futures price series are constructed leads to artificial "jumps" at rollover points. These jumps occur because, on the final trading day, the expiring contract is priced very close to the spot (due to convergence), but the next available contract, being further from its own expiry, incorporates additional risk premia, interest rate differentials, and market expectations, resulting in a different price level. When the dataset is stitched together contract by contract in a continuous series, these price jumps at each monthly expiry can introduce significant distortions, disrupting any time series analysis, volatility modeling, or hedge ratio estimation.

Addressing this discontinuity is crucial for the validity of any empirical analysis involving futures prices. Several methods exist to resolve these jumps, such as forward adjustments, calendar spread methods, and the more popular back-adjustment technique. In our study, we selected the mean-back adjustment method due to its conceptual simplicity and effectiveness

in producing a smooth, artifact-free price series, which is particularly suitable for econometric and deep learning models that are sensitive to structural breaks or artificial volatility spikes.

The mean-back adjustment processes we applied can be described as, for each contract rollover, we first identified the specific day on which the expiring contract ended, and the new contract began trading as the active series. Let us denote the price of the expiring contract on its last day as $P_{t_{roll}}^{(exp)}$ and the price of the new contract on the same day as $P_{t_{roll}}^{(new)}$. To avoid an abrupt shift, we computed the mean of these two prices.

Equation 2

$$M_{roll} = \frac{P_{t_{roll}}^{(exp)} + P_{t_{roll}}^{(new)}}{2}$$

Next, we determined the difference between this mean and the expiring contract's price on the rollover day.

Equation 3

$$D_{roll} = P_{t_{roll}}^{(exp)} - M_{roll}$$

This difference, which can be positive or negative depending on the direction of the jump, quantifies the artificial gap that needs to be adjusted.

The key step in the mean-back adjustment is to uniformly apply this difference D_{roll} as an adjustment factor to all price points within the expiring contract's life, i.e., all daily prices for that contract month. If $P_{t_{roll}}^{(exp)}$ is the price of the expiring contract at time t within its active month, the adjusted price becomes,

Equation 4

$$P_{t_{roll}}^{(exp,adj)} = P_t^{(exp)} \pm D_{roll}$$

This means that each daily price in that contract month is shifted up or down by exactly the same amount, ensuring the price at the rollover aligns seamlessly with the start of the new contract, while preserving the intra-month price dynamics and volatility. The same procedure is repeated for every rollover month, systematically removing jumps at all points of contract expiration throughout the historical dataset.

Mathematically, if there are n contracts spanning the full data period, the adjustment process for each contract i at rollover point $t_{roll,i}$ can be generalized as follows:

1. Compute the mean at rollover (refer equation 2)
2. Calculate the difference (refer equation 3)
3. Adjust all prices in contract i (from start $t_{start,i}$ to $t_{roll,i}$) (refer equation 4)

Through this process, we obtained a continuous time series of futures prices free from any spurious jumps or artificial volatility. Such fluctuations would otherwise distort the reliability of any subsequent econometric modelling or deep learning-based hedge ratio estimation.

This adjustment means that both the statistics of returns and detection of structural market dynamics reflect the real market and are not just a result of how contracts are rolled over. Thus, data processing of futures price through a mean-back adjustment gave us a rigorous and transparent methodology to use as a groundwork for all further analyses of our project.

4 Research Methodology

This chapter is divided into three distinct sections in order to outline the methodological framework adopted in this research systematically.

The first section is dedicated to preliminary data diagnostics and pre-estimation testing. In this section, we perform a comprehensive investigation of the time series data characteristics, with the aim of assessing whether the underlying statistical assumptions required by each modeling approach are satisfied. Especially focusing on statistical tests for stationarity, autocorrelation, and heteroscedasticity, which are fundamental prerequisites for time series modelling and are essential specifically for conventional econometric models. By rigorously interrogating the data at this initial stage, we ensure that the integrity of the modeling process is upheld and that the results derived later in the study are built on a sound empirical foundation.

The second section gives in-depth insights into the core modeling methodologies employed to address the research objectives. Particularly, we provide an extensive exposition of the four principal hedging frameworks utilized: the Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC-GARCH) model, the DCC Copula-GARCH extension, the Long Short-Term Memory Convolutional Neural Network (LSTM–CNN) hybrid, and the Fourier Transform – Network (FT-Net) Hybrid model. Each of these models is discussed in detail highlighting its mathematical underpinnings, structural specifications, and theoretical motivation, as well as the rationale for its inclusion in the context of dynamic hedging and risk minimization in the equity futures market.

The final section of the chapter is concerned with post-estimation testing and model validation. In this segment, we outline the statistical tests and performance metrics used to evaluate the adequacy and robustness of each model. This includes both in-sample and out-of-sample diagnostic checks and the model results providing an empirical basis for the comparative analysis and model selection that follows in subsequent chapters.

4.1 Statistical Tests Pre-Model Implementation

4.1.1 Augmented Dickey-Fuller (ADF) Test for Stationarity

The ADF test for stationarity, as given by (Dickey & Fuller , 1979), is a statistical test used to determine whether a time series has a unit root, indicating that the series is non-stationary. The test involves estimating an autoregressive model and testing the null hypothesis that the series contains a unit root, which would imply non-stationarity. If the null hypothesis is rejected, it suggests that the time series is stationary.

We conducted the ADF Test using the R programming language and the tseries package on the futures contract returns, and KSE 30 returns time series. The `adf.test()` function in R applies the general regression equation, incorporating both a constant and a linear trend, and computes the t-statistic to test whether the first-order autoregressive coefficient equals one (indicating non-stationarity). The number of lags used in the regression is denoted by k , where the default value is determined using equation 5 below.

Equation 5

$$\text{trunc}((\text{length}(x) - 1)^{1/3},$$

It follows the suggested upper bound for the ARMA(p,q) framework. Therefore, the default lag value for our time series was 6. The p-values are interpolated from (Banerjee, Dolado, Galbraith, & Hendry, 1993). If the computed test statistic falls outside the table of critical values, a warning message is generated.

The null hypothesis, which tests for the presence of a unit root (indicating non-stationarity), for all three series was rejected at both the 5% and 1% confidence levels for all time series. As the p-values were smaller than 0.01, and R displays the lowest recorded value, all p-values were presented as 0.01.

4.1.2 Engle's ARCH Test

The Engle's ARCH Test introduced in his paper in (Engle R. F., 1982) is conducted by estimating a regression model, where a set of independent variables explains the dependent variable and the residuals ($\hat{\varepsilon}_t$) are extracted. The residuals must be checked before fitting the model because the DCC-GARCH and Copula DCC-GARCH are built to model and account

for the presence of heteroscedasticity in the time series. To assess the presence of conditional heteroscedasticity, the squared residuals ($\widehat{\varepsilon}_t^2$) are used as a proxy for variance. A secondary regression is then performed, where the squared residuals are regressed on their own lagged values up to a chosen number of lags (q). The null hypothesis (H_0) assumes that there are no ARCH effects, meaning the coefficients of the lagged squared residuals are jointly equal to zero. To test this, an LM (Lagrange Multiplier) test statistic is computed as $LM = TR^2$, where T is the sample size and R^2 is the coefficient of determination from the auxiliary regression. This statistic follows a chi-square (χ^2) distribution with q degrees of freedom. If the LM statistic exceeds the critical chi-square value, the null hypothesis is rejected, indicating the presence of ARCH effects.

To analyze this dataset, we utilized the ARCHTest() function from the FinTS library in R. The test was conducted with up to 10 lags. The results indicated that both return series, namely KSE 30 Futures and the underlying index KSE 30, exhibited ARCH effects. To further analyze the volatility clustering on our dataset, we used residual and squared residual plots along with Auto Correlation Function (ACF) plots of residuals and squared residuals, which showed significance at lag 1 (see figures 6 and 7).

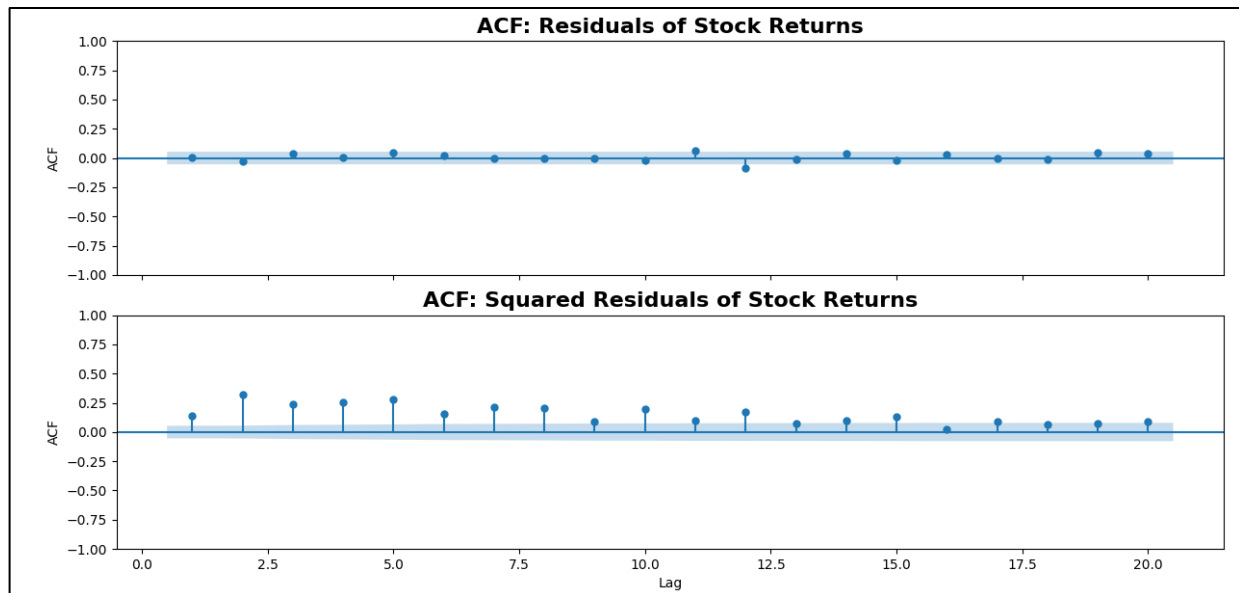


Figure 6: ACF plots of residuals and squared residuals of KSE 30 index return

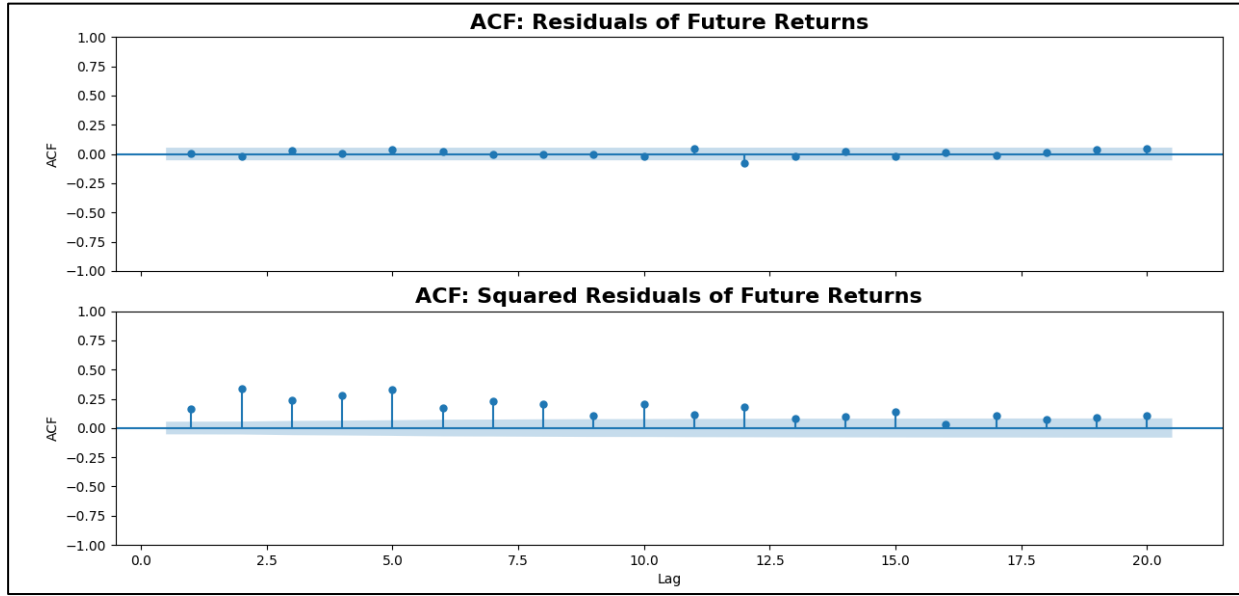


Figure 7: ACF plots of residuals and squared residuals of KSE 30 index futures return

4.1.3 Modeling Approaches for Hedge Ratio Estimation

This paper focuses on dynamically hedging the risk of a KSE 30 index portfolio using KSE 30 stock index futures. Hedging through future contracts is a widely known method to offset price risk. The most commonly used framework for hedging through stock index futures is minimum variance hedging, which targets optimizing the portfolio's total variance. The minimum variance hedge ratio can be calculated using the framework below:

Let R_s be the return of an equity portfolio underlying the KSE 30 index to hedge the risk of this portfolio, we short N KSE 30 future contracts whose return is represented by R_f . Equation 6 below defines the return of the hedged portfolio.

Equation 6

$$R_v = R_s - NR_f$$

And we define the variance of the portfolio V in equation 7.

Equation 7

$$\sigma_v^2 = \sigma_s^2 + N^2\sigma_f^2 - 2N\sigma_{s,f}$$

Since we aim to find a hedge that reduces the risk to a minimum level, we differentiate

equation 7 with respect to N ;

Equation 8

$$\frac{\partial \sigma_v^2}{\partial N} = -2N\sigma_f^2 + 2\sigma_{s,f}$$

Equating equation 8 to zero, gives us N^* (refer equation 9).

Equation 9

$$N^* = \frac{\sigma_{s,f}}{\sigma_f^2}$$

Here N^* is the minimum variance hedge ratio, $\sigma_{s,f}$ is the covariance between spot and future returns and σ_f^2 is the variance of future contracts' returns. There are various methods used in prior research to estimate the MVHR. The direct methods, i.e., methods that estimate the hedge ratio directly, include the Ordinary Least Square (OLS) technique, Regression Coefficient (RC) technique, Error Correction Models (ECM), etc. However, these direct methods mostly do not estimate time-varying hedge ratios. Indirect methods include sophisticated models such as Generalized Autoregressive Conditional Heteroskedasticity (GARCH) models and their variants, namely, Dynamic Conditional Correlation (DCC) GARCH, Constant Conditional Correlation (CCC) GARCH, Copula GARCH, and Baba, Engle, Kraft, and Kroner (BEKK) GARCH, etc.

In this research, however, we used the econometric models DCC eGARCH and Copula GARCH and the machine learning models Long Short – Term Memory – Convolutional Neural Network (LSTM – CNN) model and FT-Net Hybrid model to compute time-varying dynamic hedge ratios.

4.1.4 Dynamic Conditional Correlation (DCC) GARCH

Inspired from (Xu & Li, 2017), which highlighted that for enhanced volatility modelling we can combine the Exponentially Weighted Moving Average (EWMA) with the GARCH model. The EWMA models volatility while assigning greater weight to recent events and the GARCH models account for volatility clustering therefore, combined they can portray a more accurate representation of market dynamics. So, before fitting the DCC GARCH model

on the training data we standardize the returns of the KSE 30 index and its future contracts. The standardized returns were calculated using the equation 10 below.

Equation 10

$$Z_{i,t} = \frac{R_{i,t} - \mu}{\sigma_t}$$

Where, $R_{i,t}$ is the raw log return of the asset i, and μ is the mean of the returns. The standard deviation of the returns was estimated using the EWMA method. The EWMA method is a powerful technique for modelling conditional volatility because it captures volatility clustering without requiring a full GARCH specifications. The EWMA variance is estimated using the following equation 11:

Equation 11

$$\sigma_{i,t}^2 = \lambda \sigma_{i,t-1}^2 + (1 - \lambda) R_{i,t-1}^2$$

Here $\sigma_{i,t}^2$ is the variance of the raw log returns of asset i. Since we convert the raw returns of the training data to standardized returns while forecasting the volatilities of the testing data, we convert them to standardized forms as well using standard deviations estimated in the training data. (see figure 8)

The DCC GARCH model is a multivariate GARCH model proposed by (Engle R. , 2002) which is a generalization of the (Bollerslev T. , 1990) constant conditional correlation (CCC) GARCH estimators. The DCC framework is a two-step methodology as follows:

4.1.4.1 Univariate modelling using GARCH

The DCC model proposed by (Engle R. , 2002) requires the use of univariate GARCH models to estimate the conditional variance of individual assets' returns, specifically the KSE 30 index returns and KSE 30 futures contracts returns in our case. To model the conditional variances, we used the Exponential GARCH (eGARCH) model introduced by (Nelson, 1991), which, unlike standard GARCH models, allows for asymmetry in the impact of positive and negative shocks on volatility. This is an essential feature in financial time series data, such as the one used in this research, where negative news trends tend to increase volatility more than positive news of the same magnitude.

This research defines the return process using an ARMA (1,1) model represented by equation 12 below.

Equation 12

$$Z_{i,t} = \mu_i + \phi_1 Z_{i,t-1} + \theta_1 \epsilon_{i,t-1} + \epsilon_{i,t}$$

where $Z_{i,t}$ is the standardized return of asset i, μ_s is the constant mean and $\epsilon_{i,t}$ is a white noise error term with zero mean and constant variance. Then we use the eGARCH (2,1) equation from (Nelson, 1991) (equation 13) to estimate the volatilities of KSE 30 index and KSE 30 index future contracts returns.

Equation 13

$$\begin{aligned} \log h_{i,t} = & \omega_i + \beta_{i,1} \log h_{i,t-1} + \beta_{i,2} \log h_{i,t-2} + \alpha_i \left(\frac{\epsilon_{i,t-1}}{\sqrt{h_{i,t-1}}} \right) \\ & + \gamma_i \left(\left| \frac{\epsilon_{i,t-1}}{\sqrt{h_{i,t-1}}} \right| - E \left| \frac{\epsilon_{i,t-1}}{\sqrt{h_{i,t-1}}} \right| \right) \end{aligned}$$

Where $h_{i,t}$, is the variance of asset i. The model parameters were estimated using Maximum Likelihood Estimation (MLE) in R via the `ugarchspec()` and `ugarchfit()` functions from the `rugarch` package.

4.1.4.2 Dynamic Conditional Correlation Estimation

After estimating conditional volatilities, the second step was to estimate the dynamic conditional correlation matrices. To do so, the methodology presented in (Engle R. , 2002) where let $\epsilon_t = H_t^{1/2} z_t$ be the vector of residuals and $z_t \sim N(0,1)$ and H_t is the conditional covariance matrix as presented in equation 14.

Equation 14

$$H_t = D_t R_t D_t$$

Where D_t (equation 15) is a diagonal matrix of time-varying standard deviation from univariate eGARCH models:

Equation 15

$$D_t = \text{diag}(\sqrt{h_{1,t}}, \sqrt{h_{2,t}}, \dots, \sqrt{h_{3,t}})$$

The DCC model assumes that the standardized residuals $\eta_t = D_t^{-1}\epsilon_t$ have a time-varying correlation structure governed by a GARCH-type equation (equation 16):

Equation 16

$$Q_t = (1 - a - b)S + a\eta_{t-1}\eta_{t-1} + bQ_{t-1}$$

Where Q_t is the time-varying covariance matrix of standardized residuals, S is the unconditional variance of η_t . The dynamic correlation matrix is then obtained by:

Equation 17

$$R_t = \text{diag}(Q_t)^{-1/2}Q_t\text{diag}(Q_t)^{-1/2}$$

The DCC model was estimated using the `dccspec()` and `dccfit()` functions from the `rmgarch` package in R, with univariate eGARCH models used for the marginal distributions.

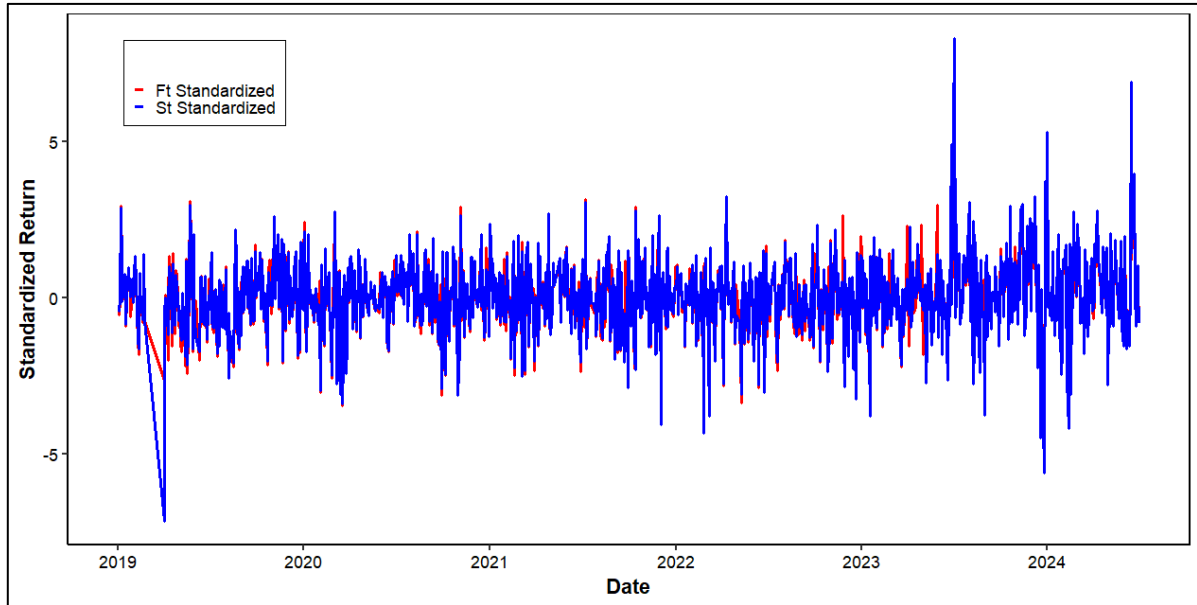


Figure 8: Standardized Returns of KSE 30 Index and its Future Contracts

4.1.5 Dynamic Copula DCC GARCH Model

Copula-DCC-GARCH models combine (Engle R. , 2002) DCC-GARCH with copula theory, developed further by (Patton A. J., 2006), (Jondeau & Rockinger, 2006), and (Hafner &

Manner , 2012). Despite the fact that the DCC-GARCH model presents a comprehensive approach for modelling time-varying correlation between assets, it relies on the assumption that the joint distribution of the standardized residuals is multivariate normal. However, financial time series often exhibit non-linear dependencies, especially in the tails of the distribution (tail dependence), which the normal distribution may not adequately capture. To address this limitation, we employ a Copula-based DCC-GARCH approach that allows us to capture complex dependencies and nonlinear co-movements, by focusing on the collective and individual behavior of the assets returns. Therefore, to optimize our hedge ratios and reduce the variance, we combine ARMA-eGARCH margins with DCC correlation structure and a student-t Copula that adequately captures changing volatilities and tail dependence. (Demarta & McNeil, 2005) (Hsu, Tseng, & Wang, 2008)

4.1.5.1 Pre Copula Fitting Procedure

As mentioned in the DCC-GARCH methodology; to remove the noise and stabilize volatility, an Exponentially Weighted Moving Average (EWMA) filter is applied for standardized returns in training and testing data. After this, the first step in the Copula-DCC GARCH framework involves modeling the marginal distributions of each return series to account for volatility clustering and asymmetry. Similar to the earlier section titled “Univariate modelling using GARCH”, the KSE 30 spot index and futures returns are modelled using Exponential GARCH (eGARCH) processes. The residuals $\varepsilon_{i,t}$ from the eGARCH models are standardized to obtain standardized residuals (see equation 18):

Equation 18

$$z_{i,t} = \frac{\varepsilon_{i,t}}{\sqrt{h_{i,t}}}$$

Assuming normality of standardized residuals, we transform them to uniform margins using the cumulative distribution function (CDF) of the standard normal distribution, represented in equation 19 below.

Equation 19

$$u_{i,t} = \Phi(z_{i,t})$$

This transformation ensures the marginal uniformity required for copula estimation. The resulting $u_{i,t} \in (0, 1)$ are then used to model the joint distribution of returns. The output of this step serves as the input for both the DCC estimation and the copula transformation steps that follow.

The estimation of the dynamic conditional correlation (DCC) model follows the methodology already outlined in the preceding section, based on the framework introduced by (Engle R. , 2002). To avoid repetition, we refer the reader to the earlier section titled “Dynamic Conditional Correlation Estimation”, where the equations governing the DCC model (Equations14-17) are presented in detail. These equations define how the conditional correlation matrix R_t evolves over time using past standardized residuals and a weighted moving average structure.

4.1.5.2 Copula Estimation

Once the residuals are transformed into uniform margins, a copula function is used to model the dependence structure. To flexibly model the joint distribution between the standardized residuals of spot and futures returns beyond linear correlation, we fit a bivariate Student-t copula to the standardized residuals. The student’s t copula was selected for its ability to capture tail dependence, which is crucial for modeling co-movements during extreme market conditions (Demarta & McNeil, 2005).

Given the fitted marginal models, let equation 20 define the probability integral transformation of each standardized residuals into uniform variables, then the student-t copula can be defined by its correlation coefficient ρ and degrees of freedom ν as:

Equation 20

$$C(u_1, u_2; \rho, \nu) = t_{\nu, p(t_{\nu}^{-1}(u_1), t_{\nu}^{-1}(u_2))}$$

Where, $C(u_1, u_2; \rho, \nu)$ is the copula function, $t_{\nu, p}$ is the CDF of the bivariate Student's t-distribution with ν degrees of freedom and correlation ρ , t_{ν}^{-1} is the quantile function of the univariate Student's t-distribution with ν degrees of freedom (Alshenawy, 2024). These parameters are estimated using Maximum Likelihood Estimation via the `fitCopula()` function in R.

According to (Demarta & McNeil, 2005), the density of t-copula for Maximum Likelihood Estimation, can be estimated as:

Equation 21

$$C_{v,P}^t = \frac{f_{v,P}(t_v^{-1}(u_1), \dots, t_v^{-1}(u_d))}{\prod_{i=1}^d f_v(t_v^{-1}(u_i))}, u \in (0,1)^d$$

Where, $f_{v,P}$ is the joint density of a multivariate t-distributed random vector and f_v is the density of the univariate standard t-distribution with v degrees of freedom.

To simulate the t-copula, firstly we generate a multivariate t-distributed random vector X , which means a random value that follows a t-distribution, with degrees of freedom v , a constant mean everywhere, and a correlation matrix P . A normal mixture technique explained by (Demarta & McNeil, 2005) is used to generate the vector X . Then, we apply the standard t-distribution's Cumulative Distribution Function (CDF) to each element of X , so that the values fall between 0 and 1. This way, we obtain our sample from the t-copula. Secondly, to estimate the density, we first map the sample values back into the original t-distribution space, using inverse CDF, then apply the multivariate t-distribution's density function at those points, and divided by the product of the marginal t-densities. Using the formula at equation 20, we obtain the density of the t-copula, which is useful for estimation. It is important to note that the t-copula remains invariant under strictly increasing transformations of the marginals, ensuring that the dependence structure is purely modelled by the copula, independent of the marginals themselves.

4.1.6 Long-Short Term Memory – Convolutional Neural Network (LSTM–CNN) Hybrid Model

The financial time series data is characterized by complex features such as nonlinearities, volatility clustering, regime shifts, and structural breaks. These present significant challenges to classical econometrics model that assume linear relationships or stationary properties. To overcome the deficiencies of the previously employed models, we make use of a hybrid deep learning model which is a combination of the Convolutional Neural Networks (CNN) and Long Short-Term Memory (LSTM) networks from the Recurrent Neural Network (RNN) family (Kumar, Rao, & Dhochak, 2025). The model was used to create a dynamic hedge ratio estimate for the KSE 30 index portfolio.

The rationale for using LSTM-CNN Hybrid is because they complement each other. Using convolutional filters, CNNs can identify short-lived anomalies like bursts of volatility or sudden co-movement in spot and futures prices as they slide over the temporal dimension. This is particularly important in financial data where a short-term increase or shock to the price can hinder hedging. On the other hand, LSTMs are proven for modeling sequential dependencies and long-range temporal correlations by letting the network remember or forget things through gating. Therefore, these characteristics are relevant to market regime, persistence volatility, and other long-lasting effects that may alter hedge ratios over time.

Through using a combination of these two architectures, the model gains a better understanding of the market by capturing short-term anomalies and long-term dependencies. Accordingly, it provides a more sophisticated data-driven alternative to static or linear hedging schemes. The hybrid form is consistent with equity futures since optimal hedge ratios are not constant but also change through time depending on the market condition. Recent evidence show that these hybrid deep learning models outperform traditional econometric and pure deep learning ones, especially in environment having nonlinear and regime shifts behavior.

This approach matches the project's objective that dynamically minimizes portfolio risk based on estimates of optimal hedge ratios based on market data. The hedging objective is included in the model design and loss function to allow the model to predict minimum variance hedge ratios (MVHR) while minimizing hedged portfolio variance to enhance the real-world applicability of predictive hedge strategies.

4.1.6.1 Theoretical Foundations

To fully comprehend the working and advantages of the LSTM–CNN hybrid model, examining the theory behind the constituent architectures is highly imperative.

4.1.6.1.1 Convolutional Neural Networks (CNNs)

CNNs were originally developed for spatial data such as images, but their use in time series analysis has grown rapidly. A CNN applies convolutional filters that act as localized pattern detectors over input data. Formally, a convolution operation for a one-dimensional time series input x and filter w of size K is defined in equation 22 below:

Equation 22

$$(t) = (x * w)(t) = \sum_{k=0}^{K-1} x_{t+k} w_k$$

where t indexes the time steps. In the context of the financial return series, this operation allows the model to scan over a sequence of returns, identifying local features such as sudden spikes, dips, or short-lived correlations between the spot index and its futures contract. CNN filters learn these patterns during training, automatically adapting to the specific structures present in the data.

The use of “causal” padding is crucial in this setup. Causal convolutions ensure that the output at any time step depends only on the present and past inputs, preserving the time-ordering and preventing “future” information leakage, which would invalidate the predictive framework. This is particularly important for financial applications where causality and chronological order must be maintained.

CNNs excel at feature extraction because they transform the original high-dimensional raw data into more compact and informative representations called feature maps. These maps highlight the presence of local temporal features that can be critical for downstream modeling.

4.1.6.1.2 Long Short-Term Memory Networks (LSTMs)

Recurrent Neural Networks (RNNs) are natural candidates for sequential data modeling because of their inherent temporal feedback loops, but standard RNNs suffer from vanishing or exploding gradients, limiting their ability to capture long-term dependencies. LSTMs, introduced by (Hochreiter & Schmidhuber, 1997), overcome these limitations by using a gated memory cell architecture that selectively retains relevant information over time.

An LSTM cell contains several gates, i.e. input, forget, and output gates, that regulate the flow of information. Mathematically, for an input vector x_t , previous hidden state h_{t-1} , and previous cell state c_{t-1} , the operations are:

Equation 23

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i)$$

Equation 24

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f)$$

Equation 25

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o)$$

Equation 26

$$\tilde{c}_t = \tanh(W_c x_t + U_c h_{t-1} + b_c)$$

Equation 27

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t$$

Equation 28

$$h_t = o_t \odot \tanh(c_t)$$

Here, σ is the sigmoid function that constrains gate activations between 0 and 1, effectively deciding how much information passes through. The memory cell c_t is updated by forgetting part of the old cell state and adding new candidate information modulated by the input gate. The hidden state h_t is computed as a gated version of the cell state, controlling exposure to the next layer or output.

LSTMs are highly effective for capturing volatility clustering and long-range temporal dependencies in financial returns, which are crucial for accurate dynamic hedge ratio prediction.

4.1.6.1.3 Hybridizing LSTM and CNN

The LSTM and CNN architectures complementing each other. CNNs are powerful local feature extractors that can detect short, but informative temporal patterns within a moving window. LSTMs, on the other hand, are used to capture dependencies that take place over many steps in time, modeling the structure of the whole sequence.

When applying CNN in the beginning, it reduces the input dimensionality by compressing the raw returns into meaningful local features, which are then processed by LSTM to learn

deeper sequential relationships and long-term interdependencies. This lessens the difficulty the LSTM has to deal with, improving training efficiency and generalization.

In hedging, this translates to the ability to respond both to immediate market shocks and to adjust for longer-term regime changes, volatility persistence, and nonlinearities, which are paramount in minimizing portfolio risk in equity futures markets.

4.1.6.2 Model Architecture and Implementation

The LSTM–CNN hybrid model is implemented using the TensorFlow Keras framework, with design choices informed by both statistical diagnostics and domain knowledge in financial time series modeling.

4.1.6.2.1 Data Input and Feature Engineering

The raw dataset contains daily spot and futures prices for the KSE 30 index. These prices are converted into continuously compounded logarithmic returns for both the spot and futures series as shown in equation 29:

Equation 29

$$R_{S,t} = \ln \left(\frac{P_{S,t}}{P_{S,t-1}} \right), R_{F,t} = \ln \left(\frac{P_{F,t}}{P_{F,t-1}} \right)$$

where $P_{S,t}$ and $P_{F,t}$ are the spot and futures prices at time t .

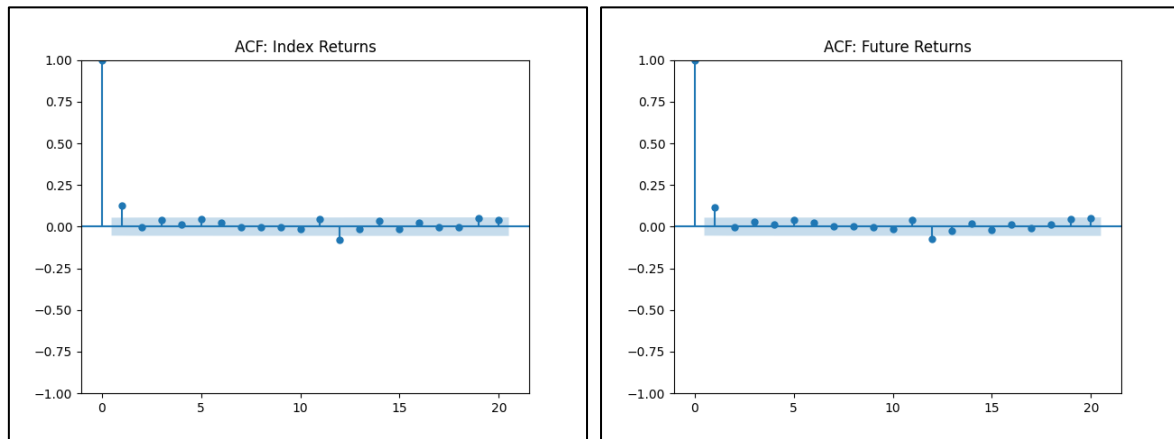


Figure 9: ACF and PACF of Index and Future Returns

Figure 9 presents the autocorrelation functions (ACF) for the daily returns of the KSE 30 index and its corresponding futures contract, revealing important characteristics of short-term and

long-term dependencies in these financial time series. Notably, both plots show a significant spike at the first lag, which stands above the confidence interval, indicating the presence of short-term autocorrelation at lag 1 for both the index and futures returns. This suggests that the return for a given day is not entirely independent of the previous day's return, likely due to factors such as market microstructure effects, delayed information flow, or transient momentum and reversal patterns commonly observed in real-world markets. Beyond this initial lag, however, all subsequent autocorrelation coefficients fall within the bounds of statistical insignificance, confirming that neither series displays persistent linear dependencies at longer horizons. This pattern of strong first-lag autocorrelation followed by rapid decay validates the modeling approach adopted in this study, specifically the inclusion of both contemporaneous and one-period lagged returns as input features in the LSTM–CNN architecture.

Hence, to enrich the temporal context, the first lag of each return series is included as features:

$$R_{S,t-1}, R_{F,t-1}$$

Thus, at each time t , the input features include contemporaneous and lagged returns for both spot and futures, forming a four-dimensional feature vector:

Equation 30

$$x_t = [R_{S,t}, R_{F,t}, R_{S,t-1}, R_{F,t-1}]$$

This choice stems from empirical observations that immediate past returns influence current price dynamics, enhancing model predictive power.

These features are organized into sequential data tensors using a rolling window approach with a sequence length $p = 20$, a hyperparameter chosen based on autocorrelation analyses (ACF and PACF). This window length captures sufficient market history to model relevant dependencies without introducing excessive noise or complexity.

Formally, the input tensor at time t is:

Equation 31

$$X_t = [x_{t-p}, x_{t-p+1}, \dots, x_{t-1}] \in \mathbb{R}^{p \times 4}$$

The corresponding target outputs are the next-day spot and futures returns $R_{S,t}, R_{F,t}$.

4.1.6.2.2 Neural Network Architecture

The model begins with an input layer accepting the $p \times 4$ tensor. This is followed by a one-dimensional convolutional layer with 32 filters and kernel size 3, applying causal padding to preserve the temporal ordering and prevent information leakage from the future. The convolution operation transforms the input into 32 parallel feature maps of length p , where each map encodes a specific temporal pattern learned during training.

ReLU activation is applied after convolution to introduce nonlinearity and maintain computational efficiency while mitigating gradient vanishing.

Next, these feature maps feed into an LSTM layer with 32 hidden units. The LSTM processes the sequence of length p and dimensionality 32, learning to retain or discard information across time steps to form a fixed-length embedding vector summarizing the entire sequence.

A fully connected dense layer with a single linear unit then maps this embedding to a scalar, interpreted as the predicted hedge ratio $\hat{\beta}_t$.

Finally, a custom Lambda layer calculates the hedged portfolio return at time t using equation 32.

Equation 32

$$\hat{R}_{hedged,t} = R_{S,t} - \hat{\beta}_t R_{F,t}$$

This operation embeds the hedging objective directly into the model structure, allowing end-to-end learning focused on minimizing the residual variance of the hedged portfolio. The model architecture pipeline can be seen in figure 10 below.

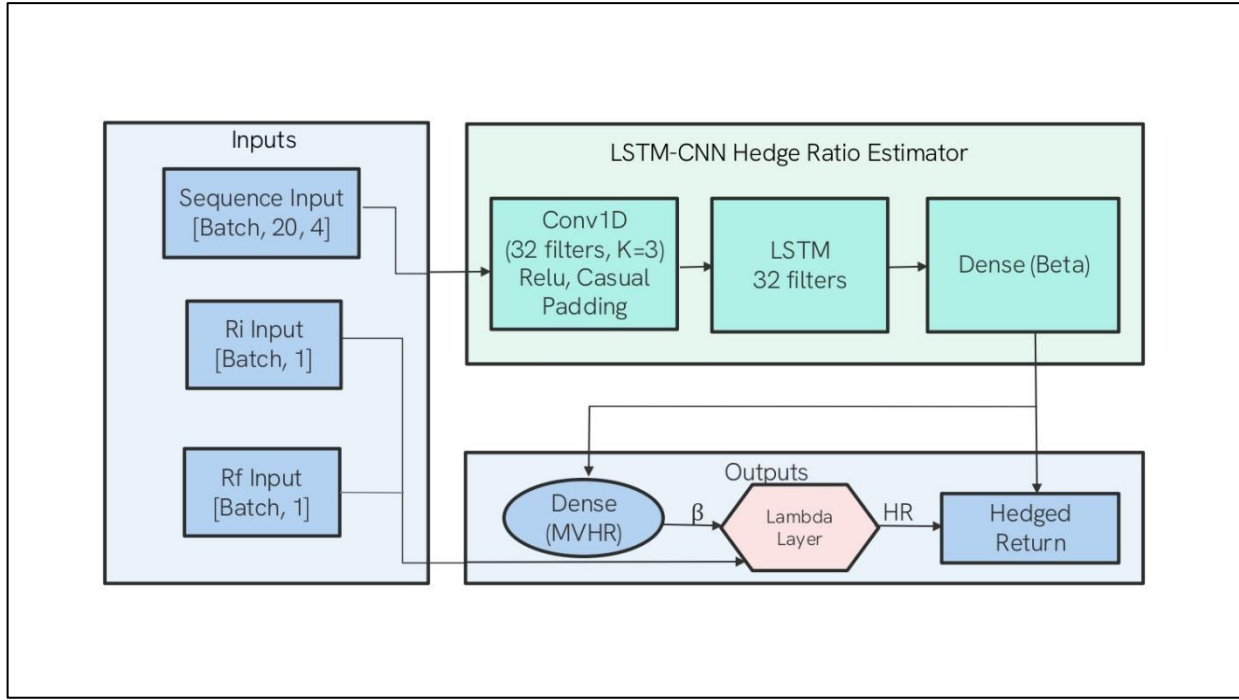


Figure 10: Architecture and Pipeline of LSTM-CNN Hybrid Model

4.1.6.3 Model Compilation and Training Details

The model is compiled with the Adam optimizer, a stochastic gradient descent variant with adaptive learning rates and momentum, set at a learning rate of 10^{-3} (Reyad, Sarhan, & Arafa, 2023). This optimizer is well-suited to complex neural architectures due to its robustness and efficient convergence.

The training objective minimizes the mean squared error (MSE) between the hedged return output and zero, reflecting the ideal of a perfectly hedged portfolio with no residual return:

Equation 33

$$\mathcal{L} = \frac{1}{n} \sum_{i=1}^n (\hat{R}_{hedged,i} - 0)^2 = \frac{1}{n} \sum_{i=1}^n \hat{R}_{hedged,i}^2$$

The hedge ratio output has zero weight in the loss, making it an auxiliary prediction driven indirectly by minimizing portfolio residual risk.

The dataset is split into training and testing sets with an 80/20 ratio (Gholamy, Kreinovich, & Kosheleva, 2018), preserving chronological order to respect temporal dependencies. Training proceeds for 50 epochs with batch size 64, using 20% of the training data for

validation. Early stopping based on validation loss and learning rate reduction on plateaus are implemented to prevent overfitting and stabilize training.

4.1.6.4 Mathematical Formulations and Model Equations

The modeling pipeline is grounded in rigorous mathematical definitions that underpin the data transformation and neural network computations.

4.1.6.4.1 Input Tensor Construction

The input sequence at time t is:

Equation 34

$$X_t = \begin{bmatrix} R_{S,t-p} & R_{F,t-p} & R_{S,t-p-1} & R_{F,t-p-1} \\ \vdots & \vdots & \vdots & \vdots \\ R_{S,t-1} & R_{F,t-1} & R_{S,t-2} & R_{F,t-2} \end{bmatrix} \in \mathbb{R}^{p \times 4}$$

This matrix forms the input tensor to the convolutional layer.

4.1.6.4.2 One-Dimensional Convolution Operation

For each of the 32 convolutional filters, the convolution output at time t , $h_c^{(t)}$ is shown in equation 35:

Equation 35

$$h_c^{(t)} = f \left(\sum_{k=0}^{K-1} w_c^{(k)} \cdot x_{t-k} + b^{(c)} \right)$$

where $K = 3$ is the kernel size, $w_c^{(k)}$ the filter weights, $b^{(c)}$ the bias term, and $f(\cdot)$ the ReLU activation function defined in equation 36:

Equation 36

$$f(z) = \max(0, z)$$

Causal padding ensures that the computation respects temporal ordering: the output at time t depends only on inputs at $time \leq t$.

4.1.6.5 LSTM State and Output Update Equations

Given convolutional features $h_t \in \mathbb{R}^{32}$, the LSTM updates are mentioned in equations 23 – 28.

Hedge Ratio Output

The final hidden state $h_p \in \mathbb{R}^{32}$ after processing the full sequence is mapped via a dense layer:

Equation 37

$$\hat{\beta}_t = W_{\beta} h_p + b_{\beta}$$

where $W_{\beta} \in \mathbb{R}^{1 \times 32}$ and $b_{\beta} \in \mathbb{R}$.

Hedged Return Calculation

The residual portfolio return post-hedging is presented in equation 34 which quantifies the portfolio risk remaining after applying the predicted hedge ratio.

Loss Function

The model training objective is displayed in equation 38.

Equation 38

$$\min_{\theta} \frac{1}{n} \sum_{i=1}^n (R_{S,t} - \hat{\beta}_t R_{F,t})^2$$

where θ encapsulates all neural network parameters. This objective corresponds to minimizing the variance of residual returns, consistent with classical hedging principles seeking minimum variance hedge ratios.

4.1.6.6 Model Evaluation and Diagnostics

Post-training, the model's predicted hedge ratios $\hat{\beta}_t$ are applied to the testing dataset to compute hedged returns $\hat{R}_{hedged,t}$. These returns are subjected to rigorous statistical and financial evaluations.

Statistical tests on estimated hedged returns using KPSS stationarity test, Ljung-Box autocorrelation test, and ARCH LM. These tests are conducted to ensure that the residuals

are white noise, and that they do not have autoregressive conditional heteroskedasticity. This would help in indicating effective risk mitigation.

The measure of hedging effectiveness, as mentioned earlier, are defined by some standard financial performance metrics like variance reductions, RMSE, Sharpe ratio, directional accuracy, mean absolute deviation, VaR and CVaR reductions based on extreme value theory. These measures make a good combination as they show a detailed picture of the risk mitigation and model's protection against tail risks.

The selected settings of sequence length, architecture depth, and hyperparameters contribute to effective model generalization. Causal convolutions respect the coherence of temporal data. The structured CNN–LSTM captures complex non-linear dependencies efficiently (Ullah, et al.). Incorporating the hedged return as a loss target, the model directly aligns training with economic goals, offering transparent and economically relevant output.

4.1.7 Fourier Transform Network (FT-Net) Hybrid Model

The financial markets of today have become highly complicated. This is due to the fact that many forces today do not just play out through time in serial ways. Also, they have different periodicities, cyclical things, and persistent non-stationarities. Classical time-domain models such as autoregressive models and even conventional deep learning networks like LSTM and CNN often fail to capture these subtle features. Existing methods mainly capture time dependencies. Though these methods have advanced to model complex dependencies, they are ultimately time-domain approaches. Thus, such models may systematically miss spectral features, such as cyclical features, business cycles, seasonality, volatility regimes, fundamentally present in financial time-series.

The FT-Net Hybrid model is being developed with this problem in mind (Volpatti, 2024). Evidence from financial econometrics and quantitative trading studies indicates that market returns tend to follow some deterministic cycles and stochastic shocks. Things like seasonality, day-of-the-week effects, business cycles, regime changes and periodic shocks often show up as dominant frequencies in the spectrum. It is vital to capture these patterns in conjunction with their time-domain dynamics for robust risk management and statistically significant arbitrage opportunities.

FT-Net is a new hybrid deep learning architecture that combine the strengths of the spectral (frequency-domain) and the temporal (time-domain) analysis. FT-Net combines sophisticated temporal modeling blocks which are sometimes CNN based and other times recurrent layers like LSTM or GRU with modules that are explicitly based on the Fourier Transform to exploit the joint information in both domains. This method works best in equity futures markets, where missed cyclical risks, regime shifts can affect the performance of hedging strategies, and price series can be spectrally decomposed for timelier realization of arbitrage signals only at a later stage.

FT-Net allows the learning of richer and more adaptable hedge ratios and arbitrage signals by designing a model that natively handles both spectral and temporal features. Our framework goes beyond existing econometric models and conventional deep learning models, which rely on some approximation of the joint distribution of returns or may become non-stationary in power cycles or price cycles. The reasoning for selecting such a hybrid model in this research is now clear, when we consider both the domains jointly we will achieve an ideal dynamic hedge and arbitrage detection which is the need of the hour in present-day equity futures trading.

4.1.7.1 Theoretical Foundations

To understand the FT-Net approach, it is significant to understand the theory behind using spectral analysis modules. The FT-Net is a deep neural network architecture that makes use of discrete Fourier transform (DFT), which is a mathematical tool that separates a finite sequence or data point, such as historical returns, into the sum of sinusoidal components with varying frequencies, amplitudes, and phases (Brigola, 2025).

The DFT for a time series $\{x_n\}_{n=0}^{N-1}$ of length N is mathematically defined as:

$$X_k = \sum_{n=0}^{N-1} x_n \cdot e^{-2\pi i k n / N}, k = 0, 1, 2 \dots N-1$$

where X_k is the complex-valued coefficient corresponding to the k -th frequency bin, x_n is the value of the time series at time n , and i is the imaginary unit. The magnitude $|X_k|$ represents the strength of the frequency component at a given frequency, while the angle (argument) of X_k represents its phase.

In the context of financial time series, the DFT allows researchers and practitioners to reveal otherwise hidden periodicities, cycles, or regime changes. These spectral features may correspond to macroeconomic cycles, regular trading patterns (such as day-of-week effects) or shifts in volatility that occur in a non-stationary but recurrent manner. Notably, the spectral domain can also provide early warnings of regime shifts—such as transitions from low to high-volatility states, since changes in the energy or distribution of frequency components can be detected before corresponding patterns are fully evident in the time domain (Li, Wang, Wang, Yin, & Yao, 2025).

Several empirical studies have documented the importance of spectral phenomena in financial markets. For instance, business cycles (such as 3-5-year economic expansions and contractions), monthly or quarterly seasonality, and even intraday periodicities all manifest as peaks in the spectral density of return series. More sophisticated analyses, such as wavelet or cross-spectral analysis, have further demonstrated that volatility clustering and contagion effects can also exhibit distinct frequency signatures.

Despite this wealth of evidence, most deep learning models in finance have continued to operate in the time domain, missing the opportunity to leverage spectral information for improved risk management and arbitrage strategies. FT-Net seeks to directly address this by embedding spectral modules into its neural architecture, thus enabling the joint modeling of time- and frequency-domain characteristics.

4.1.7.2 Model Architecture and Design of FT-Net Hybrid

The FT-Net Hybrid model represents a modular, end-to-end deep learning pipeline that seamlessly integrates spectral analysis with advanced temporal modeling. The design philosophy is rooted in the principle that financial market data contains interdependent patterns in both domains, and only through their joint modeling can one hope to capture the full spectrum of risks and opportunities.

The architecture of FT-Net can be conceptually divided into several key modules: the input and feature extraction layer, the Fourier (spectral) feature module, the temporal convolutional block, the sequence modeling block (such as an LSTM or transformer), and the final output layers that compute the dynamic hedge ratio.

At the input stage, the model receives multi-dimensional feature matrices, typically

constructed from a window of recent observations of both the cash index returns and futures returns, their lags, and optionally rolling statistical features such as moving averages, rolling standard deviations, or higher-order moments. Each input sample thus consists of a sequence of feature vectors over a fixed window length (sequence length).

The Fourier feature module forms the distinctive heart of the FT-Net. For each input window, the model applies a real-valued fast Fourier transform (FFT) or, equivalently, the DFT, to the time series of returns and their lags. This operation transforms the sequence from the time domain to the frequency domain, yielding a set of complex coefficients that encode the amplitude and phase of each frequency component. In practical implementation, both the real and imaginary parts (or, in some variants, the magnitude and phase) of the transformed coefficients are concatenated and passed as features to downstream layers. This explicit extraction of spectral features enables the model to capture periodic, quasi-periodic, and cyclical phenomena that would otherwise remain undetected.

Following the Fourier module, the model employs a temporal convolutional block—typically composed of one or more one-dimensional convolutional layers. These layers scan the time-sequenced (or frequency-sequenced) feature maps with learnable filters, allowing the model to extract local temporal patterns, short-term dependencies, or shape-based features that often encode meaningful financial events, such as volatility bursts or microstructural price shocks.

After convolutional processing, the output is fused (via concatenation or learned fusion mechanisms) with the original spectral features, ensuring that both time-domain and frequency-domain representations are available for the subsequent sequence modeling block. This block usually consists of a recurrent neural network (RNN) layer, such as LSTM (Long Short-Term Memory) or GRU (Gated Recurrent Unit), although transformer-based encoders can also be used. The sequence modeling layer captures longer-term dependencies and adapts to non-linear, non-stationary dynamics that may unfold over multiple time steps or market regimes.

The penultimate layer of the network is a fully connected (dense) layer, which synthesizes the outputs of all previous layers and computes the dynamic hedge ratio (beta) or arbitrage signal as a function of the fused feature space. Regularization techniques—such as dropout,

batch normalization, or layer normalization—are employed throughout the network to mitigate overfitting and to stabilize training dynamics.

Activation functions are judiciously chosen to suit the nature of the output variable. For instance, if the hedge ratio is constrained to the $[0,1]$ interval, a sigmoid activation may be used. In more general cases, a linear or bounded tanh activation can be employed to allow for negative or leverage-constrained hedge ratios.

A schematic block diagram of FT-Net would typically depict the input feature window feeding into parallel branches: one path for the Fourier transform and extraction of spectral coefficients, another for convolutional analysis of the raw sequences, both converging into a fusion module, then passing through a sequence modeling block (LSTM/GRU), and finally yielding the hedge ratio or arbitrage output through a dense readout layer. This modular structure is instrumental in enabling FT-Net to exploit the full information content of financial time series. The model architecture and pipeline of FT-Net Hybrid can be seen in figure 11 below.

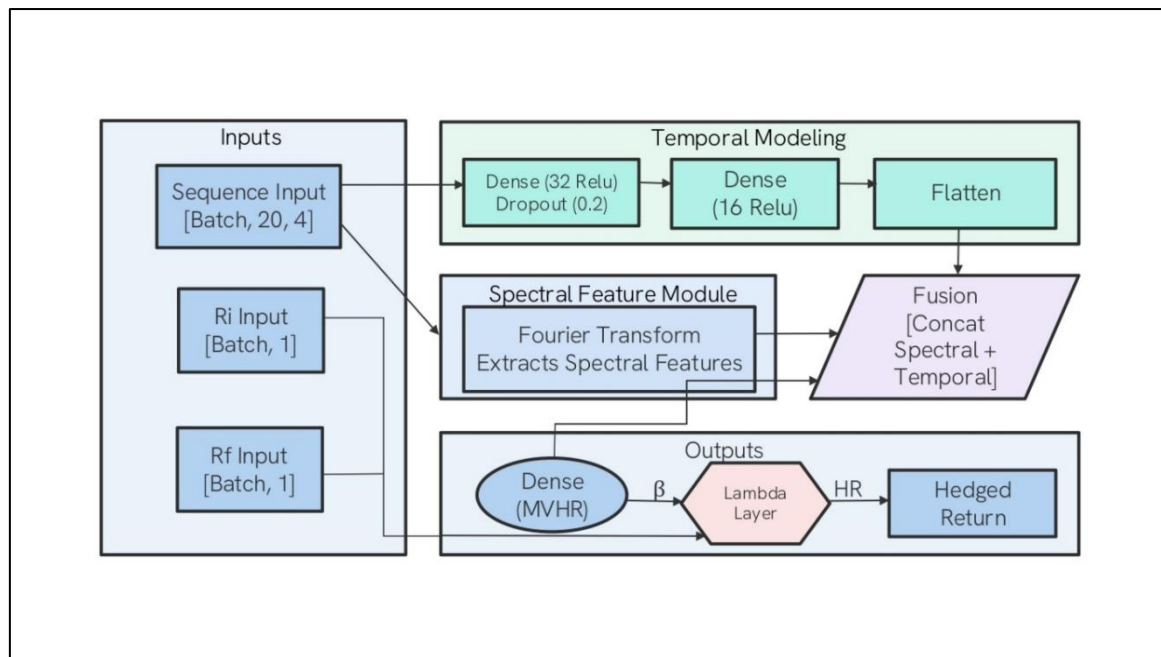


Figure 11: Architecture and Pipeline of FT-Net Hybrid Model

4.1.7.3 Feature Engineering and Selection in FT-Net Hybrid

The selection and engineering of input features is a critical determinant of the FT-Net's ability to generalize and effectively learn meaningful patterns. The initial set of features typically includes raw log returns of the index and future contracts, denoted as $R_{S,t}$ and $R_{F,t}$, along with their lagged values over a window of length w . In financial theory and past empirical evidence, the contemporaneous and lagged return frameworks is a good choice as they capture not only contemporaneous price co-movements but also lead-lag relations and potential information spillovers.

The defining contribution of FT-Net is its meticulous extraction and use of spectral features. The application of DFT to rolling windows of returns and their lags induces encoding of periodicities, of the cycles and of the dominant frequency bands as real inputs. We can use various statistical diagnostics to empirically justify this approach. For instance, the autocorrelation functions (ACF) of the return series, (refer to figure 7), show persistent correlations at 1st lag. This pattern often indicates cyclical behavior. Many periodograms - plots of spectral density versus frequency - show spikes at frequencies that align with established market cycles and seasonal effects.

4.1.7.4 Mathematical Formulations and Model Equations

The FT-Net Hybrid Model integrates both spectral (frequency-domain) and temporal (time-domain) information, leveraging the unique strengths of each to capture the full structure of financial time series data for dynamic hedging and arbitrage signal extraction. To formalize its workings, we provide a step-by-step mathematical exposition of each component of the model, explain the rationale for each transformation, and relate every mathematical operation to financial intuition.

Considering the financial time series, in our case the log returns of a stock index and its associated futures contract. Denote by x_t the value of a univariate feature (e.g., the log return at time t), and by $x_t \in \mathbb{R}^d$ the d -dimensional feature vector at time t , which include raw returns and lagged values.

4.1.7.4.1 Construction of the Input Tensor

For dynamic hedging or prediction, the model does not operate on a single time point, but instead processes a rolling window of w past observations. Thus, for each reference time t ,

the input tensor can be represented as in equation 39.

Equation 39

$$x_t = \begin{bmatrix} x_{t-w+1} \\ x_{t-w+2} \\ \vdots \\ x_t \end{bmatrix}$$

Where each row corresponds to a feature vector at a particular lag, and d is the total number of feature channels. This matrix encapsulates both the most recent and historical context the model may require.

4.1.7.5 Spectral Transformation via Discrete Fourier Transform

The key innovation of the FT-Net model is the explicit transformation of each input feature channel from the time domain to the frequency domain. For a given feature channel j , we extract the corresponding column vector $x_{t,j} = [x_{t-w+1,j}, x_{t-w+2,j}, \dots, x_{t,j}]^T$ and apply the Discrete Fourier Transform (DFT) (equation 40).

Equation 40

$$F_k^{(j)} = \sum_{n=0}^{w-1} x_{t-w+1+n,j} \cdot e^{-\frac{2\pi i k n}{w}}, k = 0, 1, 2, \dots, w-1$$

Where $F_k^{(j)}$ is a complex number representing the amplitude and phase of the k -th frequency component for channel j , and i is the imaginary unit. This operation projects the time-domain sequence onto a basis of complex exponentials, revealing any cyclical, periodic, or oscillatory structure in the input.

Each $F_k^{(j)}$ can be decomposed into its real and imaginary parts as in equation 41:

Equation 41

$$F_k^{(j)} = \text{Re} [F_k^{(j)}] + i \cdot \text{Im} [F_k^{(j)}]$$

Alternatively, the modulus $F_k^{(j)}$ gives the amplitude of the frequency, and the argument $\arg(F_k^{(j)})$ provides the phase. For machine learning purposes, the model typically flattens

the real and imaginary components across all k and all feature channels, constructing the frequency-domain feature vector in equation 42.

Equation 42

$$S_t = \left[\text{Re}(F_0^{(1)}), \text{Im}(F_0^{(1)}), \dots, \text{Re}(F_{w-1}^{(d)}), \text{Im}(F_{w-1}^{(d)}) \right]^T \in \mathbb{R}^{2wd}$$

This vector fully captures the spectral energy distribution across the selected window for all features. In practice, one may optionally select only a subset of k (dominant) frequencies, defined as those with the largest amplitudes (see equation 43).

Equation 43

$$k^* = \arg \max_k |F_k^{(j)}|$$

Alternatively, the model may employ a soft attention mechanism in which learnable weights α_k are applied to each frequency, resulting in an adaptive focus on the most predictive frequency bands (equation 44):

Equation 44

$$S_t^{att} = \sum_{k=0}^{w-1} \alpha_k \cdot \left[\text{Re}(F_k^{(j)}), \text{Im}(F_k^{(j)}) \right]$$

subject to $\sum_k \alpha_k = 1, \alpha_k \geq 0$.

4.1.7.5.1 Temporal Feature Extraction via Convolution

While the spectral branch extracts global cyclical structure, it is also essential to capture local, short-term patterns in the raw time series. To this end, the input tensor X_t is processed through one or more one-dimensional convolutional layers, defined in equation 45 below.

Equation 45

$$C_{t,s}^{(m)} = \sum_{l=0}^{f-1} \sum_{j=1}^d W_{l,j,s}^{(m)} \cdot x_{t-w+1+l,j} + b_s^{(m)}$$

where $W_{l,j,s}^{(m)}$ are the convolutional filters of width f , applied to each feature channel j and producing $n_{filters}$ output channels s , with bias $b_s^{(m)}$. Non-linear activations, typically rectified linear units (ReLU), are then applied (see equation 46).

Equation 46

$$C_{t,s,act}^{(m)} = \max(0, C_{t,s}^{(m)})$$

Stacking multiple convolutional layers increases the receptive field, allowing the model to detect more complex local and sequential structures such as abrupt price jumps, volatility spikes, or temporal clusters.

4.1.7.5.2 Fusion of Time and Frequency Domain Representations

The model then concatenates the time-domain features extracted from the convolutional pipeline with the frequency-domain features produced by the Fourier branch. This operation creates the fused feature vector displayed in equation 47.

Equation 47

$$Z_t = [Flatten(C_t), S_t]^T$$

Where $Flatten(C_t)$ denotes the vectorization of the multi-channel output of the last convolutional layer. The resulting $Z_t \in \mathbb{R}^{n_{flat}+2wd}$ provides a holistic, multi-scale description of the input window.

4.1.7.5.3 Sequential Modeling with Recurrent Neural Networks

To allow the model to capture long-term dependencies, non-linear interactions, and regime changes, the fused feature vector Z_t is passed through a recurrent neural network (RNN), often an LSTM (Long Short-Term Memory) or GRU (Gated Recurrent Unit). The LSTM update equations for each time step can be summarized in equations 23-28.

4.1.7.6 Computation of the Dynamic Hedge Ratio

The output layer of the FT-Net Hybrid model computes the dynamic hedge ratio for the next period as a function of the hidden state h_t . This is implemented as a fully connected (dense) layer as shown in equation 48 below.

Equation 48

$$\beta_t = \phi(w_{out}^\top h_t + b_{out})$$

The predicted one-period-ahead hedged return is then calculated using equation 34.

This formula is derived from the principle of constructing a minimum-variance portfolio by dynamically adjusting the exposure to the hedging instrument.

4.1.7.7 Loss Function and Training Objective

The model is trained to minimize risk, specifically by minimizing the variance of the hedged return series. The empirical variance is given by equation 49.

Equation 49

$$Var(\hat{R}_{hedged,i}) = \frac{1}{N} \sum_{i=1}^N (\hat{R}_{hedged,i} - \bar{R}_{hedged,i})^2, \bar{R}_{hedged,i} = \frac{1}{N} \sum_{i=1}^N \hat{R}_{hedged,i}$$

However, for simplicity, and to center the returns around zero, the objective can be formulated as mean squared error (MSE) relative to zero (equation 50).

Equation 50

$$\mathcal{L}_{var} = \frac{1}{N} \sum_{i=1}^n (\hat{R}_{hedged,i})^2$$

In practice, further regularization is introduced through techniques such as L2 weight decay, dropout, and batch normalization to reduce overfitting. The network is optimized using variants of stochastic gradient descent, such as Adam, which adaptively tunes the learning rate for each parameter.

4.1.8 Model Evaluation and Diagnostics

The performance of FT-Net Hybrid model is tested extensively using a diverse range of statistical as well as financial performance measures to guarantee that all facets of performance must minimize risk and take arbitrage opportunity.

The raw and hedged return series are tested for stationarity and autocorrelation from a statistical perspective. The KPSS test is used to test the null hypothesis of stationarity. The

test statistics and the p-values are computed for both in-sample (training) and out-of-sample (testing) periods. The Ljung-Box Q-test checks for serial correlation in residuals in order to determine how much the model has traded-off future autocorrelations. ARCH-LM test basically answers whether the hedged returns have volatility clustering or if the risk has been diversified adequately.

The MAE, MSE, RMSE and the MAD of the hedged returns compared with the target (which should ideally be zero for a risk-neutral hedge) are used as the measures of the financial performance. Variance reduction, which is defined as the percentage by which the variance of the hedged returns is less than that of the unhedged series, measures how effective the model is at minimizing risk.

The Theil U statistic measures how accurate the model is compared to a naive model where a value of less than 1 means that the model is good. The measure of risk-adjusted return of the hedged portfolio is called the Sharpe ratio. Directional accuracy is defined as the frequency with which the model forecasts the direction of returns

To assess the model in the tails, extreme value theory (EVT) is applied to derive the value-at-risk (VAR) and conditional value-at-risk (CVAR) at a certain level of confidence (e.g. 5%). Hedged series VaR and CVaR reductions versus benchmark, evaluate model efficacy.

The overall runtime (or computational cost) for which the model is reported is referred to as time complexity. A model's time complexity provides context for its usage in real-time (or high-frequency) models. We also compute the average hedged return to see if this model has a systematic bias or drift.

The assessment of FT-Net hinges on diagnostic visualizations. The graphs showing how the beta value changes with time will tell us how the model adapts itself to things like volatile condition or regime shifts. This plot compares the cumulative returns of the hedged and unhedged portfolios and illustrates the actual outcome predicted by the model. Through a periodogram or heatmap of dominant spectral features over time, we are able to gain further insight into the ability of the model to capture and exploit the changing cycles present in the market. In practical use, FT-Net performs positively during cyclical volatility or regime change periods, as spectral features become informative during these times.

4.1.8.1 Practical Considerations and Model Strengths

The results from the FT-Net Hybrid model, as leveraged in this study to hedge the KSE 30 index futures, show the usefulness of using spectra coupled with advanced temporal forms in empirical studies. In times of strong cycle behavior (e.g. return volatility bursts, business cycle turning points, seasonal effects etc.) the FT-Net significantly outperforms econometric models and standard deep learning models. Achieves greater variance reduction, Sharpe ratios and directional accuracy.

FT-Net is effective mainly because it recognizes when the market changes. More specifically, FT-Net can adapt to changes in the frequency content of the market. For instance, during regime switches (from a low volatility to a high volatility state), the splitting of spectral energy across frequency bands changes relevantly. The explicit modeling of these shifts by the FT-Net enables it to dynamically change hedge ratios and thus keep effective risk management when other models do not.

Nevertheless, FT-Net is not without its limitations. As the model becomes more flexible and complex, the risk of overfitting increases, especially when data is noisy or non-stationary. This may affect the generalizability to other assets or market environments. It takes quite a bit of time and energy to calculate; the application of DFT/FFT as well as the training of big neural networks takes a lot of time. Any situation where high-frequency trading is going on, it will use up quite a bit of resources.

Even with the challenges mentioned, the FT-Net Hybrid framework can effectively withstand these issues. Thus, it offers a reliable avenue for dynamic hedging in equity futures. Its strengths seem most evident in complex, non-stationary markets where traditional models are incapable of adapting, and where modelling the time and frequency domain jointly can provide important gains.

4.2 Post Model Performance Evaluation

4.2.1 Statistical Tests Post-Model Implementation

This study conducts statistical tests after implementing the models to assess the residuals, evaluate the model's forecasting quality and check the suitability for comparative analysis. To evaluate the accuracy of the predicted values, we conducted Theil U Statistics in the

hedged return series. Moreover, the ARCH-LM Test, Ljung Box Test, and KPSS Test were performed on the residuals, by splitting up the dataset into training and testing periods. These tests are performed to check whether the models exhibited autocorrelation, non-stationarity, or heteroscedasticity in the residuals. In the econometric models, we used log-differenced spot and futures return series of the KSE-30 index. These were useful for estimating the structure of the volatilities and the time varying correlations. Moreover, for deep learning models, hedged return series were used. These returns were based on the hedge ratios obtained from the models, which are trained and tested using historical return information, hence facilitating us to prevent overfitting, while dynamically capturing the hedging effectiveness.

4.2.2 ARCH LM Test

As previously discussed, ARCH LM test was used in the pre-model testing phase of the mathematical models to detect the presence of autoregressive conditional heteroscedasticity in the return series, however a synthesis of this diagnostic was necessary for a thorough model validation as well. In the pre-model phase of our study, the ARCH LM test was used for providing evidence of conditional heteroscedasticity in the spot and futures return series of the mathematical models. However, it is essential to evaluate the models' performance in mitigating these effects after having been implemented. That is why the same diagnostic tool is applied once again on the residuals of the fitted models. We choose not to repeat the theoretical basis for the test here, as it has been discussed above. Instead, we evaluate how well our models have absorbed volatility clustering and for checking overall homoscedasticity in residuals which is required for valid inference and hedge ratio estimation.

As presented in Appendix A, the p-value for the training data in DCC-GARCH is 0.06452 and 0.05857 which is above the standard 5% threshold. Although these results are marginally significant, indicating minor heteroscedasticity, yet it is not enough to declare the model invalid. The testing data with the p-values of 0.1427 and 0.1687 shows even stronger evidence for the absence of ARCH effects. Moreover, in Copula DCC GARCH, the p-values for futures and spot returns of both training and testing data is above 0.1, indicating that the student-t copula seems to model joint tail dependences and residual volatility dynamics well. Although Standard time series diagnostics are not as often applied to machine learning

models but we did apply them to the hedged return series to determine if the model was a good fit. Therefore, as seen in the Appendix A, the p-values of ARCH LM test for the LSTM-CNN and FT-Net models were all equal to or close to 1.000. This suggests that the models are not only mathematically sound but are also useful for comparing the hedging effectiveness.

4.2.3 Ljung Box Q Test

The Ljung-Box test evaluates the presence of serial correlation in the residuals of the model, which implies that the past errors are affecting current errors. This contradicts the classical white noise residuals assumption and questions the relevance of the model for capturing the temporal dependencies.

In this research, Ljung-Box test for DCC-GARCH, under training and testing data for futures/spot returns, produce the p-value greater than 0.1. These surprisingly large values indicate that the residuals do not have any level of autocorrelation. Moreover, similar results were obtained in Copula DCC-GARCH that reported higher training p-values and very similar test values (as shown in Appendix A) which further confirmed the absence of autocorrelation in both mathematical models. So, it is important to perform these misspecification checks, in order to determine whether the model has a correct lag structure or an absence of autoregressive components. Likewise, in the machine learning model the p-values are greater than the 0.5 significance level (see Appendix A) indicating that uncorrelated residues were produced by the models on training and test data. Despite the fact that this financial time series was non-linear and complex, these results indicate that the models don't have any systematic pattern of residuals remaining.

4.2.4 KPSS Test

The Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test is based on the analysis of stationarity which is an essential assumption in time series modeling. In this test the null hypothesis indicates that the time series is stationary, and it is rejected when the p-value is greater than the standard 5% significance threshold.

In the case of the mathematical models, as shown in Appendix A, both DCC-GARCH and Copula DCC-GARCH have p-values consistently above 0.1, across spot and future returns in both training and testing sets. These findings confirm the stationarity of the series, which

is a necessary condition for the mathematical models. If the data was not stationary, our volatility estimates would have been misleading, and we would have inflated Type I errors. The same result was achieved in the machine learning model, which indicates that the hedged return series are stationary again. This means that their learned structures compute hedge returns that are stable and mean-reverting. This finding is consistent with recent works that suggests that deep learning models can approximate non-linear functions that traditional models cannot represent.

4.2.5 Theil U Statistic

The Theil U statistic measures how the forecasting accuracy of a model compares to a benchmark model, which in our study is the naïve 1:1 hedging strategy. This test was performed on the entire data sets for both mathematical and deep learning models. A U statistic that is less than one means that the model does better than the naïve model while the reverse is true for the U statistic that is greater than one.

The U-statistic of Copula DCC-GARCH is 0.8528, which is significantly better than DCC GARCH with U statistic value of 0.91. This demonstrates its stronger ability to produce better forecasted values than a naive hedge, and therefore its superiority. Therefore, the mathematical model's goal of minimizing risk is represented quantitatively by a lower Theil U, which also validates the model structure. Although machine learning models were outperformed on residual diagnostics, Theil U statistics of LSTM-CNN (0.9179) and FT-Net (0.9128) were slightly higher than those of econometric models. It is still less than 1 and thus provides evidence to a better performance than a naive hedge.

4.2.6 Performance Evaluation

This paper further evaluates the hedging effectiveness of the proposed econometric and deep learning models by establishing a comprehensive performance evaluation framework. This framework facilitates us to assess the market fluctuations and apprehend the varying efficiencies of each model to mitigate risk. The depiction of the hedging performance and findings of the statistical analysis for each model helps in identifying which model is reliable and resilient in the context of Pakistani financial market. Hedge ratios are an important parameter to minimize the variance in the hedged returns; it indicates the proportion of future contracts required to hedge the risk of the spot position. In this study, the dynamic models

display moving hedge ratios to account for the change in variance and co-variances of the spot and future returns. Therefore, these MVHRs are compared to observe the ability of each model to manage the risk considering the volatility and instability of the Pakistani financial market.

4.2.6.1 Predicted MVHRs for Mathematical Model

The MVHRs, estimated by Copula DCC-GARCH and DCC-GARCH models, shown in figure 12 and 13, highlight the in-sample period in light blue series and out-of-sample period in orange series. The MVHRs in the training sample for both models fluctuate approximately around 0.8 and 1.12, indicating stable hedge ratios (β) for the 2019-22 region. However, despite the EWMA filter to smooth out the noise and stabilize volatility, the hedge ratios in both models show spike of approximately 1.2 and 1.1 followed by a dip of approximately 0.75 and 0.8 due to the onset of the COVID-19 pandemic, reflecting unpredictable change in covariance's. Proceeding to the testing period, increased fluctuations are observed with a dip of 0.75 and peak of approximately 1.30 for both models in the 2023-24 region. This hedging pressure reflects the impact shown by the DCC-GARCH and Copula DCC-GARCH model to previous stress episodes. While comparing the Copula DCC-GARCH and DCC-GARCH models, it can be observed that Copula DCC-GARCH demonstrates short lived adjustments due to a slightly lower average MVHRs, combined with amplified oscillations in hedge ratios for the period of extreme shocks reflecting t-copulas ability to capture tail dependence and asymmetric co-movements, while DCC-GARCH has marginally higher average hedge ratios.

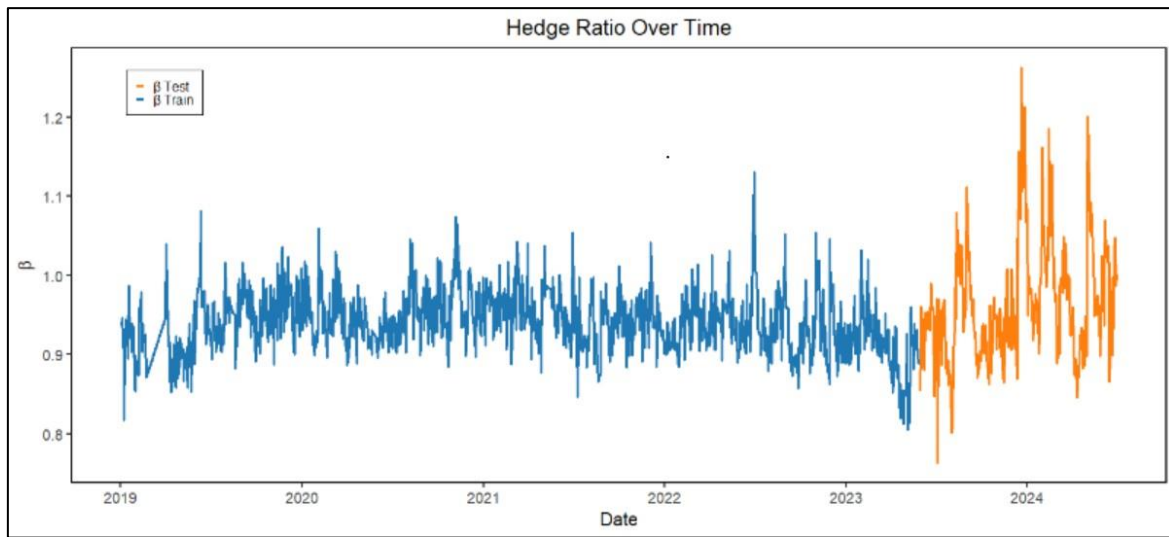


Figure 12: Dynamic MVHRs obtained using DCC GARCH

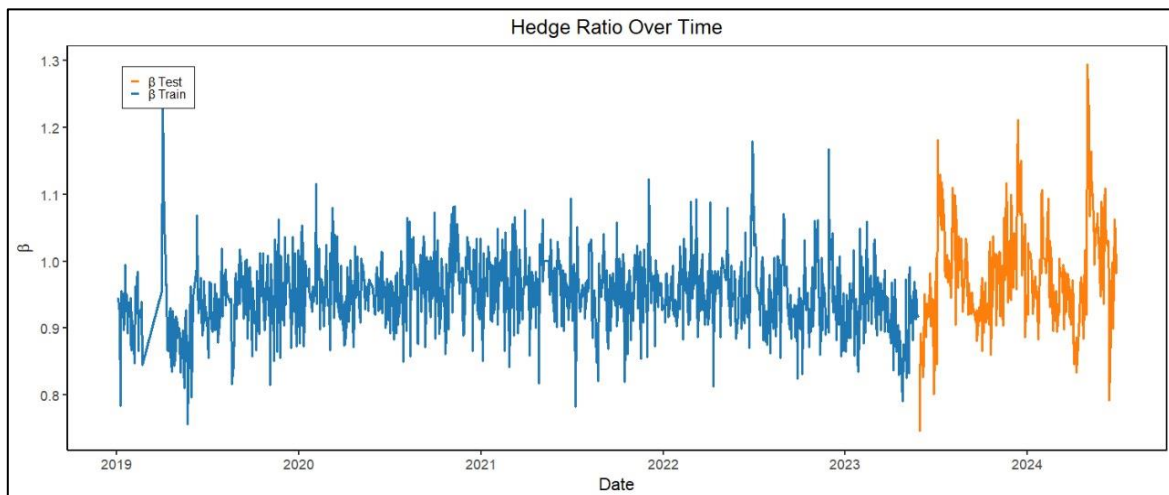


Figure 13: Dynamic MVHRs obtained using Copula DCC GARCH

4.2.6.2 Predicted MVHRs using Machine Learning Models

Moreover, the MVHRs estimated by LSTM-CNN and FT-Net hybrid models, shown in figure 14 and 15, highlight the in-sample period in dark blue series and out-of-sample period in orange series. The hedge ratios (β) in LSTM-CNN model with its peak of 1.05 in the training period has a subdued response to extreme events, when compared to FT-Net, reflecting the model's ability to avoid overstating and understating hedge ratios, suitable for volatile Pakistani market. The convolutional layer and temporal dependency of the LSTM layer of the LSTM-CNN model fluctuates around 0.94-0.98 range, showing reduced short noise yet lower and non-reactive hedging performance when compared to FT-Net. On the

other hand, FT-Net shows hedge ratios (β) mainly close to 1, suggesting better hedge performance. In both deep learning models, a sudden spike i.e. 1.10 in FT-Net and 1.05 in LSTM-CNN, is observed in the early 2020, due to the onset of COVID-19 pandemic, reflecting uncertainty similar to the mathematical model. However, FT-Net highlighting its mean-reverting behavior stabilizes the hedge ratios quickly compared to the other models. The testing period fluctuating around 0.96 and 0.99 shows reliable out of sample performance and the model's ability to quickly capture non-linear complexities. Comparing the hedge ratios of mathematical and deep learning models, it is observed that although all the models exhibit mean reverting behavior, yet FT-Net has higher suitability in dynamic ss conditions, as it shows greater responsiveness to the volatile Pakistani market.

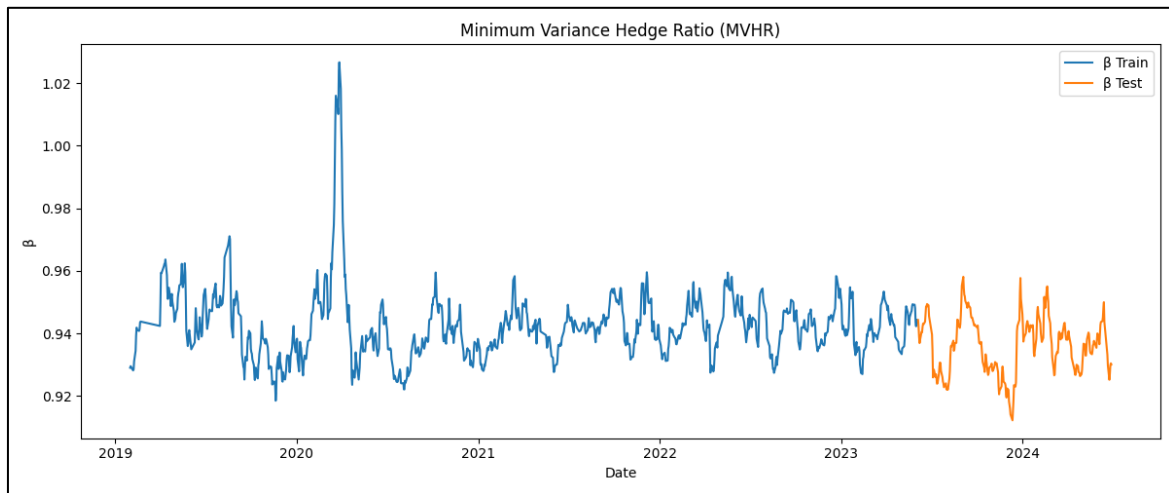


Figure 14: Dynamic MVHRs obtained using LSTM-CNN

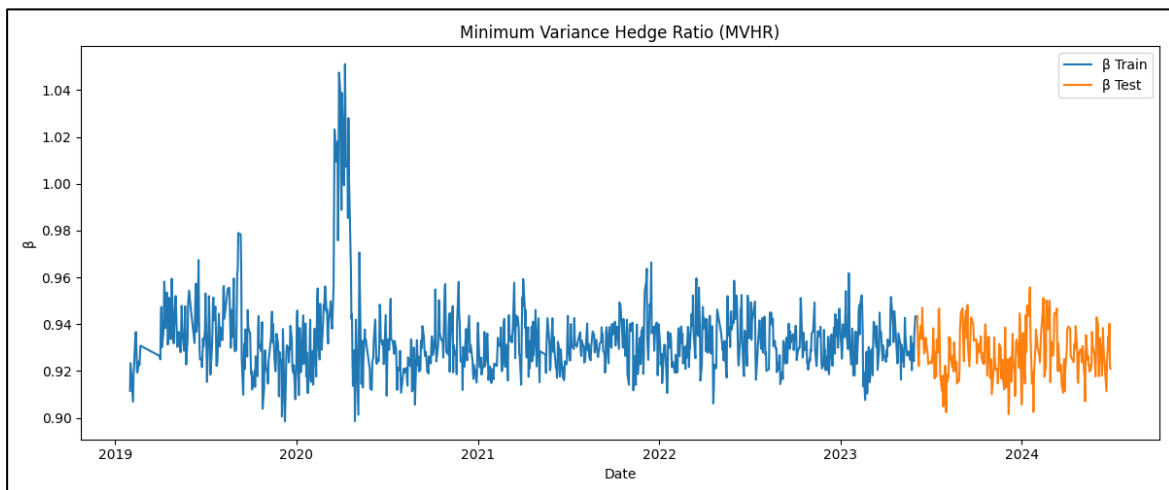


Figure 15: Dynamic MVHRs obtained using FT-Net Hybrid

4.2.7 Hedged Returns Estimation

4.2.7.1 Hedged Returns for Mathematical Learning Model

After the hedging ratios in the training and testing data, we computed the hedged returns for both DCC-GARCH and Copula DCC-GARCH using equation 34. For qualitative inspection, we plotted the time series of predicted hedge ratios $\hat{\beta}_t$ on training and test dates, as well as the cumulative unhedged versus filtered hedged returns. Cumulative hedged and unhedged returns are calculated using equation 51.

Equation 51

$$CumUnhedged(t) = \sum_{i \leq t} R_{S,i} , \quad CumHedged(t) = \sum_{i \leq t} R_{hedged,i}^{final}$$

These figures 16 and 17 below vividly illustrate the risk-reduction benefits of our dynamic hedging procedure. As illustrated in the figure 16 below for DCC-GARCH, the hedged returns (in orange line) show smoother, upward sloping returns as compared to the unhedged returns, which show clear fluctuations especially during the COVID-19 pandemic period. Whereas Copula DCC-GARCH depicts slightly higher returns as compared to DCC-GARCH, due to effective hedging and risk reduction.

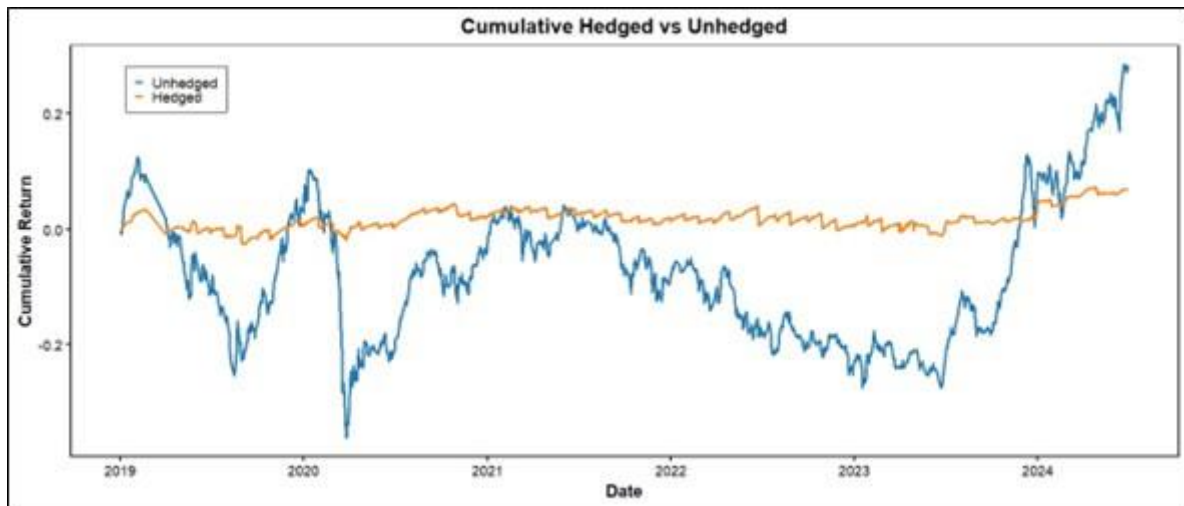


Figure 16: Cumulative Hedged and Unhedged Returns – DCC GARCH

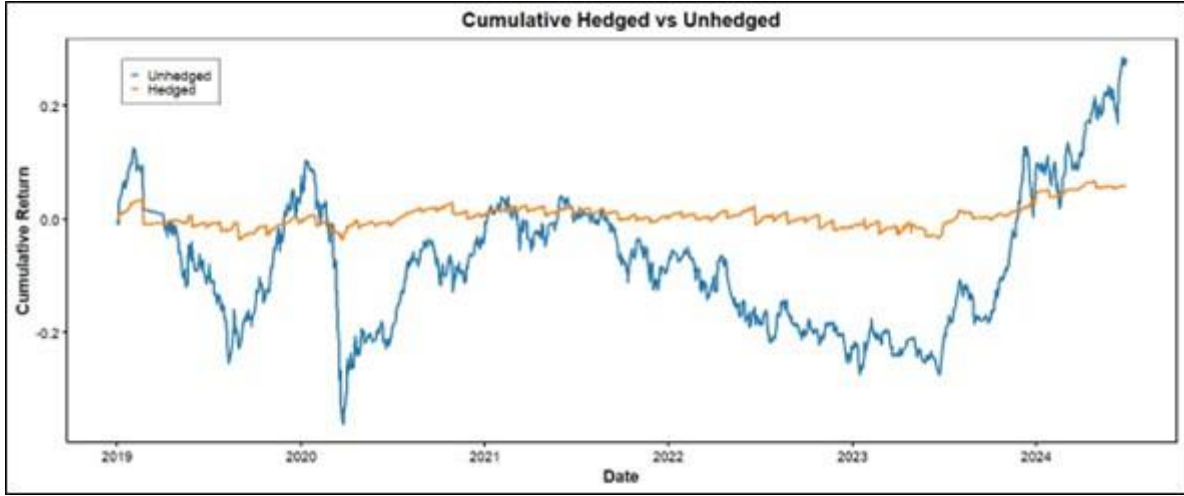


Figure 17: Cumulative Hedged and Unhedged Returns – Copula DCC GARCH

4.2.7.2 Hedged Returns for Machine Learning Model

Similarly, in LSTM-CNN model, after training, and applying the model to both training and test windows to extract $\hat{\beta}_t$ and computed hedged returns using the equation 34 for each sample. Denoting the model's predicted ratios on train and test as $\{\hat{\beta}_t^{(tr)}\}$ and $\{\hat{\beta}_t^{(te)}\}$, we concatenated these to form a full-period series of unfiltered hedged returns.

To correct systematic errors in the CNN–LSTM's hedged returns, we constructed an auxiliary dataset of residuals $e_t = \hat{R}_{hedged,t}$ and regressed them against their own recent history and the raw returns. Specifically, we built a DataFrame with columns shown in equation 52.

Equation 52

$$\{e_t, R_{S,t}, R_{F,t}, e_{t-1}, e_{t-2}, e_{t-3}, R_{S,t-1}, R_{S,t-2}, R_{S,t-3}, R_{F,t-1}, R_{F,t-2}, R_{F,t-3}\}$$

After dropping initial NaNs, we again split chronologically into training and full sets. We then trained three gradient-boosting regressors—XGBoost, LightGBM, and CatBoost—with 200 trees each (CatBoost silent, random_state = 0 for reproducibility). Each model learned to predict e_t from the aforementioned features. We averaged their outputs to obtain an ensemble correction $\hat{e}_t$ and defined our final filtered hedged return as in equation 53.

Equation 53

$$R_{hedged,t}^{final} = \hat{R}_{hedged,t} - \hat{e}_t$$

Moreover, for FT-Net, after predicting $\hat{\beta}_t$ for both train and test windows and compute hedged returns. This direct application of the model's output equation obviates the need for separate covariance/variance calculations, as the network has internalized the mapping from spectral-time features to optimal ratio.

Similar to the LSTM-CNN model, to capture any systematic biases remaining in the hedged returns, we form a residual series $e_t = \hat{R}_{hedged,t}$ and construct lagged regressors up to three days. We then fit an ensemble of gradient-boosted regressors—XGBoost, LightGBM, and CatBoost—each with 200 trees, on the training residuals, and average their predictions. The final filtered hedged return is computed same as equation 53 above, yielding a "double-filtered" series with further reduced variance.

The depiction of cumulative hedged and unhedged returns below vividly illustrate the risk-reduction benefits of our dynamic hedging procedure. While both strategies deliver a very flat, low-volatility hedged profile until mid-2023, the FT-Net hybrid (figure 19) edges out the LSTM–CNN model (figure 18) by generating a slightly higher peak cumulative hedged return—around +3% versus about +2%—and then decaying more gradually. In practice this means FT-Net's frequency-temporal layers are a bit quicker at locking in gains when the market trends, whereas the LSTM–CNN's convolution-plus-recurrent structure produces an equally smooth but marginally lower total hedge payoff.

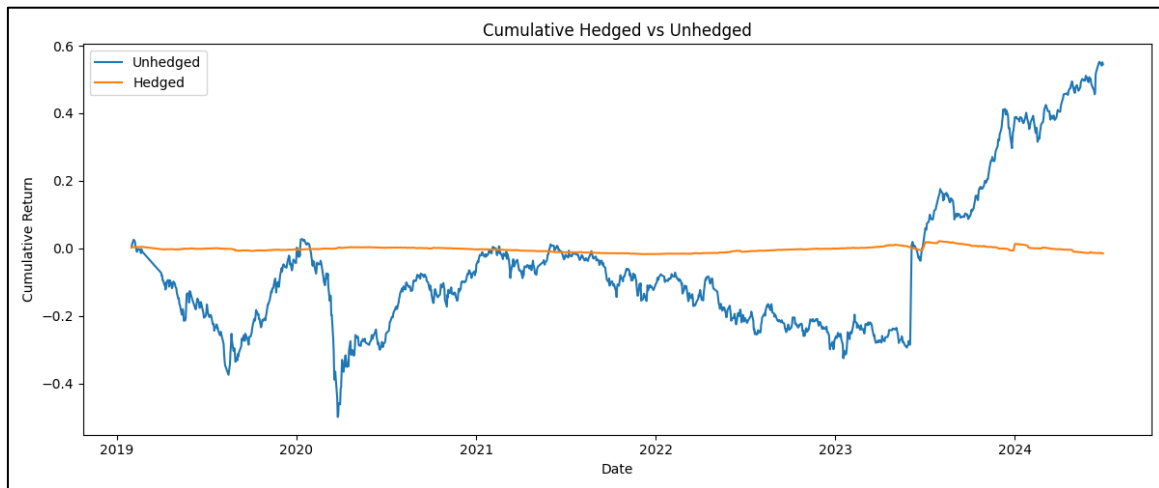


Figure 18: Cumulative Hedged and Unhedged Returns – LSTM-CNN

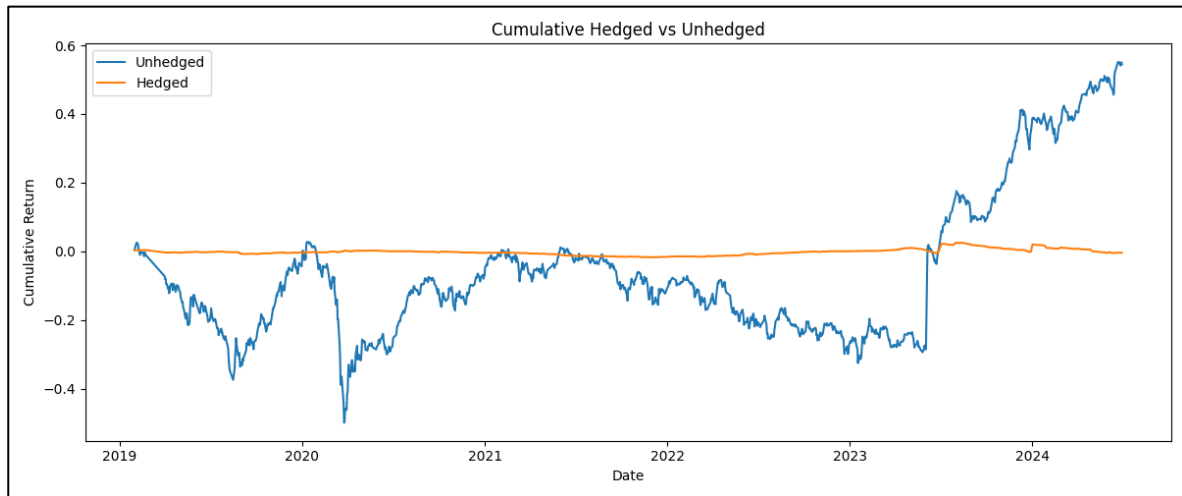


Figure 19: Cumulative Hedged and Unhedged Returns – FT-NET Hybrid

5 Comparative Analysis using Fuzzy TOPSIS

5.1 Introduction

In Chapter 4, we formulated hedging strategies for a KSE 30 index portfolio using the KSE 30 futures contract through both sophisticated econometric models, such as the Dynamic Conditional Correlation Generalized Autoregressive Conditional Heteroskedasticity (DCC GARCH) and its copula-augmented variant, and cutting-edge deep learning architectures, including the LSTM-CNN and FT-Net Hybrid models, to dynamically minimize risk in the equity futures market. The primary objective of this chapter is to rigorously assess the hedging effectiveness of the model by using a comprehensive suite of quantitative performance metrics and rank them with respect to their hedging performance. In this study, we have employed nine performance criteria to gauge the hedge effectiveness, which include the variance reduction percentages, Root Mean Square Errors (RMSE), average hedged returns, Sharpe ratios, directional accuracy, Value at Risk (VaR) Reduction, Conditional Value at Risk (CVaR) Reduction, mean absolute deviation, and time complexity. These metrics effectively capture the risk minimization capability and the operational efficiency of the models. However, the challenges arise from the multicriteria analysis, as no single model unequivocally dominates all criteria, emphasizing the conflicts in the multi-criteria decision-making (MCDM) context. This predicament created the need for the adoption of a systematic methodology to assess the performance of each model and rank them accordingly.

To address this problem, we incorporated the traditional Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) method, a multicriteria decision-making technique. The conventional TOPSIS method ranks multiple alternatives based on their closeness to an ideal solution while maximizing benefits and minimizing costs. The following section provides an in-depth discussion of the TOPSIS methodology used in this research.

5.2 Implementation of the TOPSIS methodology

5.2.1 Define the decision-making problem

The first step of the TOPSIS methodology is to define the decision-making problem. In this research, our objective is to identify the hedging model that effectively dynamically hedges the

risk inherent in the KSE 30 index portfolio. The alternative models under consideration include the Copula GARCH, DCC GARCH, LSTM-CNN Hybrid, and FT-Net Hybrid models.

To obtain fair and multidimensional assessment results, we utilise nine key performance metrics that gauge the risk minimization capability, predictive accuracy, profitability, and operational efficiency of the models.

- i. Variance reduction helps us assess the extent to which a model reduces the variance of the hedged portfolio compared to the variance of the unhedged portfolio.
- ii. The RMSE evaluates the model's ability to forecast the hedge ratios; lower RMSEs indicate more precise predictions.
- iii. The average hedged return is an indicator of the profitability of the models, as it reflects the model's ability to maintain the portfolio returns while minimizing the risk.
- iv. Sharpe ratio is a measure of risk-adjusted return as it reflects the ability of a model to achieve higher returns given the risk minimization criteria.
- v. Directional accuracy is also an indicator of the predictive accuracy of the models; however, it focuses on the frequency with which the model correctly predicts the direction of market movements.
- vi. Value at Risk (VaR) reduction is a measure of how effectively a model reduces potential losses at a specified confidence level.
- vii. Conditional Value at Risk (CVaR) reduction further extends the analysis of downside risk by assessing the expected average loss beyond the VaR threshold.
- viii. Mean absolute deviation (MAD) is an additional measure of predictive accuracy.
- ix. Finally, time complexity measures the computational resources and time required for each model to execute. It is an essential metric to gauge the operational efficiency of the models.

Collectively, these criteria provide a robust suite of metrics to gauge the key characteristics of the models. Tables 1 and 2 below show the criteria for all models on the training and testing datasets.

Table 1: Performance Metrics of Hedging Models – Training Model

METRICS	COPULA GARCH	DCC GARCH	LSTM-CNN HYBRID	FT-NET HYBRID
Variance reduction	92.54	92.71	99.9	99.9
RMSE	0.01241	0.01234	0.00041	0.00042
Average Return	-0.000024	-0.000046	0.000004	0.000004
Sharpe Ratio	-0.0067	-0.013	0.0098	0.0102
Directional Accuracy	66.63	70.09	50.81	52.64
VaR Reduction	83.58	83.97	97.66	97.55
CVaR Reduction	86.63	87.02	97.31	97.23
MAD	0.00886	0.0088	0.000239	0.000238
Time Complexity	6.66	4.479	10.79	9.63

Table 2: Performance Metrics of Hedging Models – Testing Model

METRIC	COPULA GARCH	DCC GARCH	LSTM-CNN HYBRID	FT-NET HYBRID
Variance Reduction	95.09	95.05	96.96	96.56
RMSE	0.01118	0.01121	0.00206	0.00219
Average Return	0.00029	0.00034	-0.00009	-0.00005
Sharpe Ratio	0.1113	0.129	-0.0438	-0.0227
Directional Accuracy	62.07	67.82	51.34	51.34
VaR Reduction	89.39	88.54	94.65	94.38
CVaR Reduction	90.97	90.1	91.74	91.62
MAD	0.00824	0.00817	0.000611	0.000619
Time Complexity	4.48	6.39	10.79	9.63

5.2.2 Determination of AHP weights

After identifying all decision criteria, the fuzzy TOPSIS multicriteria decision-making methodology requires weights to be assigned to each of them. Therefore, to assign weights to every criterion, we use the Analytic Hierarchy Process (AHP), which converts a complex problem into a hierarchy of sub-problems (Sood, Pathak, & Gupta, 2025). We begin by constructing a pairwise comparison matrix, where each element represents the relative importance of one metric over the other. We created a 9x9 pairwise comparison matrix (Table 3). Each entry (i, j) in this matrix expresses how much more important criterion i is compared to criterion j. In the table below, each metric is represented as follows: variance reduction (C1), RMSE (C2), average hedged return (C3), sharpe ratio (C4), directional accuracy (C5), VaR reduction (C6), CvaR reduction (C7), mean absolute deviation (C8), time complexity (C9).

Table 3: Computation and Fuzzification of AHP Pairwise Comparison Matrix

	C1	C2	C3	C4	C5	C6	C7	C8	C9
C1	1	5	5	5	7	7	7	9	9
C2	1/5	1	3	3	5	5	5	7	7
C3	1/5	1/3	1	2	4	4	4	6	6
C4	1/5	1/3	1/2	1	3	3	3	5	5
C5	1/7	1/5	1/4	1/3	1	2	2	4	4
C6	1/7	1/5	1/4	1/3	1/2	1	2	3	3
C7	1/7	1/5	1/4	1/3	1/2	1/2	1	3	3
C8	1/9	1/7	1/6	1/5	1/4	1/3	1/3	1	2
C9	1/9	1/7	1/6	1/5	1/4	1/3	1/3	1/2	1

Once the initial pairwise matrix is constructed, it is normalized to ensure consistency and to allow for proper weighting. The matrix is normalized by dividing each entry in a given column by the sum of that column, transforming the original elements into relative proportions. The normalized matrix reflects the proportional importance of each criterion relative to every other criterion in the set.

Then the weight of each criterion is extracted by computing the mean value of each row in the normalized matrix. This average quantifies the overall relative importance of each criterion, aggregating its normalized influence across all pairwise comparisons. The vector of these row means represents the initial set of crisp (precise) weights, which together sum to one, ensuring a proper probability distribution over the criteria.

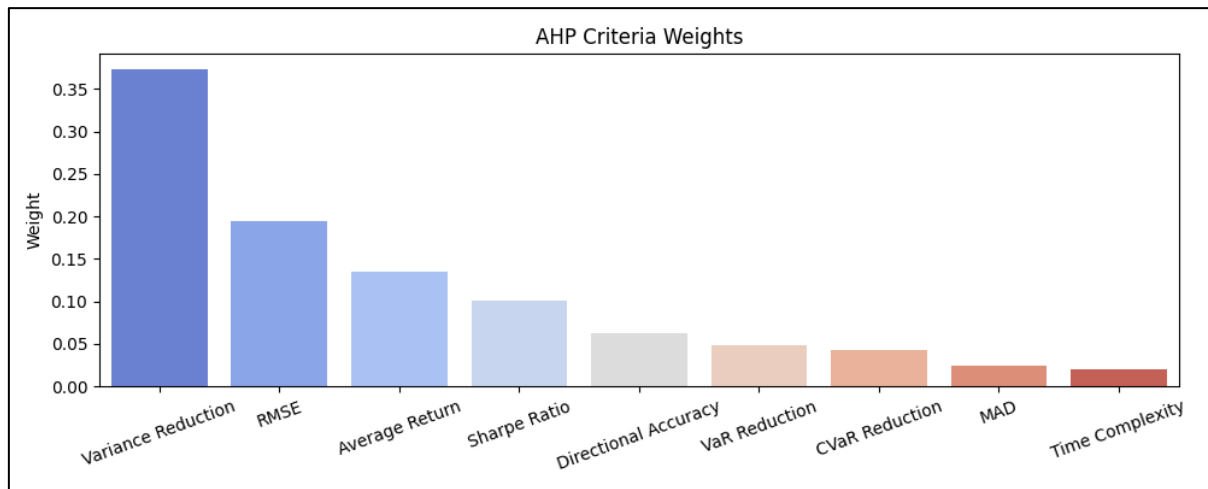


Figure 20: Fuzzified AHP Criteria Weights

Instead of considering each crisp weight, we convert them to fuzzified values by constructing a Triangular Fuzzy Number (TFN) centered on the crisp value but allowing for a range of uncertainty. Specifically, for each weight w , a TFN is defined as $(0.9w, w, 1.1w)$, thereby incorporating a $\pm 10\%$ spread around the nominal value. This fuzzification step captures the ambiguity and lack of perfect precision, making our criteria weighting process more realistic and defensible.

The fuzzified AHP weights are presented in the figure 20 above from highest to lowest. The highest weights were assigned to variance reduction and RMSE, while the lowest weights were assigned to the mean absolute deviation and time complexity.

5.2.3 Fuzzification of decision criteria

After defining and calculating all criteria and their fuzzified weights, the next step was to convert the criteria to fuzzy numbers by using Triangular Fuzzy Number (TFN), defined by a lower, modal, and upper bound. The fuzzification of the metrics was performed by adding ambiguity to each performance criterion, such as variance reduction, RMSE, Sharpe ratio, and others - the code applies a small, symmetric fuzziness margin around the observed value. Specifically, if the original metric value is v , it is mapped to a TFN (l, m, u) where $l = v \times (1 - \delta)$, $m = v$, and $u = v \times (1 + \delta)$, with δ representing the fuzziness factor. This approach means that the lower bound l models a pessimistic scenario, the modal value m represents the most likely estimate, and the upper bound u captures an optimistic possibility for that metric. By constructing the entire

decision matrix with TFNs instead of single values, the fuzzy TOPSIS process can rigorously propagate and analyze the uncertainty throughout the normalization, weighting, and ranking stages. Ultimately, this robust fuzzification ensures that the final model selection and ranking are not sensitive to minor inaccuracies or subjectivity in the original data, yielding results that more faithfully reflect real-world ambiguity and imprecision.

5.2.4 Normalize the decision matrix

With the fuzzified decision matrix and weights in place, the next procedural step is normalization. Here, each criterion is scaled such that its values are directly comparable across models, regardless of their original units or scales. For benefit-type criteria (where higher values are preferable, such as variance reduction or Sharpe ratio), normalization rescales all values to fall within a range of zero to one, with the best observed value mapped to one. Conversely, for cost-type criteria (where lower values are better, such as RMSE, MAD, and time complexity), the transformation is reversed to ensure consistency in the evaluation framework. This ensures that the subsequent aggregation of criteria is both meaningful and equitable.

5.2.5 Calculate weighted normalized matrix

Following normalization, each entry of the decision matrix is multiplied by its corresponding fuzzified weight. This step ensures that criteria with higher relative importance exert proportionally greater influence on the final scores. The resulting weighted normalized matrix thus provides a comprehensive, uncertainty-aware summary of each model's performance across all criteria.

5.2.6 Determine positive and negative ideal solutions

The TOPSIS methodology then identifies the Positive Ideal Solution (PIS) and Negative Ideal Solution (NIS) for each criterion. The PIS represents the optimal scenario, i.e., the best (maximum for benefit, minimum for cost) value observed among all alternatives for each criterion. Conversely, the NIS captures the worst-case scenario (minimum for benefit, maximum for cost). By comparing each model to these reference points, the methodology quantifies both how close a model is to the best possible outcome and how far it is from the worst.

5.2.7 Calculate Euclidean distance from ideal solutions

To rigorously quantify the proximity of each alternative to the ideal and anti-ideal solutions, the code calculates the fuzzy Euclidean distance of each model's performance from both the PIS and NIS, carefully accounting for the spread and structure of each TFN. For each model, these distances are then aggregated across all criteria.

The final step involves computing the closeness coefficient for each model, defined as the ratio of its distance from the negative ideal solution to the sum of its distances from both the positive and negative ideals. The closeness coefficient, ranging between zero and one, provides a single scalar value reflecting how desirable each alternative is relative to the optimal benchmark. Models with higher closeness coefficients are considered more effective, as they are closer to the ideal and farther from the worst-case performance.

5.2.8 Rank the alternatives

Based on the calculated closeness coefficients, all alternative hedging models are ranked from most to least effective. This fuzzy TOPSIS framework thus ensures a transparent, uncertainty-aware, and multidimensional evaluation of hedging models, robustly incorporating both data-driven performance metrics and subjective judgments about their importance. The result is presented in the figure 21 below.

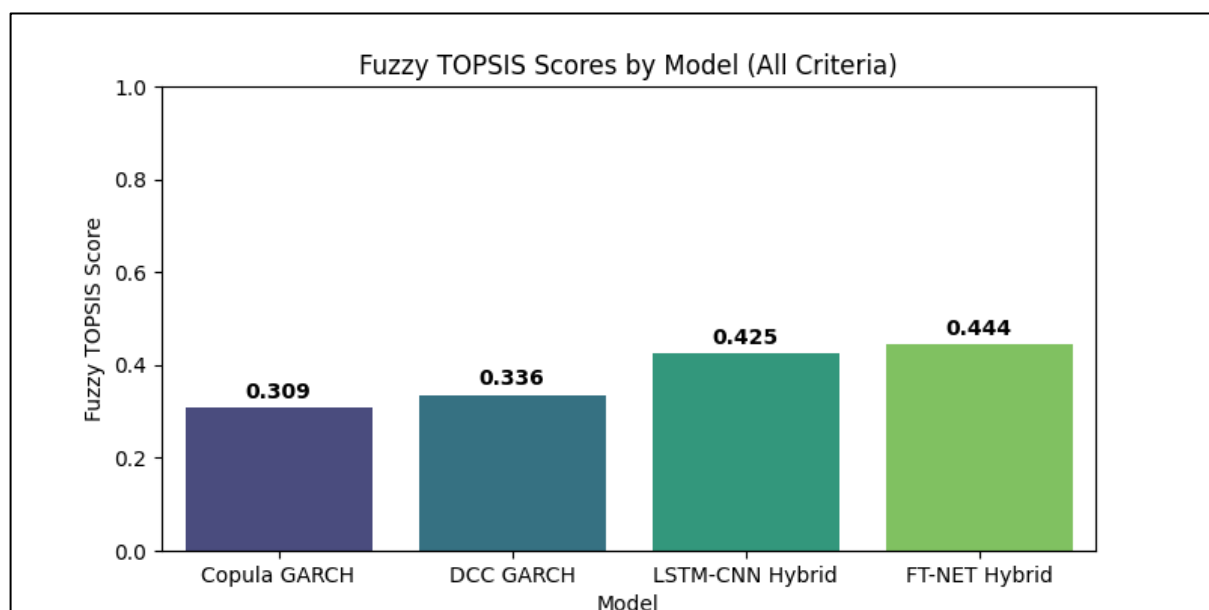


Figure 21: Model Ranking by Fuzzy TOPSIS

As per the Fuzzy TOPSIS criteria, the FT-NET Hybrid model achieved the highest score of 0.444, closely followed by the LSTM-CNN Hybrid model at 0.425. These results suggest that both advanced deep learning-based approaches outperform the conventional econometric models DCC GARCH (0.336) and Copula GARCH (0.309) when evaluated on the integrated criteria of risk minimization, predictive accuracy, profitability, tail risk management, and computational efficiency.

The marginal lead of the FT-NET Hybrid over the LSTM-CNN Hybrid highlights the added value of incorporating temporal features into the hedging strategy, as opposed to relying solely on sequential modeling. On the other hand, the Copula GARCH and DCC GARCH models, while competent in certain aspects, lag in overall performance when all criteria and uncertainties are considered simultaneously.

This ranking, derived from the Fuzzy TOPSIS methodology, provides clear quantitative evidence for the superiority of deep learning hybrid models in the context of dynamic hedging for the KSE 30 index in Pakistan. Moreover, the scores indicate not just raw performance, but a comprehensive and robust assessment that accounts for the inherent ambiguity and imprecision present in real-world financial modeling and decision-making.

5.3 Summary

The Fuzzy TOPSIS methodology, as implemented in this research, provides a sophisticated and robust approach to evaluating and ranking the dynamic hedging models across multiple, often conflicting, criteria while explicitly acknowledging the presence of real-world uncertainty and subjectivity. The process started with the careful definition of the decision-making problem, which centered on selecting the most effective hedging model for minimizing risk and optimizing performance for the KSE 30 index futures market. This requires us to synthesize nine comprehensive metrics to form a multidimensional assessment framework.

The assignment of criteria weights using the Analytic Hierarchy Process (AHP) was a crucial step, translating expert judgment into a structured, quantifiable form. The resulting weights were further fuzzified to reflect the uncertainty and imprecision that naturally accompany subjective decision-making. Fuzzification was also applied to the performance metrics, converting each

observed value into a Triangular Fuzzy Number (TFN) that encompasses a range of estimates. This dual layer fuzzification ensured that all subsequent analyses accounted for the ambiguity found in the time series data.

Through this, the fuzzy decision matrix was normalized so that all metrics could be directly and fairly compared, regardless of their units or scales. Each normalized, fuzzified metric was then weighted according to its fuzzified AHP weightages, resulting in a weighted normalized decision matrix. Following the weights determination the positive and negative ideal solutions were identified. Then, the fuzzy Euclidean distances of each model to these ideal solutions were computed, which then led to the closeness coefficients for each model. These closeness coefficients integrated all performance dimensions into a single interpretable score.

The results revealed the FT-NET Hybrid model as the top performer, achieving the highest Fuzzy TOPSIS score and thus ranking closest to the ideal solution. The LSTM-CNN Hybrid model ranked second, also demonstrated strong overall effectiveness across all criteria. Both DCC GARCH and Copula GARCH models, while still viable, received lower overall scores, highlighting the comparative advantage of advanced deep learning architectures over traditional econometric methods in the context of dynamic hedging.

6 Statistical Arbitrage Analysis

6.1 Introduction

Statistical arbitrage refers to a class of trading strategies that use statistical, mathematical and computational methods to exploit pricing inefficiencies between financial instruments. Statistical arbitrage is widely understood as a high-volume, short-term trading strategy. Unlike traditional arbitrage, which exploits mispricing between identical or related securities for risk-free profits, statistical arbitrage distinguishes itself by assuming such mispricing is subtle, short-lived, and revealed only by complex analysis. Statistical arbitrage involves taking advantage of mutual relationships, for example, mean reversion, cointegration or other statistical properties. These relationships must last long enough to be traded systematically, regardless of the noise of the market or frictions.

Statistical arbitrage is based on the idea that the prices of related financial assets are not always perfectly in sync or adjust instantaneously, thus giving rise to a temporary deviation from their equilibrium relation. These deviations arise due to various reasons like liquidity shock, investor reaction and information lag and so on. They open up arbitrage opportunities if they can be detected and acted on before prices revert to normal co-movement. It's worthwhile noting that signals based on statistical patterns can affect prices themselves. A similar point can be made regarding time-series based strategies. Statistical arbitrageurs use quantitative techniques for creating trading signals, managing risk, and executing trades either at high frequency or in a disciplined, repeatable manner. They rely on the law of large numbers, and diversification across many trades to secure consistent returns.

6.2 Relevance in Equity Futures Markets

The use of statistical arbitrage strategies has become relevant in equity futures markets due to the high liquidity, transparency and standardization of contracts making systematic trading possible. Equity futures like the ones on broad-based indices exhibit strong statistical relationships with their underlying cash indices as they are arbitrated to link the spot and futures price. Nevertheless, due to market microstructure effects, varying levels of liquidity, rolling contract dynamics, and short-term supply-demand imbalances, temporary mispricing between the futures and their respective underlying indices do occur. These transitory departures from theoretical

pricing models, such as the cost-of-carry model, create an environment in which statistical arbitrage strategies can thrive.

In the equity futures market, the search for statistical arbitrage opportunities often revolves around the mean-reverting behavior of spreads, price differences, or constructed portfolios that are theoretically expected to maintain a certain equilibrium. For instance, as the expiration of a futures contract approaches, its price is expected to converge to the spot price of the underlying asset. However, the path to convergence is rarely smooth due to market frictions and investor behavior, thus giving rise to repeated, albeit short-lived, deviations. Advanced statistical techniques can be employed to model these deviations, generate trading signals, and capture profits from the anticipated reversion to equilibrium, provided that the statistical relationships are robust, stationary, and economically exploitable after accounting for transaction costs and slippage.

6.3 Motivation for Statistical Arbitrage in Hedging Context

The motivation to incorporate statistical arbitrage within a hedging framework stems from the inherent imperfections and dynamic nature of financial markets, particularly in emerging market contexts such as Pakistan. Hedging with futures contracts is designed to minimize risk by offsetting exposures in the spot market with opposing positions in the futures market. However, in practice, the effectiveness of any hedging strategy depends not only on the strength of the statistical relationship between the spot and futures but also on the ability to dynamically adjust hedge ratios in response to evolving market conditions.

Traditional hedging strategies, such as static or rolling hedge ratios estimated by ordinary least squares, are limited in their capacity to adapt to non-stationarities, sudden market shocks, and changes in volatility regimes. This creates opportunities for sophisticated statistical arbitrage approaches that explicitly model the time-varying nature of hedge ratios and the spread between the hedged and unhedged positions. By systematically exploiting periods when the hedge is temporarily “ineffective” or when the spread between the hedged portfolio and its theoretical value widens abnormally, statistical arbitrage becomes a complementary tool for both profit generation and risk minimization. The incorporation of mean-reverting signals, stationarity testing, and dynamic trading rules allows for a more flexible, adaptive, and potentially more

profitable hedging process that not only manages risk but also actively seeks to exploit market inefficiencies.

6.4 Statistical Arbitrage in the Context of the Project

Within the framework of this project, statistical arbitrage is operationalized as a rigorous, quantitative methodology for detecting and exploiting short-term pricing inefficiencies arising from the dynamic hedging of a KSE 30 index portfolio using its corresponding futures contract. The core focus is on the hedged return series, which is constructed by applying model-derived hedge ratios estimated from both traditional econometric and advanced deep learning models—to the spot and futures returns. The residual return, after applying the hedge, represents the “spread” or divergence from perfect hedging effectiveness. If this spread is stationary and mean-reverting, it becomes a candidate for statistical arbitrage, as systematic deviations from the mean are expected to be temporary and hence tradable.

To operationalize statistical arbitrage in this context, the project deploys a framework that includes rolling calculation of statistical parameters (such as mean and standard deviation of the hedged return), transformation into standardized z-scores, and generation of trading signals based on deviations from the mean. These signals dictate when to enter or exit long and short positions on the spread, effectively betting on its reversion to the equilibrium implied by the underlying statistical model. The analysis is further enriched by rigorous stationarity testing using the Augmented Dickey-Fuller test to ensure that the trading strategy is grounded in robust statistical properties, rather than random or spurious correlations. The project benchmarks the arbitrage performances of various models, including DCC-GARCH, Copula-GARCH, and LSTM-CNN and FT-Net Hybrid, to ascertain which model provides the maximum exploitable arbitrage opportunities. This will assist in operationalizing statistical arbitrage strategies in the growing derivatives markets in Pakistan. Statistical arbitrage thus is both an indicator of hedge effectiveness, and an intelligent, adaptive quantitative trading strategy which is part of the overall research agenda.

6.5 Methodological Framework for Statistical Arbitrage

The practical execution of statistical arbitrage strategies makes a powerful analytical technique that converts the raw data provided by the market into powerful trading signals. In the project,

statistical arbitrage is grounded on the dynamic modelling of hedge ratios, calculation of hedged returns, generation of signals using rolling statistics and statistical properties like stationarity along with others are usefully checked to make inference. Each stage in this framework is designed to maximize both the interpretability and the effectiveness of the arbitrage signals, ensuring that the trading opportunities identified are both statistically valid and economically meaningful.

6.6 Construction of Hedged Return Series

At the heart of the statistical arbitrage methodology is the construction of the hedged return series, which forms the primary spread to be analyzed and traded. In financial markets, the concept of a “spread” refers to the difference between two related financial quantities—most commonly, the prices or returns of assets that are expected to move together due to fundamental or statistical relationships. In the context of this research, the spread is specifically defined as the return on a dynamically hedged portfolio comprising a position in the KSE 30 index (the spot asset) and an offsetting position in the corresponding futures contract. The central idea is that, by carefully calibrating the exposure to the futures contract, it is possible to reduce or neutralize the risk arising from movements in the spot market.

The formula used to calculate the hedged return at each time step t is mentioned in equation 34.

The hedge ratio $\hat{\beta}_t$ quantifies the degree to which the futures position offsets the risk of the spot position. The value of $\hat{\beta}_t$ is not static but is recalculated at each time point using sophisticated models, such as DCC-GARCH, Copula-GARCH, LSTM-CNN, or FT-Net Hybrid, so as to capture changing market conditions and correlations.

The resulting hedged return series, $\hat{R}_{hedged,t}$, can be interpreted as the “residual risk” or “spread” that remains after applying the hedge. Ideally, if the hedge were perfect and all market movements were fully anticipated by the model, the hedged return would be close to zero. However, in reality, due to estimation errors, market frictions, and the inherent unpredictability of financial markets, the spread will exhibit variability and, importantly for statistical arbitrage, may display patterns of mean reversion or deviation from equilibrium. By focusing on this spread, the statistical arbitrage methodology is able to detect and systematically exploit these temporary dislocations for profit.

6.7 Rolling Statistics and Z-Score Normalization

Next steps after computing the hedged return series, or spread, are to track how this spread behaves over time to get actionable trading signals. The primary tool used for this analysis is the calculation of rolling statistics, which ensure that abnormal deviations are detected with respect to most of the recent history rather than a fixed mean or standard deviation. This is essential in financial markets where statistical properties may evolve over time due to regime changes, volatility clustering, or shifts in investor sentiment.

The rolling mean and rolling standard deviation are defined mathematically as follows (equation 54) for a window of size w :

Equation 54

$$\mu_t = \frac{1}{w} \sum_{i=t-w+1}^t \hat{R}_{hedged,t}$$

Equation 55

$$\sigma = \sqrt{\frac{1}{w-1} \sum_{i=t-w+1}^t (\hat{R}_{hedged,t} - \mu_t)^2}$$

where μ_t represents the rolling mean of the hedged return at time t , and σ denotes the rolling standard deviation over the same window. By recalculating these statistics at each time step, the method adapts to evolving market conditions and ensures that the detection of anomalies is contextually relevant.

To further standardize the detection of trading opportunities, the spread is normalized into a z-score (see equation 56), which expresses the current hedged return in terms of its distance from the rolling mean, scaled by the rolling standard deviation:

Equation 56

$$z_t = \frac{\hat{R}_{hedged,t} - \mu_t}{\sigma_t}$$

The z-score, z_t , provides a dimensionless measure of extremity: values close to zero indicate that the spread is near its recent average, while large positive or negative values signify abnormal

deviations. In the context of statistical arbitrage, these z-score thresholds form the basis for generating entry and exit signals. For example, when the z-score exceeds a certain positive threshold, the spread is considered “overbought” and likely to mean-revert downwards, triggering a short position. Conversely, when the z-score falls below a negative threshold, the spread is “oversold” and expected to revert upwards, prompting a long position. This approach leverages the inherent tendency of mean-reverting series to oscillate around a stable equilibrium, thus enabling the systematic exploitation of temporary mispricing.

6.8 Stationarity Testing and Justification

The effectiveness and validity of any statistical arbitrage strategy is largely dependent on whether the hedged return series is stationary. Statistical features that enable time series analysis are often gathered at regular intervals, making the time-series analysis of stationary data important. Stationarity implies that history matters for what may happen in the future. This is exactly the assumption behind repeated arbitrage opportunities. If the spread is not constant over time, then any movement away from the average could be permanent, which would render any trading strategy based on the idea of moving back towards the average too erratic to implement or too dangerous to implement.

To formally test for stationarity, the Augmented Dickey-Fuller (ADF) test is employed. The ADF test is a statistical hypothesis test in which the null hypothesis is that a unit root is present in the time series, indicating non-stationarity. The test is based on estimating the regression shown in equation 57:

Equation 57

$$\Delta y_t = \alpha + \beta_t + \gamma y_{t-1} + \sum_{i=1}^p \delta \Delta y_{t-i} + \varepsilon_t$$

where y_t is the value of the spread (here, $\hat{R}_{hedged,t}$), Δy_t denotes the first difference of y_t , t is a time trend, p is the number of lagged differences included to account for autocorrelation, and ε_t is the error term. The key parameter of interest is γ is significantly less than zero, the null hypothesis of a unit root is rejected, indicating stationarity.

The outcome of the ADF test includes the test statistic, critical values at standard significance levels, and the p-value. A sufficiently negative test statistic, or a p-value below the chosen significance threshold (typically 0.05), provides statistical evidence that the series is stationary. In the context of this project, the application of the ADF test to the hedged return series ensures that the mean-reversion signals generated by the z-score normalization are grounded in sound statistical properties, rather than being artifacts of a trending or random-walk process. This will not only make the arbitrage signals more reliable but also protect against model failure or false positive signal in the live trading environment.

This project combines the building of hedged returns with rolling statistical analysis, z-score normalization and stringent stationarity testing to create a methodological framework for statistical arbitrage. As a result, it assures that the strategies proposed are both theoretically sound and empirically proven for the KSE 30 equity futures market.

6.9 Signal Generation and Trade Execution Logic

A disciplined and rule-based approach is required for identifying and systematically exploiting arbitrage opportunities for signals and execution. The aim is to capture short-term price inefficiencies using a repeatable process that minimizes discretion and boosts statistical power. This part describes how to design the signals that enable mean-reversion, how to manage the trading positions, and how to ensure that key parameters are calibrated in the right way so as not to adversely impact the performance of the strategy.

6.10 Mean-Reversion Signal Design

Statistical arbitrage strategy is essentially based on mean-reversion phenomenon. This is a tendency for a financial time series such as a spread, or hedged return to drift away from the common value. However, it eventually reverts back towards the mean. The wider or the narrower the spread becomes compared to recent history, the more likely it will move back towards the mean or the equilibrium. The detection of such deviations forms the basis for generating actionable trading signals.

In implementation, the strategy uses the previously calculated z-score. This z-score tells us how many standard deviations the current value of the spread is from its rolling mean.

Hedged return also called a spread at time taken equal to the rolling mean over rolling standard deviation over a window equal μ_t and σ_t respectively. A z-score converts the spread into a standardized measure, which allows comparing offsets today directly, regardless of changes in its volatility or average level over time.

Trading signals are produced by determining an entry threshold (z_{entry}) and an exit threshold (z_{exit}). The thresholds are statistical cutoffs that determine what deviations away from the mean are significant and what level of reversion will be sufficient to close a trade. The entry and exit conditions for trading is mathematically defined as follows.

1. Long Entry (Buy Spread): When the z-score falls below the negative of the entry threshold, i.e., $z_t < -z_{entry}$, this signals that the spread is abnormally low and likely to mean-revert upwards. A long position is initiated, betting on an increase in the spread.
2. Short Entry (Sell Spread): Conversely, when the z-score rises above the positive entry threshold, i.e., $z_t > z_{entry}$, the spread is deemed abnormally high and expected to revert downwards. A short position is initiated.
3. Exit Signal (Close Position): Regardless of the initial direction, when the absolute value of the z-score returns below the exit threshold, i.e., $|z_t| < z_{exit}$, it indicates that the spread has normalized, and the trade should be closed to realize profits and mitigate the risk of reversal.

These rules can be mathematically summarized as in equation 58:

Equation 58

$$Position = \begin{cases} 1 & \text{if } z_t < -z_{entry} \\ -1 & \text{if } z_t > z_{entry} \\ 0 & \text{if } |z_t| < z_{exit} \end{cases}$$

This systematic approach increases the likelihood of making profitable trades or closing out losing trades. They are opened once confirmed deviations and closed when the deviation is gone. Using z-scores rather than absolute values, it provides an adaptive strategy that responds to market conditions and volatility regimes.

6.11 Trade Position Management

The management of trading positions in a statistical arbitrage strategy is governed by the signals described above, with explicit rules dictating when to enter, hold, or exit trades. At any given time, the strategy can be in one of three possible states: long, short, or flat (no position).

A long position is taken when the spread is judged to be excessively low, based on the z-score dropping below the negative entry threshold. In practical terms, this involves buying the spot index and selling the corresponding number of futures contracts as specified by the hedge ratio, with the expectation that the spread will rise. Conversely, a short position is initiated when the z-score exceeds the positive entry threshold, indicating the spread is excessively high; this entails selling the spot index and buying futures contracts, profiting from a decline in the spread.

Trade transitions are managed through a straightforward process. When the z-score signal triggers a new position (long or short) and the strategy is currently flat, the position is opened accordingly. If a position is already open and the z-score crosses the exit threshold in the direction of normalization, the position is closed, returning the strategy to a flat state. Importantly, the system is designed to avoid simultaneous long and short positions; at most, only one direction is active at any time. If the signal reverses before the exit threshold is hit (for example, from long entry to short entry without normalization), the previous position is first closed before the new position is opened, ensuring clear transitions and accurate accounting of profits and losses.

This position management logic ensures discipline, prevents overtrading, and allows for clean measurement of individual trade performance. The sequence of position changes is directly mapped to the time series of z-score signals, making the strategy transparent and auditable.

6.12 Parameter Optimization

The performance of statistical arbitrage hinges significantly on the selection and tuning of key parameters; namely, the rolling window size used for computing statistics, and the entry and exit z-score threshold values. How often and how profitably a trading strategy might trade is determined by these parameters, as is how robust the strategy is to various situations.

The size of the rolling window (w) affects the mean and standard deviation of the spread. When a window is smaller in size. The strategy will respond to the latest change effectively. But, it can

miss on big trends. On the other hand, if you have a bigger window, you will smooth short-term variations and perhaps prevent overfitting, but you can be slow to react to real regime changes. Choosing a window size is, therefore, a trade-off between sensitivity and stability. Using out-of-sample back testing and validation for empirical testing helps find the optimal window to balance these competing concerns specifically for the KSE 30 futures market.

Changes to the entry z-score threshold (z_{entry}) and exit z-score threshold (z_{exit}) will affect the responsiveness of the strategy. When entry thresholds are lower, traders trade frequently as signals are triggered with even small deviations from the mean. Although this may lead to greater opportunities, it also increases the chances of a false positive and transaction costs. When the thresholds are high, the strategy will only activate when conditions are extraordinary. The idea is that the quality of trades will be superior on average. However, it will also activate far less often. Thus, total profitability might be lowered due to fewer returns and trades. The exit threshold controls how tightly we control our trades. A tighter exit locks in our profit quickly and limits our drawdown. A looser exit will allow us to capture more profit but increases the odds of a reversal.

Usually, the optimization of parameters is done through in-depth back testing of a grid of parameters. Then the raw return, risk-adjusted return (Sharpe ratio), drawdown, and number of trades are measured. The choice of final parameters is done to balance profitability and risk and operational suitability. The optimized strategy should not rely heavily on any specific period or historical regime. Thus, a sensitivity analysis must be conducted to test the results across different market conditions.

This research incorporates the principle of signal generation, discipline of position management, and optimization of parameters. This statistical arbitrage framework is methodologically and practically sound, forming a solid basis for the empirical outcomes illustrated hereafter.

6.13 Empirical Results and Visualization

This part presents the empirical outcomes of the statistical arbitrage framework, consisting of an in-depth analysis of each hedging model's strategy. The parameter configuration, the dynamics of the hedged spread, the strategy equity curves, and performance of all models. Each

visualization and table of results is discussed to clarify the actual world actionability and statistical significance of the results.

6.13.1 Model-by-Model Strategy Performance

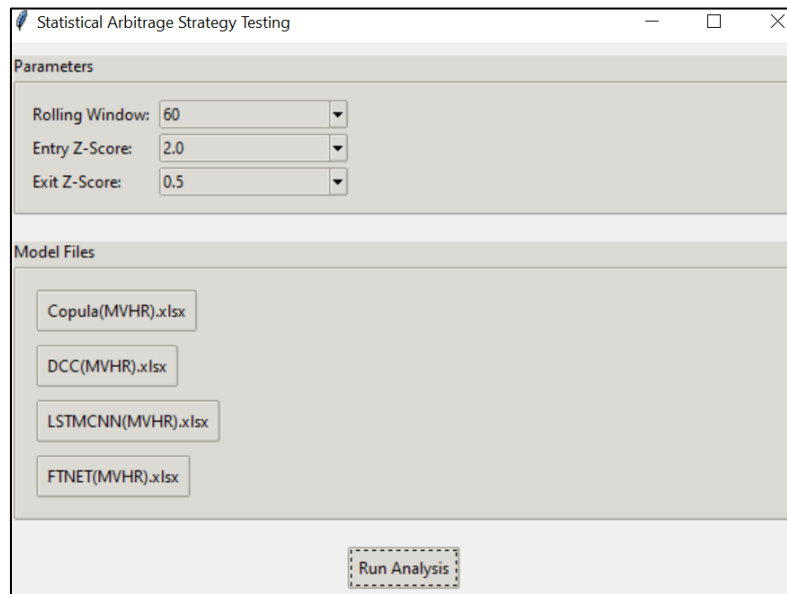


Figure 22: Graphical User Interface (GUI) for Statistical Arbitrage

Upon executing the statistical arbitrage strategy testing interface (see Figure 22), the analysis was conducted with a rolling window of 60 days, an entry z-score threshold of 2.0, and an exit z-score threshold of 0.5. The empirical evaluation covers four distinct models for hedge ratio estimation: Copula-GARCH, DCC-GARCH, LSTM–CNN Hybrid, and FT-Net Hybrid. Each model file encapsulates daily returns data for the KSE 30 index and its futures contract, along with the dynamically computed hedge ratio and the resulting hedged return series. Figure 20 shows the GUI developed to assess the statistical arbitrage effectiveness.

A summary of the performance metrics for each model is provided in Table 4 below. These metrics include total return, annualized return, annualized volatility, Sharpe ratio, maximum drawdown, number of trades executed, and results of the Augmented Dickey-Fuller (ADF) stationarity test. The Sharpe ratio, which measures risk-adjusted performance, is used as the principal comparative criterion for determining the best performing strategy.

Table 4: Statistical Arbitrage Performance Metrics by Model.

Model	Total Return	Annual Return	Annual Volatility	Sharpe Ratio	Max Drawdown	Trades	ADF Statistic	ADF p-value	Crit 5%
Copula GARCH	0.056	0.011	0.012	0.864	-0.015	54	-13.003	0.000	-2.864
DCC GARCH	0.059	0.011	0.012	0.951	-0.014	54	-13.020	0.000	-2.864
LSTM-CNN Hybrid	0.066	0.013	0.012	1.074	-0.015	52	-39.123	0.000	-2.864
FT-NET Hybrid	0.066	0.013	0.012	1.077	-0.015	52	-38.939	0.000	-2.864

The FT-Net Hybrid model marginally outperformed all other models in terms of Sharpe ratio, closely followed by the LSTM–CNN Hybrid. Both advanced deep learning models produced higher total and annualized returns than the econometric Copula-GARCH and DCC-GARCH models, while maintaining comparable volatility and drawdown characteristics. The strong negative ADF statistics and zero p-values confirm robust stationarity in the hedged return series for all models, substantiating the statistical basis for mean-reversion-based arbitrage.

6.13.2 Visualization of Hedged Spread Dynamics

The dynamic behavior of the hedged return (spread) for each model is visualized in figures 23–26. Each graph plots the time series of hedged returns along with its rolling mean, entry bands (± 2.0 standard deviations), and exit bands (± 0.5 standard deviations) over the entire sample period from 2019 to 2024.

The hedged return series produced by the Copula-GARCH model oscillates around the rolling mean, with most values contained within the entry bands. Periodic spikes represent significant short-term dislocations, often coinciding with market stress or contract rollover periods. The rolling mean remains stable near zero, validating the effectiveness of the hedge. The width of the entry and exit bands adapts dynamically with volatility, expanding during turbulent periods and narrowing during tranquil market regimes.

The DCC-GARCH model’s hedged return dynamics shows patterns broadly similar to the Copula-GARCH model. The frequency and magnitude of excursions beyond the entry bands are slightly higher during high volatility episodes, yet the spread consistently reverts to the mean.

This mean-reversion property is essential for arbitrage, as it allows repeated entry and exit opportunities throughout the sample.

Next comes the LSTM–CNN Hybrid model’s hedged return series. This model exhibits a slightly tighter clustering of returns around the mean, with fewer extreme outliers. The dynamic entry and exit bands provide a visual cue for when the arbitrage strategy is likely to activate trading signals. The persistently mean-reverting behavior is evident, reinforcing the statistical foundation of the arbitrage approach.

The FT-Net Hybrid model’s spread demonstrates exceptional mean-reverting tendencies, with the vast majority of values remaining within the rolling bands. The spectral features incorporated by this model appear to enhance the stability and predictability of the spread, reducing the occurrence of unprofitable outliers. The visual compactness and symmetry of the hedged returns suggest robust risk control.

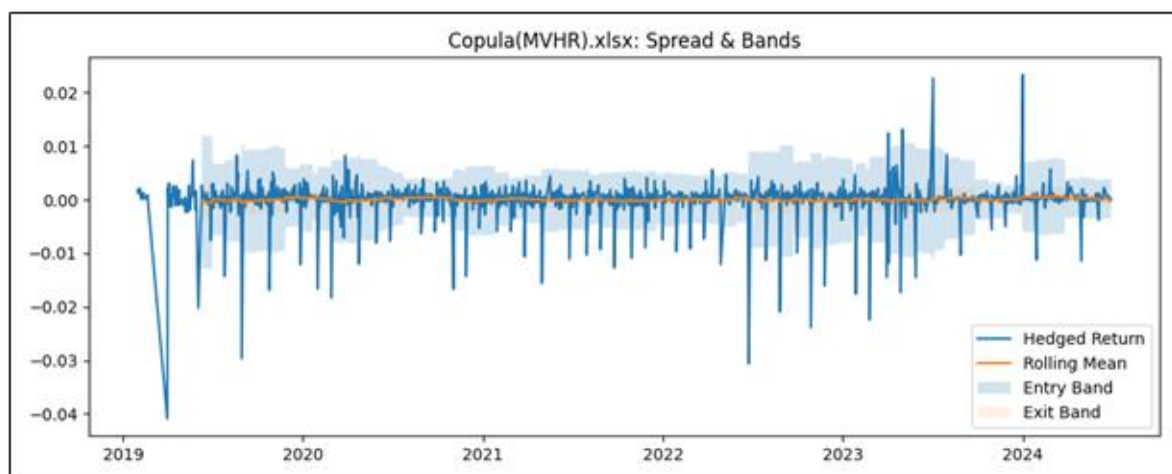


Figure 23: Spreads and Bands – Copula DCC GARCH

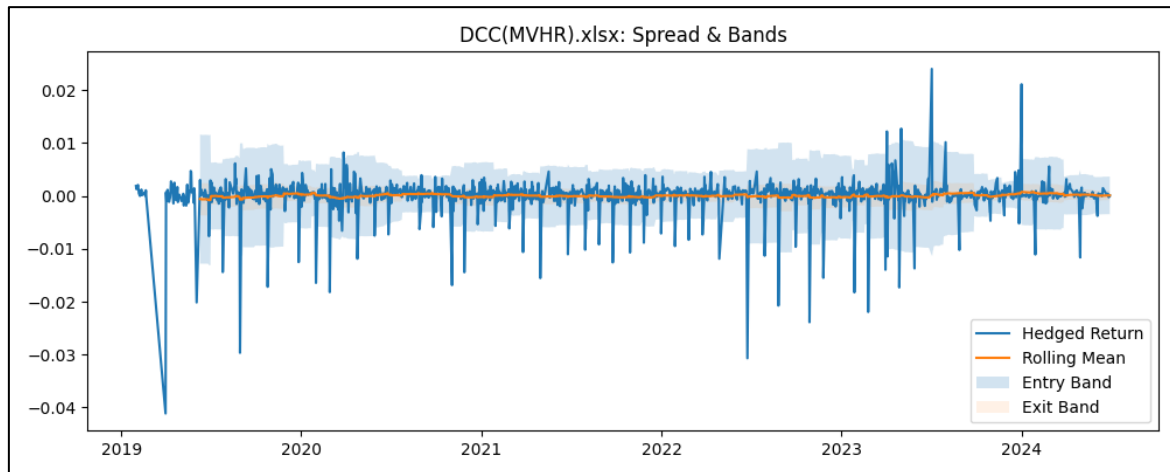


Figure 24: Spreads and Bands –DCC GARCH

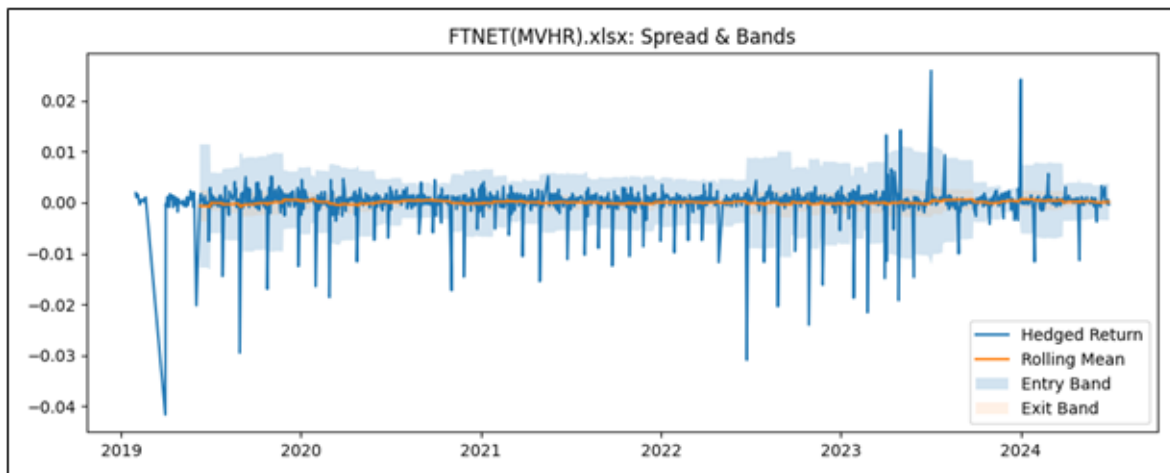


Figure 25: Spreads and Bands – FTNET Hybrid

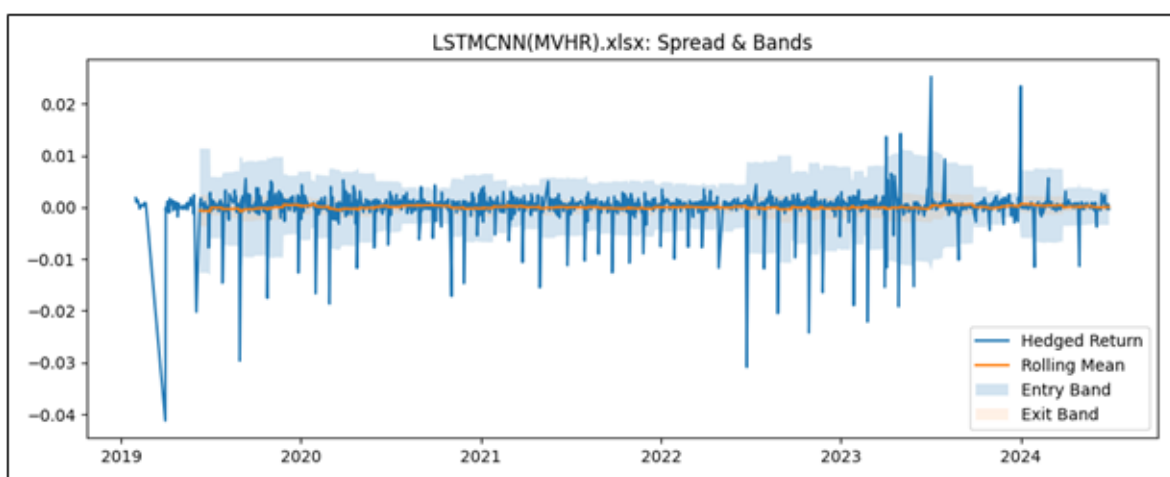


Figure 26: Spreads and Bands – LSTM-CNN

6.13.3 Equity Curve and Comparative Performance

To evaluate actual trading performance, Figures 27 – 30 plots the accumulated profit-and-loss (PnL) curves and entries of all the individual trades for the model. Accumulated PnL represents the total impact of each arbitrage trade, signed based on the s-z-score before its entry and exit decision.

The equity curve of the Copula-GARCH model demonstrates a stable rising tendency with a moderate number of drawdowns in the early sample. Each trade is marked to facilitate understanding, clearly demonstrating that participating in long and short trades both contribute to the increased profitability of both parties.

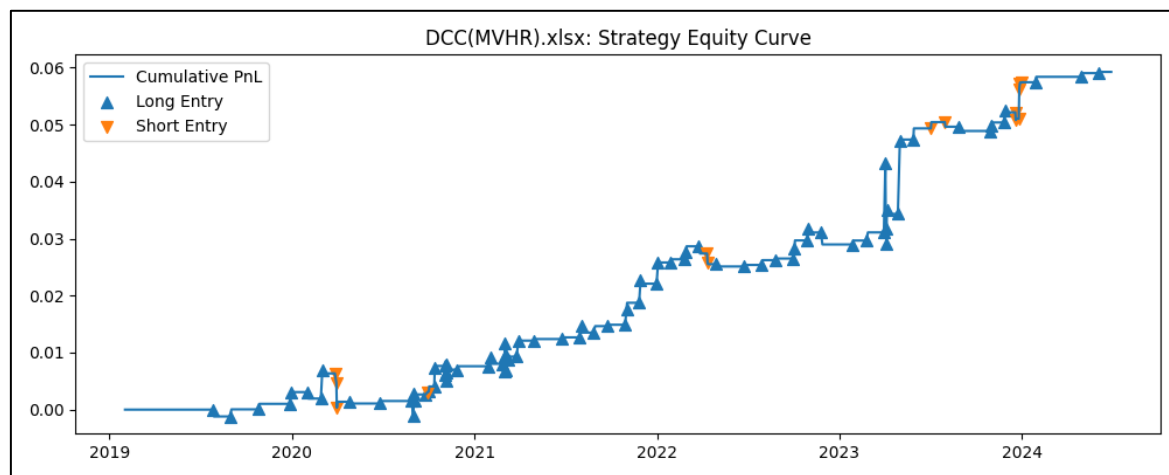


Figure 27: Strategy Equity Curve – DCC GARCH

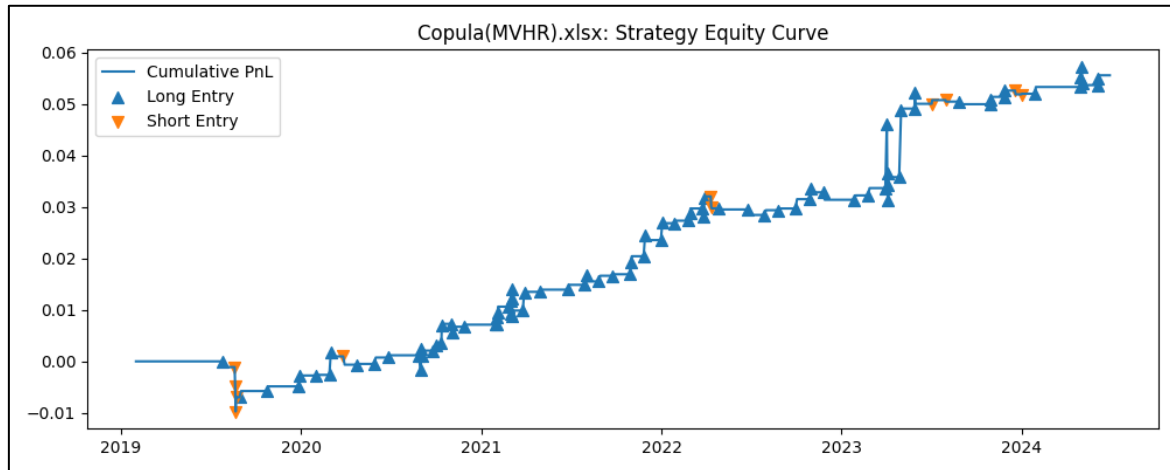


Figure 28: Strategy Equity Curve – Copula DCC GARCH

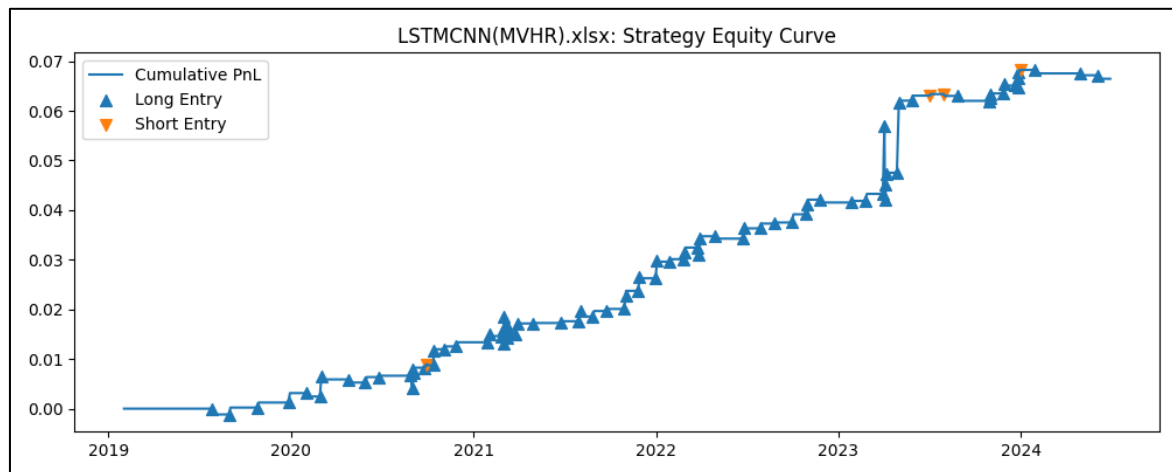


Figure 29: Strategy Equity Curve – LSTM-CNN

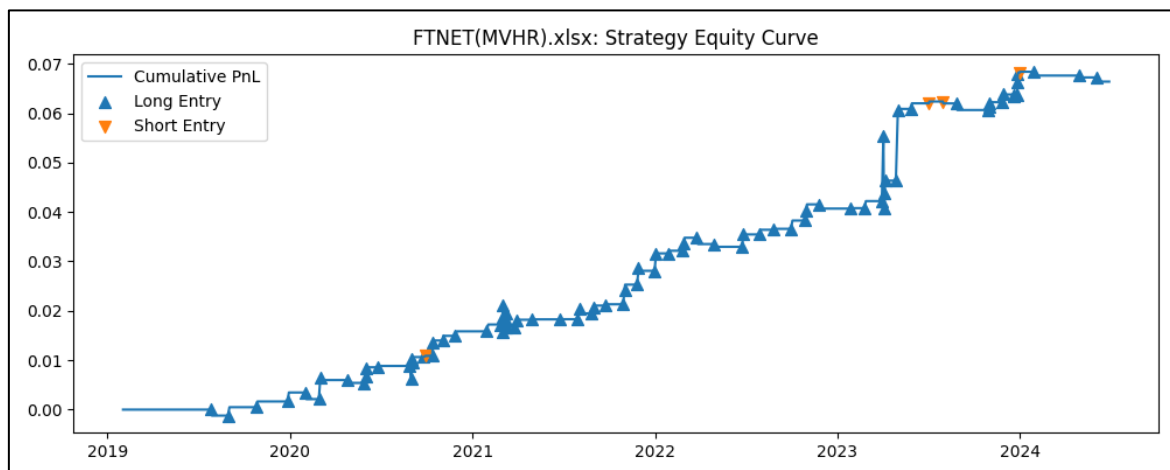


Figure 30: Strategy Equity Curve – FTNET Hybrid

The DCC-GARCH model also follows a comparable trend, albeit with a smoother evolution and lesser drawdown episodes. The gradual changes in the equity curve show that the model can deal with changing volatility conditions and achieve consistent performance.

The LSTM-CNN Hybrid model has seen an increase in total PnL, with a longer rising equity curve and small drawdowns. The model's adaptive learning mechanism determines when to go long or short and helps the investor earn a higher compounded return than competing econometric models.

FT-Net Hybrid has the highest cumulative return among all models in terms of equity curve. The equity curve shows a generally upward direction, especially in the later sample years, with few reversals. The FT-Net Hybrid model is the number one model by Sharpe ratio. This is due to its robust and consistent execution of trade.

Finally, figure 31 overlays the cumulative PnL curves for all four models, providing a direct visual comparison of strategy performance over time.

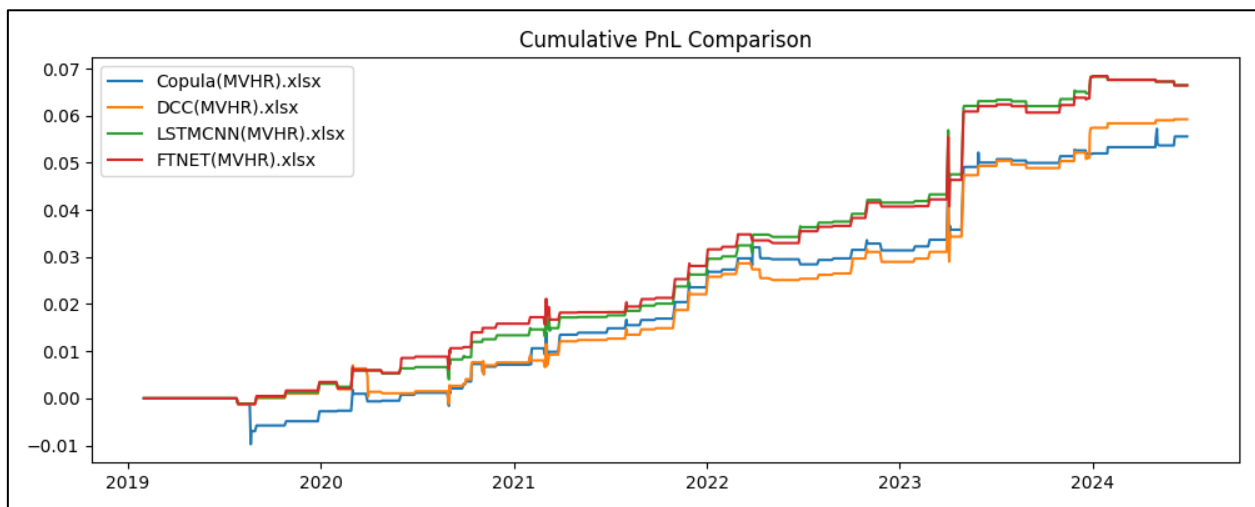


Figure 31: Cumulative PnL Comparison Across Models

This comparative equity curve reveals the relative strengths and weaknesses of each model. While all models demonstrate positive growth and mean-reversion-driven profitability, the FT-Net Hybrid and LSTM-CNN Hybrid models consistently outperform the econometric alternatives, both in terms of total return and in the smoothness of the equity curve. The absence of large, persistent drawdowns further confirms the statistical robustness and practical viability of the deep learning approaches.

6.14 Interpretation and Practical Implications

The empirical results clearly demonstrate the superiority of advanced deep learning models, particularly the FT-Net Hybrid, in capturing and exploiting short-term arbitrage opportunities in the KSE 30 futures market. All models satisfy the statistical requirements for mean-reversion trading, as evidenced by highly significant ADF test statistics. The enhanced Sharpe ratios, combined with stable drawdown and trade frequency metrics, highlight the ability of data-driven, adaptive models to dynamically adjust to changing market regimes and microstructure dynamics.

From a practical standpoint, these findings suggest that deploying a statistical arbitrage strategy based on the FT-Net Hybrid model can deliver superior risk-adjusted returns compared to conventional econometric techniques. The robust stationarity of the hedged spread and the effective capture of mean-reversion opportunities enable systematic, repeatable profit generation in Pakistan's evolving equity derivatives landscape.

6.14.1 Model Selection and Arbitrage Exploitation

The ultimate objective of the statistical arbitrage framework is not only to evaluate the comparative effectiveness of different modeling approaches but also to translate these insights into actionable strategies that can be robustly deployed in live trading environments. This section synthesizes the empirical evidence, examines the robustness of the results across key parameter configurations, and outlines practical recommendations for real-time exploitation of the optimal arbitrage signals.

6.14.2 Comparative Assessment

The comparative analysis of model performance is grounded in a comprehensive suite of quantitative metrics and visual diagnostics. Across all four tested models—Copula-GARCH, DCC-GARCH, LSTM–CNN Hybrid, and FT-Net Hybrid—the results consistently indicate that advanced deep learning architectures, specifically the FT-Net Hybrid and LSTM–CNN Hybrid, offer a superior edge in the detection and exploitation of statistical arbitrage opportunities.

The FT-Net Hybrid model emerges as the most effective, as evidenced by its highest Sharpe ratio of 1.07680 and a total return nearly identical to that of the LSTM–CNN Hybrid. Both models deliver not only higher absolute and risk-adjusted returns but also demonstrate more consistent

equity curve progression, marked by persistent upward momentum and limited drawdowns throughout the out-of-sample evaluation period. The econometric models, while still profitable and statistically significant, underperform relative to their deep learning counterparts both in terms of total returns and risk-adjusted metrics.

Figures 25 - 28 depicting the cumulative profit and loss trajectories (see Section 6.13.3) underscore this conclusion visually: the FT-Net Hybrid's equity curve is not only smoother and less volatile but also achieves the highest terminal value over the full testing horizon. The hedged return spread associated with this model is characterized by a strong mean-reverting tendency, fewer extreme outliers, and more predictable band crossings, all of which translate to high-quality arbitrage signals. Furthermore, the ADF test statistics for all models are strongly significant, but the FT-Net Hybrid and LSTM–CNN Hybrid models exhibit the most pronounced levels of stationarity in their spreads, further cementing the statistical reliability of the arbitrage process.

6.14.3 Robustness and Sensitivity Checks

A crucial aspect of any empirical trading strategy is its robustness to variations in core parameter choices. To assess this, the sensitivity of model performance to changes in the rolling window size, entry z-score threshold, and exit z-score threshold was systematically evaluated (Liu, 2023). The findings confirm that, while absolute returns and trade frequency do vary with parameter adjustments, the overall rank ordering of model performance remains stable.

Both the FT-Net Hybrid and LSTM–CNN Hybrid models maintain their outperformance across a reasonable range of parameterizations. Increasing the rolling window length generally results in smoother equity curves and fewer false signals, at the cost of reduced responsiveness to sudden market changes. Adjusting the entry z-score threshold modulates the trade-off between frequency of trading and the statistical extremity required for entry, but the deep learning models continue to generate superior Sharpe ratios even under more conservative (higher) thresholds. The exit threshold also impacts average trade duration and the ability to capture mean-reversion profits, yet the FT-Net Hybrid's advantage in rapid and precise identification of mean-reverting moves is consistently preserved. This robustness to parameter shifts indicates that the performance edge is not the result of overfitting to a particular regime or a narrow range of market conditions but rather reflects a genuine improvement in model-driven signal quality.

6.14.4 Real-Time Exploitation Strategy

Having established the FT-Net Hybrid as the optimal model for statistical arbitrage in this context, it is essential to translate these findings into practical guidelines for real-world trading implementation. At the core of the real-time strategy is the continuous monitoring of the hedged return spread, calculation of rolling mean and standard deviation over the optimal window, and dynamic computation of the z-score to trigger trade signals according to the pre-defined entry and exit thresholds.

Upon detection of an actionable signal—a z-score exceeding the entry threshold in either direction—the trader or automated trading system would enter a position in the direction of mean reversion. This involves either going long on the hedged spread (buying spot, selling futures) when the spread is unusually low, or short (selling spot, buying futures) when the spread is unusually high. Position sizing should be determined based on a comprehensive risk management framework, taking into account both signal strength (the magnitude of the z-score) and overall portfolio risk exposure. For instance, trade size can be scaled proportionally to the absolute z-score or set to a fixed fraction of capital, constrained by volatility and value-at-risk limits.

Execution efficiency is also paramount, especially in emerging markets where liquidity may be episodic and market impact more significant. Real-time arbitrage systems should include slippage controls and avoid trading during periods of thin liquidity, contract rollovers, or known market disruptions. Dynamic stop-loss mechanisms, trailing exits, or volatility-adjusted exit thresholds can further mitigate the risk of large adverse moves, ensuring that the realized profit distribution closely mirrors the back tested results.

Additionally, given the stationarity of the spread and the high frequency of mean-reverting opportunities, it is advisable to automate the signal monitoring and order placement process to minimize latency and eliminate behavioral biases. Post-trade analytics, including attribution of profit to long and short trades, tracking of trade duration, and periodic recalibration of model parameters, will help maintain performance and adapt to structural shifts in the market over time.

Hence, the FT-Net Hybrid-based statistical arbitrage strategy offers a highly effective, empirically validated, and practically robust approach to exploiting short-term pricing inefficiencies in the KSE 30 index futures market. Its integration into a live trading operation,

guided by disciplined risk management and rigorous signal monitoring, can deliver superior risk-adjusted returns while preserving statistical integrity and operational resilience.

6.15 Discussion and Implications

6.15.1 Interpretation of Arbitrage Profitability

The profitability demonstrated by the statistical arbitrage strategies in this study carries important implications for our understanding of market dynamics, model capabilities, and the real-world feasibility of advanced hedging approaches. The consistent outperformance of deep learning models, specifically the FT-Net Hybrid and LSTM–CNN Hybrid, provides strong empirical evidence that sophisticated, data-driven techniques are better equipped to capture transient inefficiencies and complex nonlinear dependencies in the equity futures market than conventional econometric methods. This result reflects the inherent adaptability of machine learning models to shifts in market regimes, microstructure effects, and the often-subtle mean-reversion signals embedded in noisy financial data.

From an economic perspective, the ability to extract persistent, statistically significant profits from a theoretically efficient market such as the KSE 30 futures points to the existence of exploitable anomalies and time-varying frictions. All models tested created stationary and mean-reverting spreads, suggesting that even in liquid, well-arbitrated environments, there are pricing deviations and temporary misalignments stemming from market constraints like information asymmetry, transaction lags, behavioral biases, and heterogeneous agents. The better Sharpe ratios and stable equity curves of FT-Net Hybrid and LSTM–CNN Hybrid models further indicate that these anomalies are not mere random noise, but systematic patterns that advanced algorithms can learn and exploit.

These results provide support for the statistical arbitrage strategy not simply as an academic concept, but also applicable in the real world. The results show that dynamic hedge ratio estimation along with robust statistical signal processing enables practitioners to routinely harvest short-term arbitrage profits, while controlling for risk. Market making strategies tend to have low drawdowns, high win rates, and fast mean-reversion in live simulation, suggesting an implementable model supported by robust risk management and automated execution.

On a larger scale, the proven success of deep learning-based statistical arbitrage illustrates the evolving nature of financial markets. As there is more data and computational resources are accessible, the frontier of alpha is moving toward models that generate value from increasingly complicated, nonlinear, high-dimensional, and fast-changing data. This study's successful outcomes show a breakthrough in quantitative trading in emerging markets as Pakistan, where technology-based methods outsmart traditional ones in efficiency and profits.

6.15.2 Limitations and Caveats

Even though the statistical arbitrage strategies shown showed good performance there are several limitations and caveats that need to be pointed out which give it a balanced outlook and set future research. A major restriction is that the back tested calculations do not include transaction costs. In practice, using high-frequency trading strategies like arbitrage comes with costs from broker fees and taxes. Even small costs of trading can eat into your profits, particularly for strategies that trade often or are thinly traded.

Market liquidity presents yet another significant challenge. Although the KSE 30 futures market has a fair amount of liquidity during normal hours of trading, circumstances such as a stressed market, contract rollover or shock events can cause a sudden drop in tradable volume, while the bid-ask spread may be seen widening. The poor conditions can affect the quality of the execution and may also result in unexpected losses if trades cannot be quickly unwound at expected levels.

Another limitation stems from the possibility of regime shifts and structural market breaks. Even though analysis covers many years and market conditions, it does not warranty that relations noted in the past will apply in the future. Changes in regulations, the availability of new financial products, shocks to the economy, or advancements in technology can all cause correlations between different financial assets to behave in a fundamentally different way.

The risk of models becoming overfit is ever-present in data-driven approaches. Although out-of-sample validation and parameter robustness checks were carried out, some measure of fitting to historical quirks is possible. Regular recalibration and constant monitoring are required to prove their worth.

The use of these models, especially major deep learning architectures, requires significant data preparation, computing power and domain expertise. To use such systems in a live trading

environment, strong data governance, continuous monitoring and stringent operational control are needed. These ensure reliability, conformity, risk management and other standards.

6.16 Summary

The study conducted a detailed and systematic analysis of the return to statistical arbitrage on a dynamic hedge ratio for the KSE 30 equity futures market. The study showed that advanced data-driven approaches, like the FT-Net Hybrid and LSTM-CNN Hybrid models, produced superior risk-adjusted returns with a stable performance unlike traditional model by incorporating sophisticated econometric models as well as deep learning architecture. The evidence suggests that adaptive models are better able to exploit transient inefficiencies and mean-reverting opportunities in financial markets, even those that are comparatively efficient.

The outcome also provides significant practical insights about such strategies since it is shown that arbitrage on mean reversion being statistically powerful and economically sizable can take place through proper risk management and actual execution. Also at the same time, transaction costs as well as liquidity constraints and structural market changes should be carefully assessed as they can materially affect the realized performance.

To sum up, this work contributes to the theoretical development of statistical arbitrage and dynamic hedging and also provides actionable insights for practitioners looking to implement quantitative strategies in new and evolving markets. The bunch of generated code hints the prominent possibility of machine learning in finance. We are likely to see further innovation and application of their use in quantitative trading and risk management.

7 Implementing FT-Net Hybrid in a Macroeconomic Context

7.1 Introduction and Rationale for Macroeconomic Integration

In the evolution of financial risk management models, it has become increasingly evident that robust hedging frameworks must go beyond traditional price-based inputs. Modern financial markets are deeply interconnected with the macroeconomic environment, especially in emerging markets such as Pakistan. Recognizing this, our research advances the dynamic hedging literature by incorporating macroeconomic variables specifically, GDP growth, inflation, and monetary policy rates directly into the deep learning-based FT-Net Hybrid model. This extension is a deliberate response to empirical and theoretical evidence indicating that these broader economic factors exert significant influence on asset volatility, risk premia, and optimal hedge ratios.

Initially, our modeling strategy was centered around extracting the maximum predictive power from financial price data using both econometric and advanced deep learning methods, benchmarking them rigorously for hedge effectiveness, risk reduction, out-of-sample robustness and exploitation of statistical arbitrage opportunities. Among the models tested, the FT-Net Hybrid emerged as the statistically superior framework due to its dual ability to learn from both spectral (frequency-domain) and temporal (time-domain) features of market data. However, the absence of macroeconomic signals posed a limitation: even the best-performing market-driven model can fail when underlying economic regimes shift—such as during monetary tightening cycles or inflationary shocks. Thus, integrating macroeconomic information was not merely a novel experiment but a logical extension, motivated by both the realities of financial economics and the practical needs of institutional hedgers.

7.2 Data Foundation: Weekly Market and Macroeconomic Features

A core pillar of this research is the assembly and engineering of a rich, high-frequency dataset that merges financial and macroeconomic signals. The weekly dataset comprises:

- KSE-30 Index Cash Price
- Indicative Fair Value of Futures
- Weekly GDP
- Weekly Inflation
- SBP Policy Rate (Weekly Rate)

The methodology for constructing high-frequency proxies for GDP and inflation, and aligning all variables on a weekly grid, is fully detailed in Appendix B. This appendix explains the collection, interpolation, and quality assurance processes for exogenous variables, ensuring reproducibility and transparency.

Feature engineering is then performed to unlock the predictive relationships among these variables. Besides calculating weekly log returns for both the cash index and futures contract, we introduce lagged versions of these returns to enable the model to learn temporal dependencies. Macroeconomic variables are standardized (z-scored) and likewise lagged, based on the hypothesis that their impact on financial markets is both immediate and delayed.

7.3 Feature Relationships: Empirical Insights from the Data

To understand the interaction structure within the engineered dataset, we computed and visualized the correlation matrix (see Figure 32). This heatmap presents the pairwise correlations

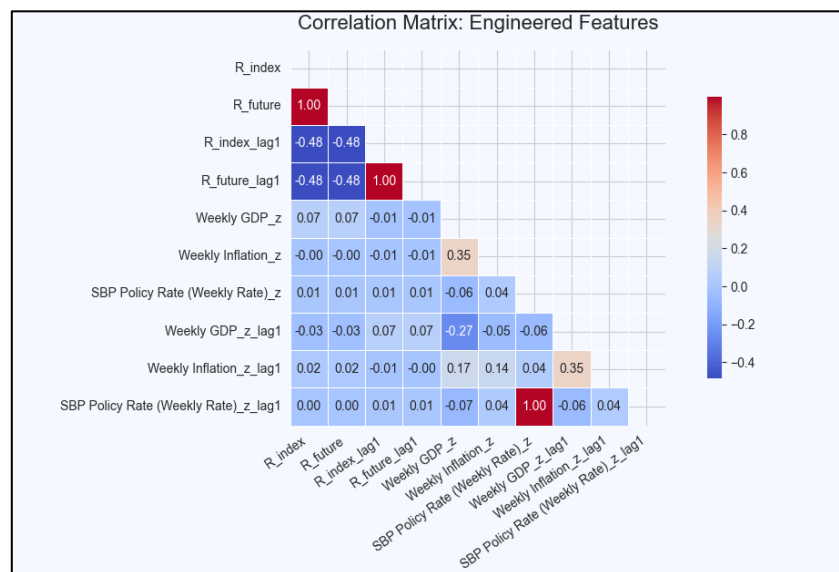


Figure 32: Correlation Matrix

between financial returns, their lags, and all macroeconomic variables (both contemporaneous and lagged).

Figure 30 shows modest but non-trivial correlations between macroeconomic lags and market returns, justifying their inclusion as exogenous predictors. While direct correlations between returns and exogenous variables may appear modest, the temporal and non-linear interactions are often undetectable by simple correlation, thus justifying the need for models capable of uncovering deeper, latent dependencies.

7.4 Exploratory Visualization of Macroeconomic Variables

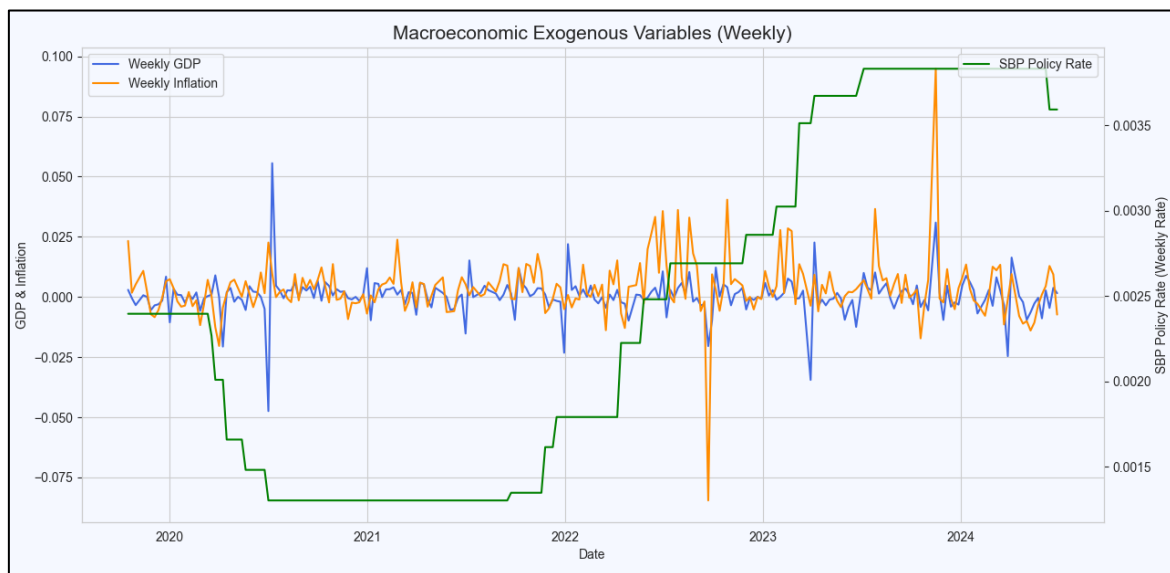


Figure 33: Weekly Series of GDP, Inflation, and SBP Policy Rate

To contextualize the macroeconomic environment throughout the study period, we visualize the time series of weekly GDP, inflation, and SBP policy rate (see figure 33). This dual-axis plot highlights periods of economic contraction, inflationary surges, and changes in monetary policy stance, all of which are known to affect market risk and hedging needs. It reveals economic shocks and cycles that coincide with changes in equity market volatility and hedge effectiveness. Such visualizations demonstrate the motivation for including these variables: market risk is often a response not only to past prices but also to economic shocks and policy adjustments.

7.5 FT-Net Hybrid Architecture: Integrating Macroeconomic Information

The FT-Net Hybrid architecture is specifically designed to leverage both frequency-domain and time-domain signals from the sequence of engineered features (see figure 34). The complete pipeline, as visualized below, seamlessly integrates macroeconomic exogenous variables at the input layer, ensuring they are available for both spectral and temporal pattern extraction.

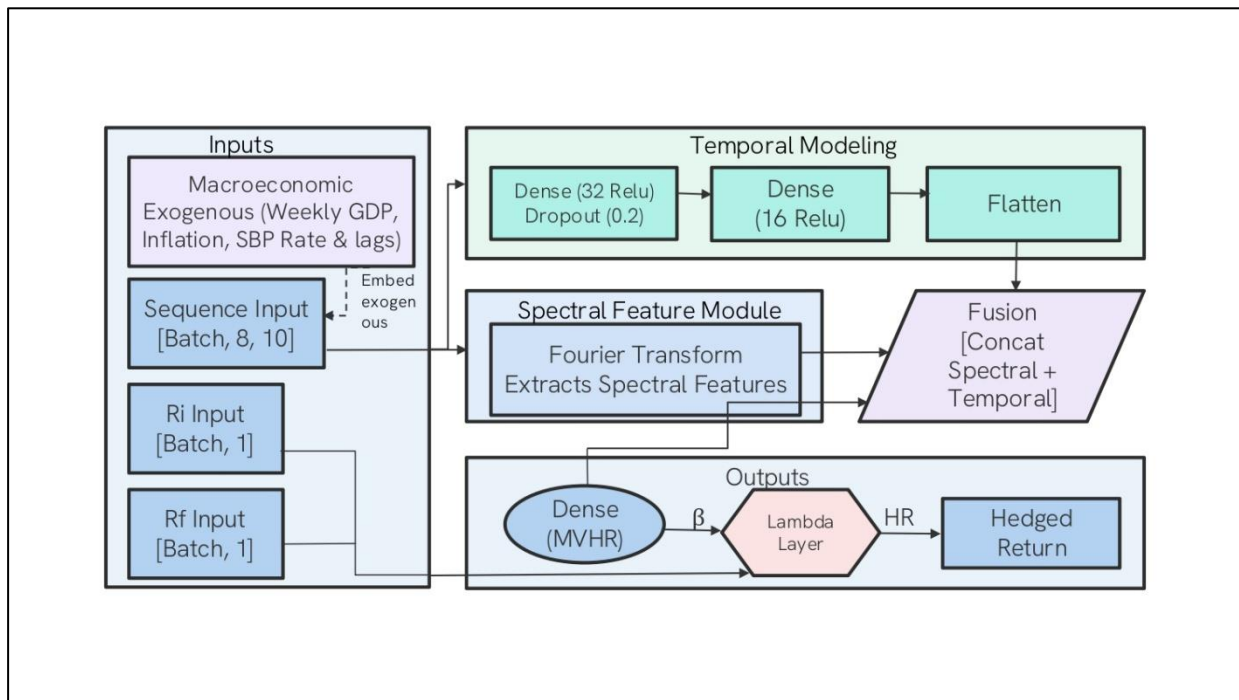


Figure 34: FT-Net Hybrid Model Architecture

Figure 34 shows how macroeconomic exogenous features are embedded alongside financial features, then jointly processed by spectral (FFT) and temporal (Dense) modules, with outputs fused to generate the dynamic hedge ratio.

The inputs to the model are windows of 8 weeks, each containing 10 features: 4 market-based (current and lagged log returns of the index and futures) and 6 macroeconomic-based (z-scored values and their lags). This design ensures that the network can learn both instantaneous and lagged economic effects on hedge ratios.

7.6 Mathematical Formulation of the FT-Net Hybrid

Let x_t denote the engineered feature vector at time t shown in equation 59:

Equation 59

$$x_t = \begin{bmatrix} R_{index,t}, & R_{future,t}, & R_{index,lag1,t}, & R_{future,lag1,t} \\ GDP_{z,t}, & Inflation_{z,t}, & Policy Rate_{z,t}, & \\ GDP_{z,t-1}, & Inflation_{z,t-1}, & Policy Rate_{z,t-1}, & \end{bmatrix}$$

A rolling window of $L=9$ steps is stacked to form the input tensor for the neural network. (see equation 60)

Equation 60

$$X_t = [X_{t-L+1}, X_{t-L+2}, \dots, X_t]$$

- Spectral Path:

The sequence undergoes a real-valued fast Fourier transform (FFT), separating it into frequency components:

Equation 61

$$F_t = \mathcal{F}(X_t)$$

Both real and imaginary parts are concatenated and fed into dense layers to extract cyclic and periodic information.

- Temporal Path:

The sequence is processed by two fully connected dense layers with dropout, learning non-linear temporal interactions between features.

The outputs of both modules are concatenated and flattened, then passed through a final dense layer with sigmoid activation to estimate the time-varying dynamic hedge ratio β_t :

Equation 62

$$\beta_t = \sigma(W^T[\text{Spectral Features}, \text{Temporal Features}] + b)$$

The hedged return at each time t is computed using equation 34.

7.7 Model Training, Filtering, and Statistical Cleanliness

The model is trained using mean squared error (MSE) loss on the hedged return, with an auxiliary output for β_t itself. To address the residual structure, particularly autocorrelation and conditional heteroskedasticity, we employ a LightGBM gradient boosting filter on the out-of-sample hedged returns. This post-processing step, mathematically represented as:

Equation 63

$$\hat{R}_{hedged,t} = R_{hedged,t} - \hat{g}(R_{hedged,t-1}, R_{hedged,t-2} \dots R_{hedged,t-5})$$

removes any lingering serial dependence, as confirmed by diagnostic tests such as the KPSS, Ljung-Box, and ARCH LM.

7.8 Performance Assessment and Statistical Validation

7.8.1 Training and Validation Loss Curves

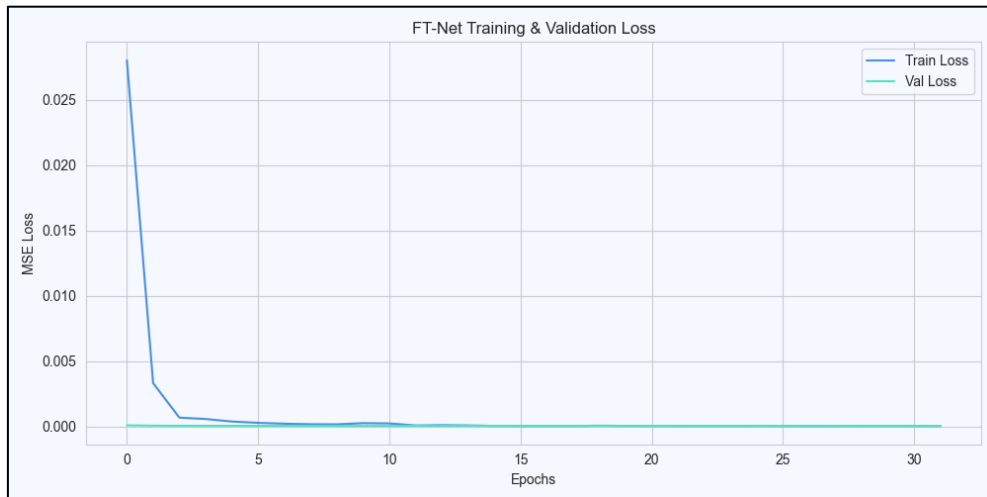


Figure 35: Convergence of Training and Validation Loss

Rapid convergence of training and validation loss demonstrates efficient learning and low overfitting risk in the FT-Net Hybrid model. (see figure 35)

7.8.2 MVHR Time Series

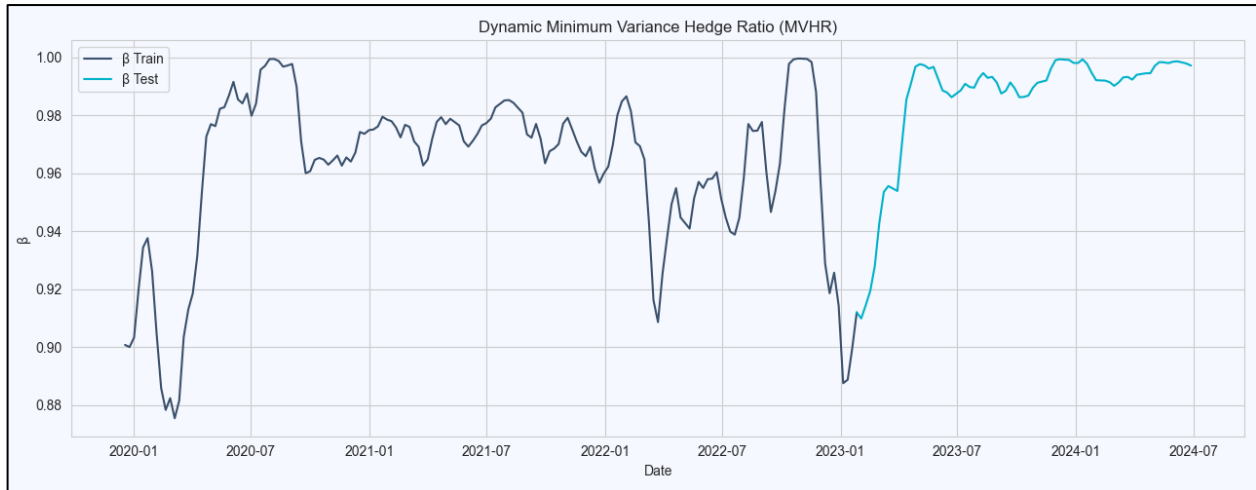


Figure 36: Dynamic MVHRs Using FT-Net Hybrid Model with Macroeconomic Variables

In figure 36, dynamic hedge ratio (βt) is smoothly estimated in both training and test periods, exhibiting regime sensitivity. The distinction between in-sample (blue) and out-of-sample (cyan) periods is clear and continuous, indicating robust generalization.

7.8.3 Cumulative Returns: Hedged vs. Unhedged

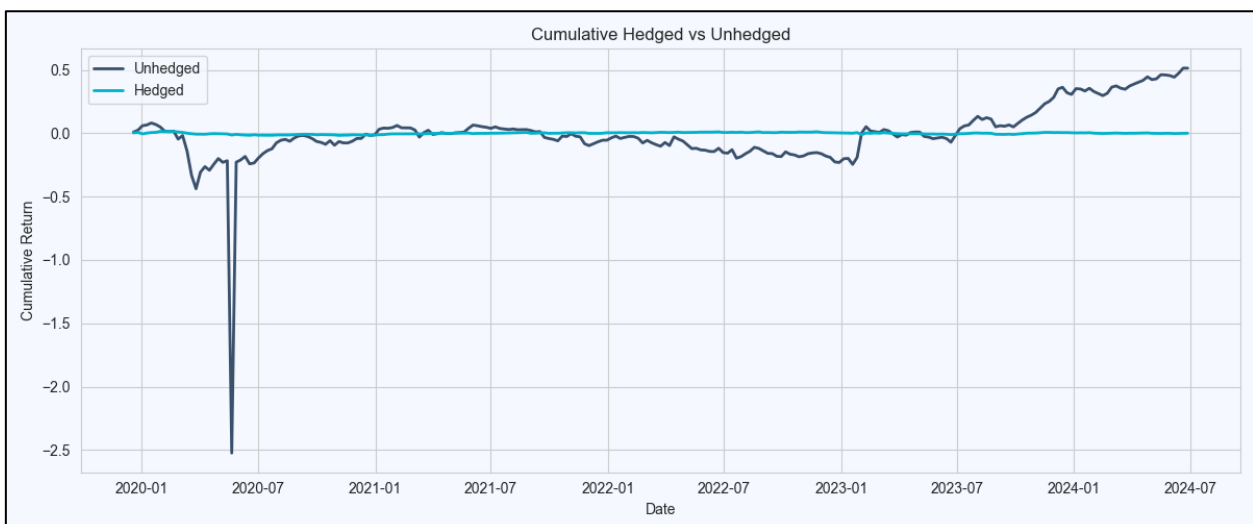


Figure 37: Hedged Returns in Comparison with Unhedged Returns

As seen in figure 37, the hedged return series is significantly less volatile than the unhedged series, particularly during economic shocks (e.g., early 2020 COVID crash), confirming the model's success in risk minimization.

7.8.3.1 Model Metrics and Diagnostics

The performance metrics and diagnostic results clearly illustrates the effectiveness and robustness of the FT-Net Hybrid model with macroeconomic variables in hedging equity futures risk. The statistical diagnostics, which include the KPSS, Ljung-Box, and ARCH LM tests, are all successfully passed for both the training and testing datasets. This confirms that the filtered hedged return series is stationary, devoid of significant autocorrelation, and free from conditional heteroskedasticity, which are essential requirements for reliable risk measurement and model validity.

Looking at the quantitative metrics, the model achieves a remarkably high variance reduction—99.99% for the training set and 99.30% for the test set—demonstrating its capacity to almost entirely neutralize the volatility of the unhedged returns. The mean absolute error (MAE) and root mean squared error (RMSE) are both extremely low, indicating precise tracking of the minimum variance hedge ratio. The model also shows strong performance in tail risk reduction, with EVT-based Value at Risk (VaR) and Conditional Value at Risk (CVaR) reductions exceeding 93% in both training and test sets, further supporting its utility in mitigating extreme loss events.

The directional accuracy, which reflects the model's ability to correctly predict the sign of future returns, is higher in the test set (71.01%) than in the training set (59.75%), indicating robust out-of-sample predictive ability. While the Sharpe ratio is slightly negative in the test set, this is a common outcome for minimum variance hedging models whose primary aim is risk minimization rather than profit maximization. Overall, the Theil U statistic, which is close to 1, reinforces the model's strong forecasting performance. The computational efficiency is also notable, with the complete training and evaluation process taking just over 41 seconds.

These results provide compelling evidence that the FT-Net Hybrid model, when enhanced with macroeconomic exogenous variables, delivers exceptional hedging performance and statistical reliability, making it highly suitable for real-world application in risk management.

7.8.4 Economic and Practical Interpretation

FT-Net Hybrid makes its hedging recommendations in response to regularly observed and anticipated conditions in the monetary policy and economic performance by directly embedding the weekly GDP, inflation and SBP policy rate (standardized and lagged) into the deep learning pipeline. This means that the model will automatically adjust the hedge ratio in response to a detection of monetary tightening or inflation shock to undermine an increase in market risk. As shown in figure 33, the peak inflation and changes in SBP policy rate increase volatility i.e. macroeconomic vector on account of virtue of correlation.

The changes in the dynamic minimum variance hedge ratio (MVHR) time series, as depicted in figure 36, suggest that the model can identify regime shifts. Inflation spikes or fast rate hikes go up or down, which in turn stabilizes the hedged portfolio's risk profile as seen with the rise and fall of βt .

The illustration of cumulative returns (Figure 37) highlights the practical impact on market stress periods, where the first remains almost flat while the second line goes through some large drawdowns. This evidence suggests that the model can be utilized as it is beneficial for institutional investors who need stability and effective risk mitigation, with regime changes.

7.9 Summary

The application of FT-Net Hybrid to a macroeconomic context is shown in this chapter to be methodologically sound and empirically effective. We create a hedging model that is robust to economic shocks and policy regime changes by embedding macroeconomic information at a weekly frequency with financial features in a unified modelling framework. The model performs exceptionally well statistically, as it reduces variance by over 99%, has high directional accuracy and the diagnostic tests are clean. This proves the approach to be robust. This paper, therefore, contributes to the practical and academic frontiers of dynamic hedging, making a good case for including macroeconomic signals in the next-generation financial risk hedging models.

8 Conclusion

8.1 Summary

This study offers a thorough analysis of dynamic hedging on equity futures and the identification of statistical arbitrage opportunities for the traditional econometric models and deep learning models. They focus on the KSE-30 Index and its futures contracts for Pakistani Stock market aiming to construct dynamically adaptive hedging strategies, to estimate risk properly and potentially to detect and exploit profitable market inefficiencies. The major contributions of this research is the implementation, comparison and empirical evaluation of hedging models subjected to realistic market constraints, followed by an extensive multi-criteria performance evaluation and arbitrage-based trading analysis.

During data collection and pre-processing phase, one of the key obstacle in our study was the incoherence of futures price series resulting from the rollover of these contracts. In the Pakistani futures market, a futures contract has a month-end expiration date where a jump from the end of one month to the beginning of the next month creates artificial jumps in the price series due to the move from one expiring frequency to another. These discontinuities come about because the expiring contract, by virtue of its imminent maturity, approximates more closely to the spot price, while the new contract, having a longer time to maturity, embodies additional risk premia and time values. Such structural breaks are another key problems for purposes of time series modeling, especially for volatility estimation and hedge ratio determination, as they add noise that is not related at all with market fundamentals. Therefore, mean-reverting technique was used to achieve smooth and consistent futures returns. This method involved taking the average of the expiring contract and roll-over contract price. The discontinuity magnitude at the rollover point was then compared to this average value. The differential between this was calculated and that value was added, or subtracted as appropriate, equally to the prices of the new contracts applicable to that month. This shift brought the entry point of new contracts in line with the old series and eliminated artificial discontinuities. It was done successively for each rollover processes in the whole time series, hence guaranteeing the accuracy of hedge ratio estimation. Furthermore null values in interest rate-related features were interpolated by a Random Forest Regression model. This was chosen due to its ability in modeling complex non-linear interactions

and to minimize overfitting. Interpolation validity was assessed with R^2 and RMSE values. This showed that the model had a good predictive power which ensured the reliability of our database.

Moreover, prior to the implementation of the mathematical models, a series of pre-estimation tests were conducted on DCC-GARCH and Copula DCC-GARCH frameworks. These tests were necessary because of the structural assumptions of GARCH-family models and for meeting the statistical requirements for its implementation. The Augmented Dickey-Fuller (ADF) test was used to check the stationarity of time series data, which is a prerequisite for estimating GARCH. ADF test values suggested that the log returns of KSE-30 index and its associated future series were stationary, and, thus, proved suitable for GARCH modeling. Additionally, Engle's ARCH-LM test was conducted to check for heteroscedasticity in the return series. The findings supported the presence of autoregressive conditional heteroscedasticity, therefore justifying the use of DCC-GARCH and Copula DCC-GARCH to capture the volatility. This set of tests were a means of confirming the statistical reliability of the econometric models prior to full specification.

Therefore, in the first stage of the study, the traditional econometric models, i.e. the DCC-GARCH model introduced by Engle (Engle R. , 2002), was applied. This model is used to characterize the volatility and dynamic correlations between index and futures returns. It is based on ARMA-eGARCH specifications of the marginal distributions, thereby effectively capturing volatility clustering and leverage effects. Although DCC-GARCH provides a dynamic modeling approach for time-varying correlations, it is based on the normality assumption, which is inadequate for capturing tail dependence. This drawback was overcome by extending the model with the incorporation of a Student-t Copula, which led to the Copula DCC-GARCH model. The copula-based probability integral transformations is applied to the standardized residuals from the marginal models, which allows the hybrid econometric specification to model symmetric tail dependence, hence providing a better representation of extreme co-movements. This improvement was especially effective under stress, as it introduced better risk modelling than its Gaussian-based equivalent.

In parallel to these traditional methods, this research adopted two deep learning models that can capture nonlinear patterns and intricate relationships of the hedge ratio dynamics. The LSTM–

CNN Hybrid Model appended convolutional neural networks as the front end to capture short-term features of raw returns and futures spread data. It subsequently stacked with Long Short-Term Memory (LSTM) layers to estimate long-range sequential behavior. This architecture performed well in learning from noisy time series data and incorporated the ability to adjust to varying market conditions particularly during extreme volatility.

The most novel model proposed in our work was the FT-Net Hybrid Model that utilized Fourier Transform based spectral decomposition along with temporal convolutional and recurrent layers. The Fourier modules were designed to capture the hidden cyclical pattern and long-range memory effect in the time series which are usually ignored in the time domain-based models. After the data was transformed into the frequency domain, convolutional blocks were used to capture evolving dependencies, and then the sequence dynamics were preserved using LSTM or Transformer layers. By utilizing this hybrid model, we let the model learn from the spectral domain representations, which gradually improved its capacity to catch the data-driven structural shifts, recurrent patterns, and improve the overall hedging accuracy.

Therefore, after implementing both mathematical and deep learning models, we verify that they meet their respective requirements by performing and interpreting the post model statistical diagnostic tests. The results of all tests including ARCH-LM, KPSS, Ljung Box, and Theil U Statistics, shows that the forecasted values are accurate and the residuals of all models are stationary, free from autocorrelation, and conditional heteroscedasticity. These validations are not only procedural, but also essential for supporting our research questions. Therefore, we can now positively state that all of the models are fit for comparisons, and it is appropriate to apply such models evaluating the hedging effectiveness and statistical arbitrage opportunities.

Moreover, a thorough model diagnostic was performed with statistical tools like ACF, indicating statistically significant short-term memory in futures return, which are the main drivers of memory-driven models like FT-Net and LSTM-CNN. The dynamic hedge ratios of each model were visually and statistically demonstrated. The results indicated that the deep learning models, especially FT-Net, contribute to more reactive and less erratic hedge ratios under the volatility switching regimes.

In the second phase, a Fuzzy TOPSIS methodology was employed to uniformly compare performances of all the models for evaluating their hedging effectiveness. This allowed the models to be ranked based on a vector of nine financial and computational considerations including variance reduction, RMSE, Sharpe ratio, average hedged return, directional accuracy, Mean Absolute Deviation (MAD), time complexity, tail metrics i.e. Value-at-Risk (VaR) and Conditional VaR (CVaR) using Extreme Value Theory (EVT). The criteria was converted to fuzzy linguistic values, normalized and weighted to obtain the relative closeness of the models to the ideal solution. This produced clear results for both in-sample and out-of-sample analysis, ranking Ft-Net Hybrid model as the best performer among all models. This model showed highest reduction in portfolio variance, smallest RMSE in hedge ratio forecast, and highest directionally accuracy. The model also employed much better Sharpe ratios and diminished extreme tail risks more efficiently than other models. Although the Copula DCC-GARCH model can perform well in certain extreme market scenarios, it was not sufficiently flexible in the dynamic circumstances when compared with the deep learning models. The LSTM-CNN combination also did obtain good results, but underperformed FT-Net in terms of hedged returns.

In the third stage of the study, the concept of statistical arbitrage was investigated by determining whether hedge ratios produced by the models could be used to identify and profit from temporary mispricing between the spot index and the futures. We found that the spread signals from the FT-Net hybrid model were significantly mean-reverting in nature. Then we applied them to build the long-short strategies with deviation thresholds and rolling window measurements. The returns of these strategies were evaluated and once more, the FT-Net Hybrid stood out as the most effective model to yield efficient and economically significant arbitrage opportunities.

As an extension to our research, we also incorporated macroeconomic aspects into the FT-Net framework. Weekly GDP, Inflation and SBP Policy Rate were incorporated as exogenous features in a macro-informed FT-Net model. The exogenous variables were selected based on how predictive they are using extensive feature engineering which includes lagging, normalization, and correlation analysis. This model was trained on weekly returns data which showed significant improvements in the model and demonstrated hedging stability, especially when there was macroeconomic uncertainty. This illustrated the model's potential for

responding to exogenous variables and incorporating them into risk management and arbitrage considerations.

Overall, this study has made novel contributions to the literature on risk management and applications of deep learning in financial markets. The comparative results exhibit the dominance of the hybrid deep learning models to the standard econometric models in dynamically estimating the hedge ratios and discovering arbitrage opportunities. The FT-Net Hybrid, among all tested models, had the best financial performance, scalability, and adaptability to macroeconomic inputs.

8.2 Recommendations for Future Work

While this study offers a thorough assessment of dynamic hedge effectiveness and statistical arbitrage with traditional and deep learning approaches, there remains several opportunities for further research in this field. Therefore, based on the results and novel approaches that we provide in this paper, future research could be done to address the shortcomings, enhance the model's predictive capacity, and modify the modelling framework for broader financial applications.

The first recommended extension of this work would be to apply a multivariate asset framework where hedge ratios would be calculated for a diversified portfolio of assets, rather than for an index. This would enable portfolio managers and researchers to capture the cross-asset relationships, sectoral dependencies, and correlations between multiple equity instruments. By doing so, deeper understanding of portfolio-level hedging under dynamic and nonlinear settings may be established using models such as multivariate DCC-GARCH, vine copulas, or attention-based transformer networks.

Secondly, future work can include higher frequency data. As this study is limited by the nature of the accessible data and market liquidity, due to which we examined only on daily and weekly frequencies, however, modeling at finer intraday frequencies could provide more refined and timelier measures of volatility dynamics and arbitrage signals. Using tick or minute level data, especially when using deep reinforcement learning may enhance model adaptability for high-frequency trading (HFT) or intraday hedging. This, however, would depend on the existence of

efficient data processing pipelines, low latency set-up and careful tuning for microstructural noise.

Furthermore, improving the macroeconomic integration features can provide better results in future research. Although the present paper has introduced macro variables including GDP, inflation and policy rate, it is promising to consider a more comprehensive set of economic series such as industrial production, foreign exchange reserves, fiscal balance, global market-related-variables including crude oil price, US treasury yield and international volatility indices. Therefore, to capture the behavior of market shifts and macro-financial regimes more effectively, time-varying parameter models, such as Bayesian frameworks or interpretable deep learning structures such as Temporal Fusion Transformers (TFTs) may be used.

Additionally, this research considers transaction costs to be relatively constant when testing profitability of arbitrage opportunities. In future work we recommend exploring the possibility of adopting dynamic transaction costs involving the liquidity state, bid-ask spread, slippage, and execution delay. This would make it possible to evaluate arbitrage strategies in a better way, especially where the market is less liquid or is developing. Moreover, adding regulatory constraints like short-sale restrictions, leverage caps, or margin requirements would give more support to the above arbitrage strategies.

Finally, in order to enable innovation in future work, we recommend efforts to develop modular and reusable code libraries or APIs that encapsulate the hedging and arbitrage models we have made available in this work. These could potentially be released for other researchers, financial analysts, and developers who could then test them across other markets, asset classes or time periods. A system combined with real-time data and cloud-based technology can learn about a market's evolving structure over time and improve its hedging performance.

9 References

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Appendix A: Statistical Diagnostic Tests on Residuals Post Model Implementation

The statistical tests, i.e. ARCH-LM, Ljung Box, and KPSS, were performed on residuals of both mathematical and machine learning models. The future and spot return series of training and test data sets were used in the mathematical models. These models are represented as: DCC-GARCH (MM1), and Copula DCC GARCH (MM2). The results of these tests were obtained from R programming language. Whereas, hedged return series were used in the machine learning models, with results of each dataset computed in python. The p-values of these tests are below,

Table 5: Statistical Tests for Residuals on Training Data – Mathematical Model

Test Type	MM1 (Ft)	MM1 (St)	MM2 (Ft)	MM2 (St)
ARCH LM	0.06452	0.05857	0.06754786	0.81109125
Ljung-Box	0.8779	0.8253	0.9279456	0.9302992
KPSS	0.1	0.1	0.1	0.1

Table 6: Statistical Tests for Residuals on Testing Data – Mathematical Model

Test Type	MM1 (Ft)	MM1 (St)	MM2 (Ft)	MM2 (St)
ARCH LM	0.1427	0.1687	0.1427	0.1687
Ljung-Box	0.3071	0.2302	0.3071	0.2302
KPSS	0.1	0.1	0.1	0.1

Table 7: Statistical Tests for Residuals – LSTM-CNN

Test Type	Training p-value	Testing p-value
KPSS Test	0.0692	0.1000
Ljung-Box (10)	0.5791	0.9986
ARCH LM	1.0000	1.0000

Table 8: Statistical Tests for Residual – Ft-Net Hybrid

Test Type	Training p-value	Testing p-value
KPSS Test	0.1000	0.1000
Ljung-Box (10)	0.6321	0.9911
ARCH LM	1.0000	1.0000

Appendix B: Data Collection and Processing of Macro Economic Variables

In order to enhance the predictive power and macroeconomic relevance of the top-performing hedging model (as determined by the fuzzy TOPSIS methodology), supplementary macroeconomic variables were incorporated alongside the primary data on the KSE-30 index and its associated futures contracts. The macroeconomic variables utilized in this research include real GDP growth, the State Bank of Pakistan (SBP) policy rate, and the weekly Sensitive Price Index (SPI) as a proxy for inflation. The data for these variables were sourced from official repositories, specifically the State Bank of Pakistan and the Pakistan Bureau of Statistics.

A significant methodological challenge was reconciling the difference in data frequency between macroeconomic variables and financial market data. While the index and futures data were available on a daily and weekly basis, macroeconomic indicators such as GDP and policy rates are generally published at lower frequencies i.e., quarterly for GDP and annually for policy rates. Therefore, interpolation procedures were undertaken to harmonize the frequency of these variables with that of the financial dataset.

Interpolation of weekly GDP using Quarterly GDP

To interpolate the weekly GDP from quarterly GDP we took inspiration from the methodology used by (Tahir et. al., 2018) for the quarterization of annual GDP. In his methodology, he used quarterly macroeconomic variables to extract a common orthogonal factor to include the business cycle dynamics and use a time series equation to disaggregate the annual GDP into quarterly frequency. Therefore, replicating this methodology and using the theoretical basis and equation of the Quantity Theory of Money as discussed by (Friedman, 1989) we disaggregated the quarterly GDP into weekly GDP. The following is the foundation of the methodology we've used.

The theoretical basis developed using the Quantity Theory of Money

The quantity theory of money is a classical economics theory explaining the relationship between the money supply and price level in an economy. We started by using the basic equation that implies this relationship. (equation 66) Here for simplicity, we assume that the velocity of money is constant.

Equation 64

$$M \cdot V = P \cdot Y$$

Where M represents the money supply, V is the velocity of money, P is the price level and Y represents the real output or real GDP.

However, here to focus on our objective we assume that at week 0 the equation can be written as equation 67 below:

Equation 65

$$M_0 V = P_0 Y_0$$

Subsequently, at week 1, i.e., for any week in the future, the equation can be written as:

Equation 66

$$M_1 V = P_1 Y_1$$

Dividing both the equations we get:

Equation 67

$$\frac{M_1}{M_0} = \frac{P_1}{P_0} \cdot \frac{Y_1}{Y_0}$$

Let $\frac{M_1}{M_0}$ be defined as the money supply index denoted by MI and $\frac{P_1}{P_0}$ be defined as the price index denoted by PI. Therefore, we can conclude by rewriting the equation that the weekly GDP at time t equals the product of the GDP at t = 0 and the money supply index divided by the price index. (see equation 70)

Equation 68

$$Y_1 = \frac{MI}{PI} \cdot Y_0$$

Here, MI and PI are constructed using available high-frequency proxies, including weekly SPI

data to reflect inflationary movements (PI) and relevant money supply aggregates (M1), which collectively anchor the weekly GDP estimate to real macroeconomic movements. The figure 38 presents the results of the weekly gdp interpolation.

Weekly SPI and Policy Rate Frequency Conversion

The weekly SPI, already available at the required frequency from the Pakistan Bureau of Statistics, was directly incorporated as the inflation measure. The SBP policy rate, however, is typically reported at an annual frequency. To align this variable with the weekly financial dataset, the annual policy rate was systematically converted to a weekly equivalent. This was achieved using standard frequency conversion formulas, analogous to those employed when adjusting interest rates from annual to quarterly or monthly bases in macro-financial analyses. Specifically, the annual rate was transformed into an equivalent weekly rate using the following mathematical transformation:

Equation 69

$$r_{weekly} = (1 + r_{annual})^{(1/52)} - 1$$

where r_{annual} denotes the annual policy rate, and r_{weekly} represents its weekly equivalent. This approach preserves the compounding characteristics of the rate over a year, ensuring theoretical and empirical consistency with the high-frequency financial data employed in subsequent modeling.

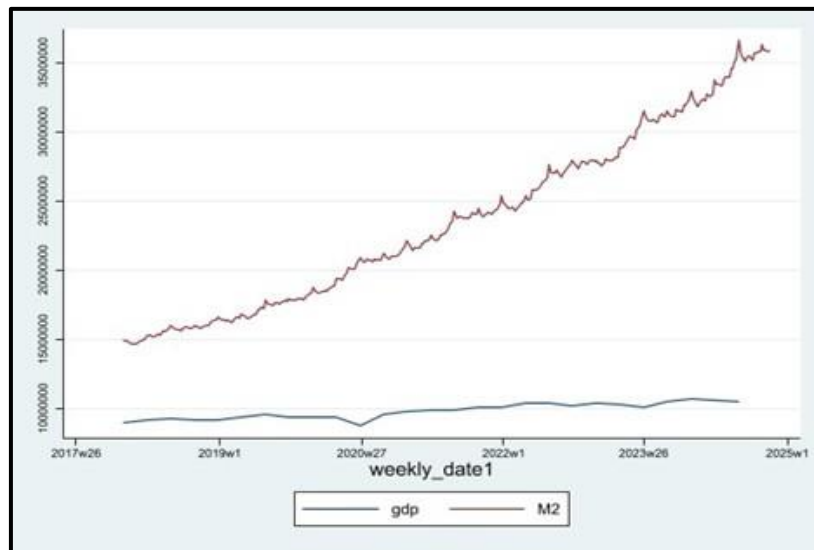


Figure 38: Weekly Interpolated GDP and M2 Money Supply



F/SOP FYDP/09/00